Biochemistry



Section C *Enzyme*

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Section C - Enzyme

C1C2 Introduction to Enzymes

C3 Enzyme Kinetics

C4 Enzyme inhibition

C5 Regulation of Enzyme activity

C1C2 Introduction to Enzymes

1. **Enzymes were among the first biological macromolecules to be studied chemically.**

   1.1 Much of the early history of biochemistry is the history of enzyme research.

1.1.1 Biological catalysts were first recognized in studying animal food digestion and sugar fermentation with yeast (brewing and wine making).

1.1.2 Ferments (i.e., enzymes, meaning in "in yeast") were thought (wrongly) to be inseparable from living yeast cells for quite some time (Louis Pasteur)

1.1.3 Yeast extracts were found to be able to ferment sugar to alcohol (Eduard Buchner, 1897, who won the Nobel Prize in Chemistry in 1907 for this discovery).

1.1.4 Enzymes were found to be proteins (1920s to 1930s, James Sumner on urease, "all enzymes are proteins", John Northrop on pepsin, chymotrypsin and trypsin, both shared the 1946 Nobel Prize in Chemistry).

1.1.5 Almost every chemical reaction in a cell is catalyzed by an enzyme (thousands have been purified and studied, many more are still to be discovered!)

1.1.6 Proteins do not have the absolute monopoly on catalysis in cells. Catalytic RNA were found in the 1980s (Thomas Cech, Nobel Prize in Chemistry in 1989).

2. The most striking characteristics of enzymes are their immense catalytic power and high specificity.

2.1 Enzymes accelerate reactions by factors of at least a million.

2.1.1 Most reactions in biological systems do not occur at perceptible rates in the absence of enzymes.

2.1.2 The rate enhancements (rate with enzyme catalysis divided by rate without enzyme catalysis) brought about by enzymes are often in the range of 10^8 to 10^{20})

2.1.3 For carbonic anhydrase, an enzyme catalyzing the hydration of CO_2 ($H_2O + CO_2 \rightarrow HCO_3^- + H^+$), the rate enhancement is 10^7 (each enzyme molecule can hydrate 10^5 molecules of CO_2 per second!)

table 8–5

Some Rate Enhancements Produced by Enzymes

Cyclophilin	10^5
Carbonic anhydrase	10^7
Triose phosphate isomerase	10^9
Carboxypeptidase A	10^{11}
Phosphoglucomutase	10^{12}
Succinyl-CoA transferase	10^{13}
Urease	10^{14}
Orotidine monophosphate decarboxylase	10^{17}

2.2 Enzymes are highly specific both in the reaction catalyzed and in their choice of substrates (i.e., reactants).

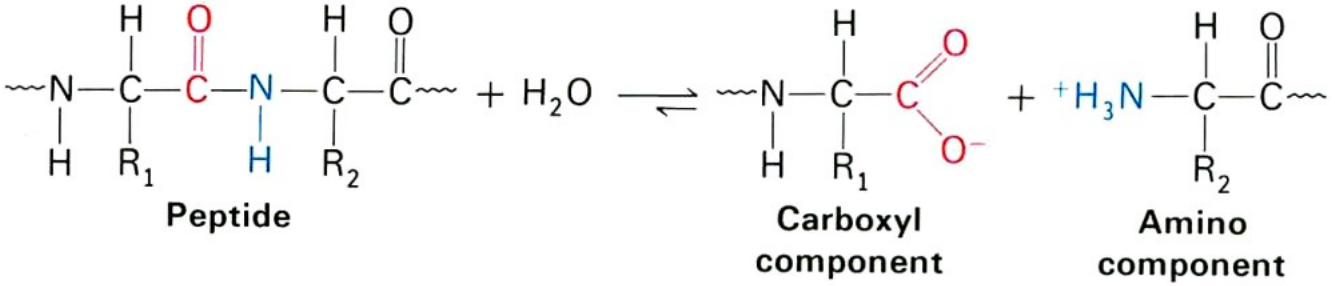
2.2.1 Enzymes exhibit various degrees of specificity in accord with their physiological functions :

Low specificity: some peptidases, esterases, and phosphatases.

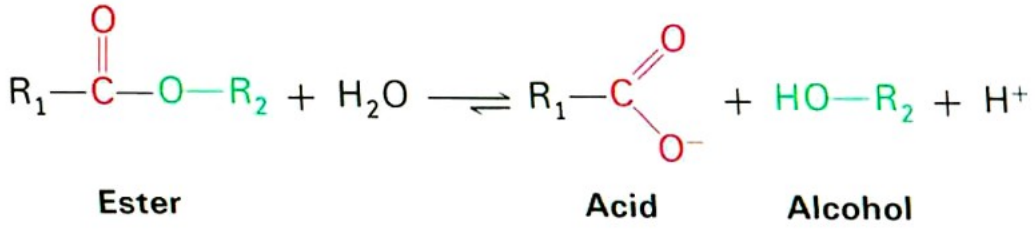
Intermediate specificity: hexokinase, alcohol dehydrogenases, trypsin.

Absolute or near absolute specificity: Many enzymes belong to this group, and in extreme cases, stereochemical specificity is exhibited (i.e., enantiomers are distinguished as substrates or products).

Let us consider *proteolytic enzymes* as an example. The reaction catalyzed by these enzymes is the hydrolysis of a peptide bond.



Most proteolytic enzymes also catalyze a different but related reaction, namely, the hydrolysis of an ester bond.



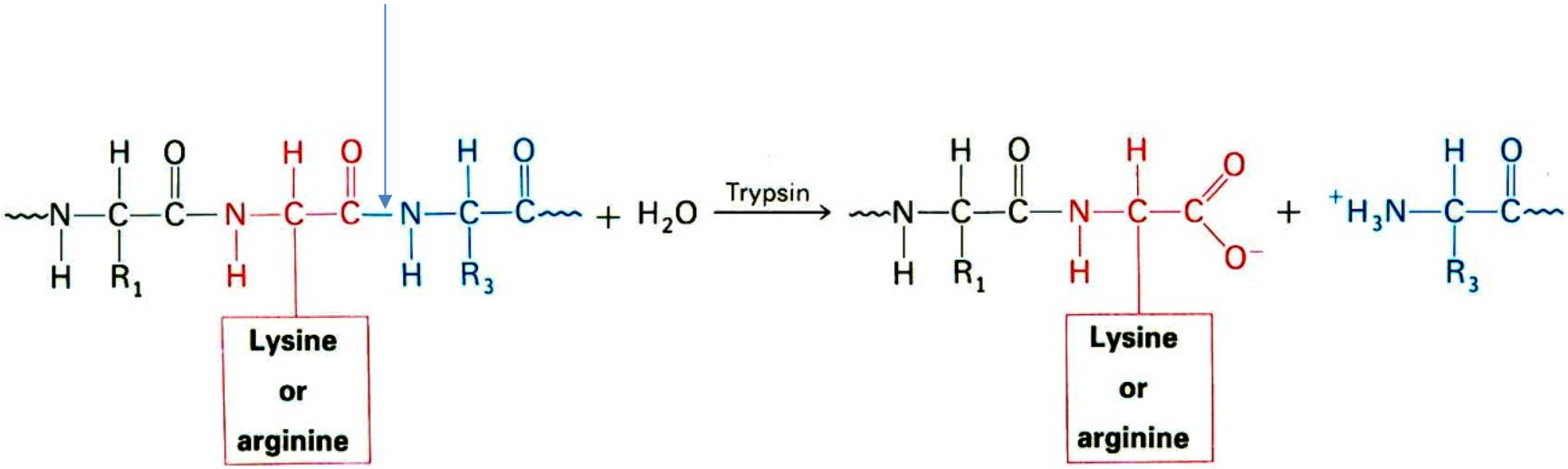
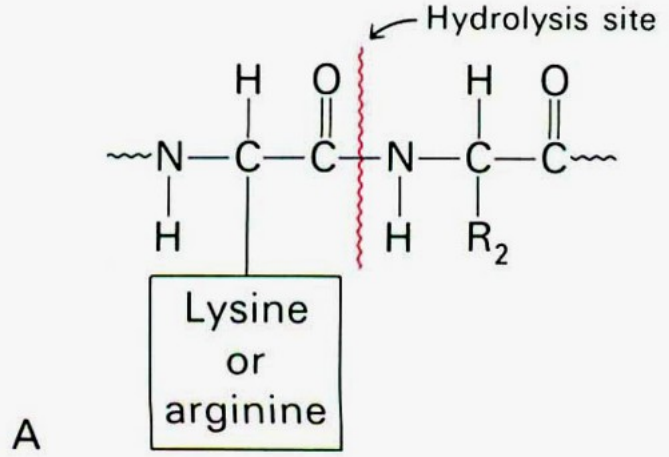


Figure 3-21
Trypsin hydrolyzes polypeptide on the carboxyl side of arginine and lysine residues.



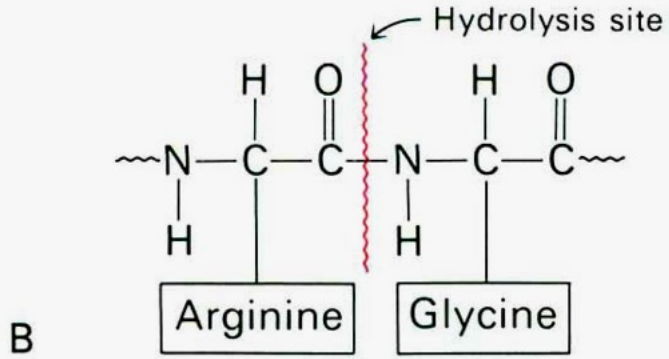


Figure 8-1
Comparison of the specificities of (A) trypsin and (B) thrombin. Trypsin cleaves on the carboxyl side of arginine and lysine residues. Thrombin cleaves Arg-Gly bonds in particular sequences only.

thrombin
n. 凝血酶

2.3 Most enzymes are proteins.

2.3.1 Some enzymes require no other chemical groups other than their amino acid residues for activity.

2.3.2 Some enzymes require the presence of cofactors, small nonprotein units, to function. Cofactors may be inorganic ions or complex organic molecules called coenzymes. A cofactor that is covalently attached to the enzyme is called a prosthetic group.

2.3.3 Prosthetic groups could be inorganic metal ions (e.g., Fe^{2+}, Mg^{2+}, Mn^{2+}, Zn^{2+}) or complex organic or metalloorganic molecules.

2.3.4 A complete catalytically active enzyme (including its prosthetic group) is called a holoenzyme.

2.3.5 The protein part of an enzyme (without its prosthetic group) is called the apoenzyme.

2.3.6 Many vitamins, organic nutrients required in small amounts in the diet, are precursors of coenzymes.

酶的组成

单纯蛋白酶类：在已发现的酶中，象脲酶、蛋白酶、淀粉酶、脂酶和核糖核酸酶等一般水解酶都属于简单蛋白质，这些酶只是由氨基酸组成，此外不含其他成分，其活性仅仅决定于它的蛋白质结构。

结合蛋白酶类：另一些酶如转氨酶、碳酸酐酶、脱氢酶等均属于结合蛋白质。这些酶其蛋白质部分称之为酶蛋白，非蛋白质部分称为辅因子（或辅助因子）。酶蛋白与辅因子单独存在时，均无催化活力，或活性很弱，结合在一起组成全酶后，才表现有明显的催化作用。

全酶 = 酶蛋白 + 辅因子（金属离子、辅酶、辅基）

（有活性） （无活性） （无活性）

酶的组成

辅因子：辅酶、辅基、金属离子

辅酶：与酶蛋白结合较松，可透析除去
辅基：与酶蛋白结合较紧，不能透析除去
} 小分子有机物

金属离子：与酶的结合，有的松，有的紧

辅酶、辅基往往是由维生素参与形成的小分子有机物

酶的组成

维生素：

维生素是维持机体生命活动不可缺少的一类小分子化合物，它既不是生物体构成成分，也不是能量物质，之所以对生命活动如此重要，是因为维生素是辅酶或辅基的组成成分，参与体内代谢过程。

维生素 { 脂溶性维生素
水溶性维生素

酶的组成

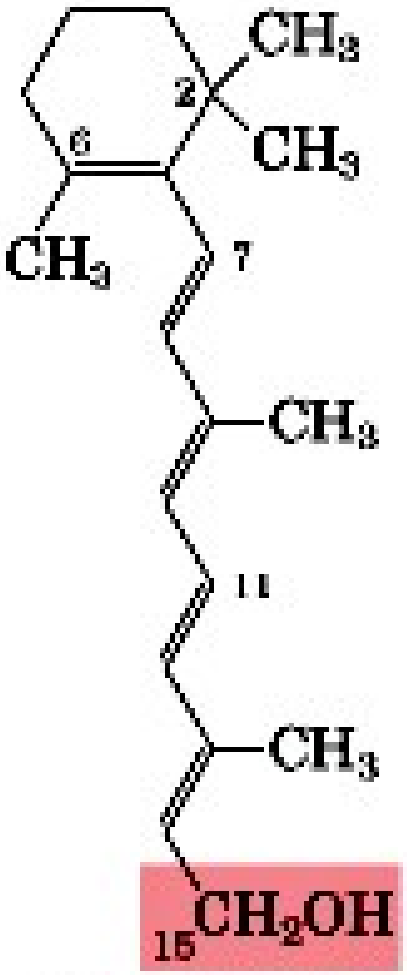
维生素：

脂溶性维生素

种类	化学本质	功能
维生素A	不饱和一元醇	缺乏得干眼病、夜盲症
维生素D	类固醇衍生物	与Ca、P代谢有关
维生素E	生育酚	与生育有关，抗氧化
维生素K	奈醌衍生物	凝血维生素

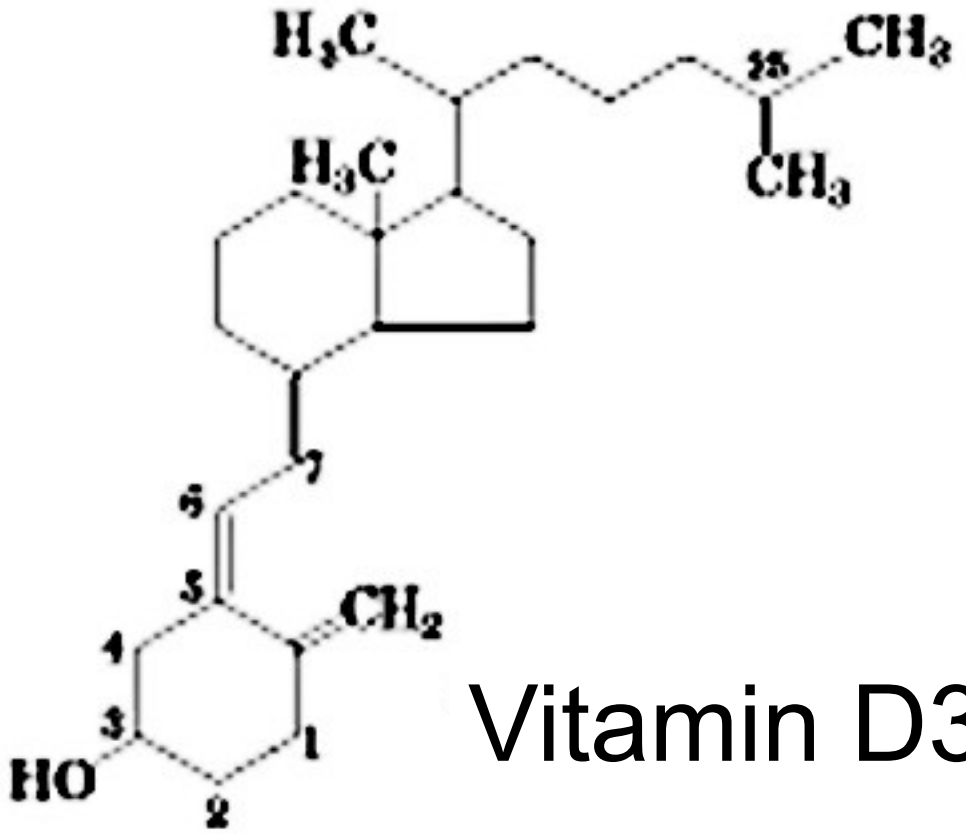
脂溶性维生素：

A、D、E、K



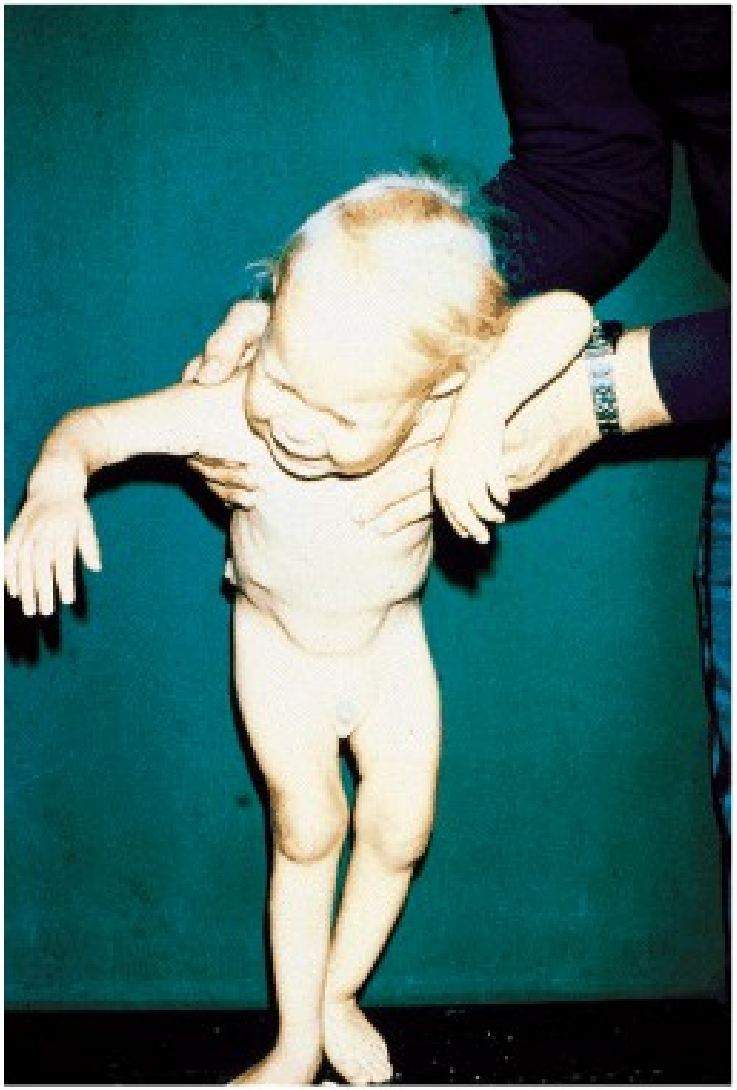
Vitamin A_1
(retinol)
(b)

视黄醇



Vitamin D3

甾醇衍生物

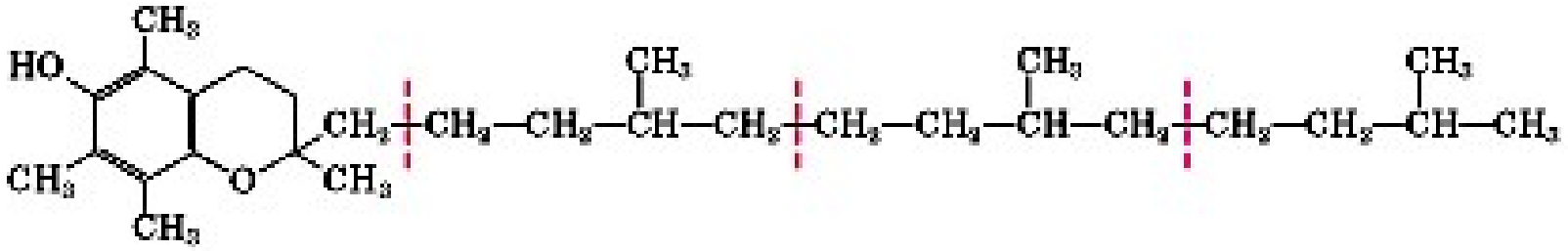


Before vitamin D treatment

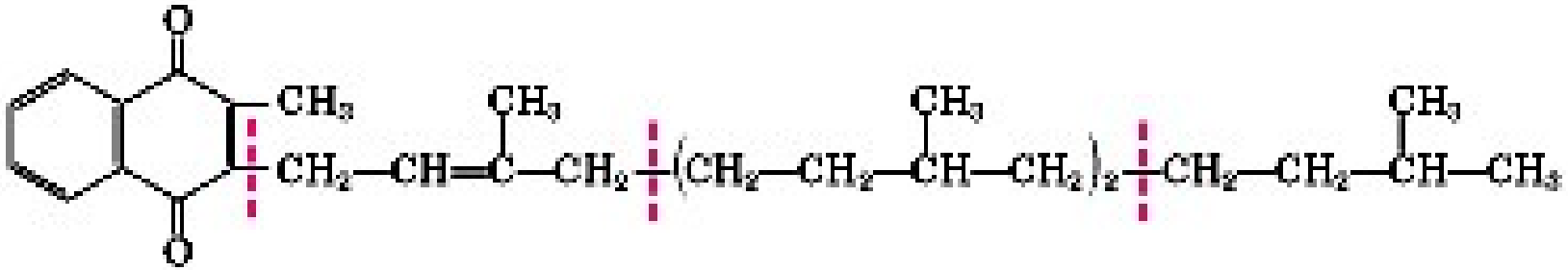


After 14 months of vitamin D treatment

(b)



Vitamin E　　生育酚



Vitamin K　　凝血维生素

酶的组成

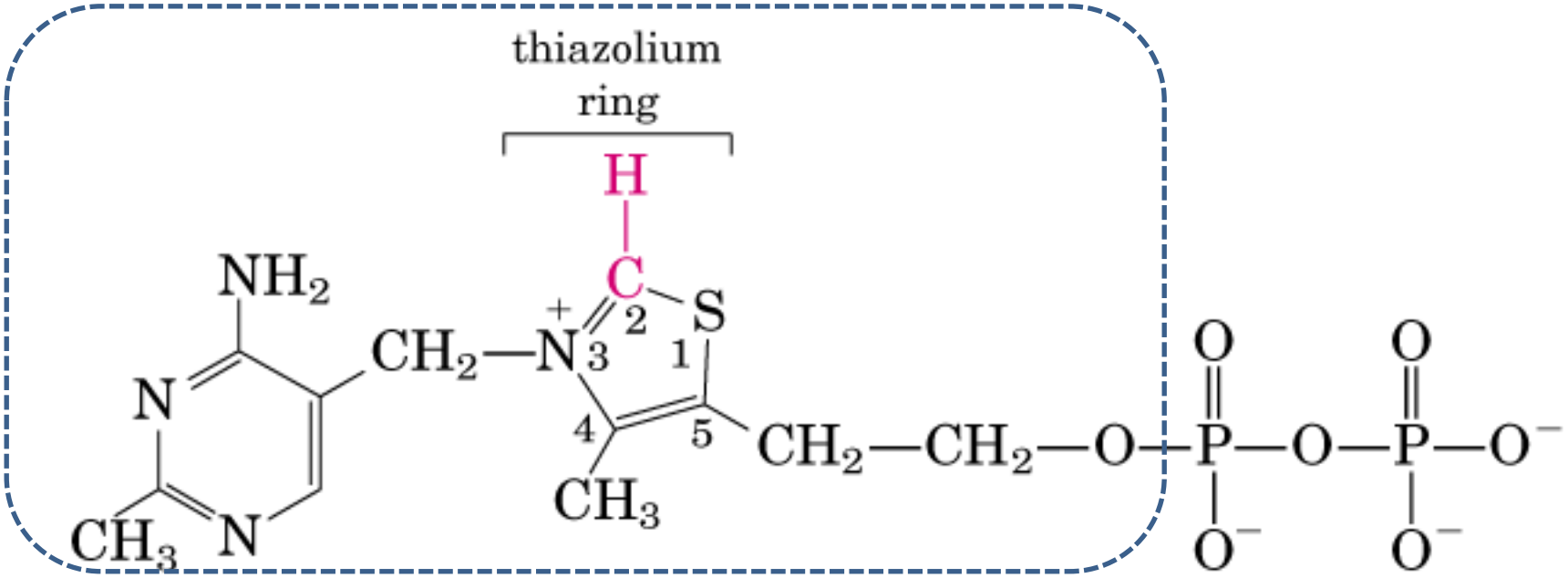
维生素：

水溶性维生素：

种类	别名	存在	缺乏病
B_1	硫胺素	谷物种子	脚气病
B_2	核黄素	小麦、青菜、蛋黄、肝脏	口角炎
B_3	烟酸（烟酰胺）	肉类、谷物、花生	赖皮病
B_5（Vpp）	泛酸（遍多酸）	自然界中广泛	未发现典型缺乏病
B_6	吡哆素	动、植物分布广泛	皮炎
B_7（V_H）	生物素	动、植物分布广泛	未发现典型缺乏病
B_{11}	叶酸	青菜、肝脏	恶性贫血
B_{12}	氰钴胺素	植物、鱼、肉、蛋	恶性贫血
Vc	抗坏血酸	新鲜果蔬	坏血病
硫辛酸		肝、酵母	未发现典型缺乏病

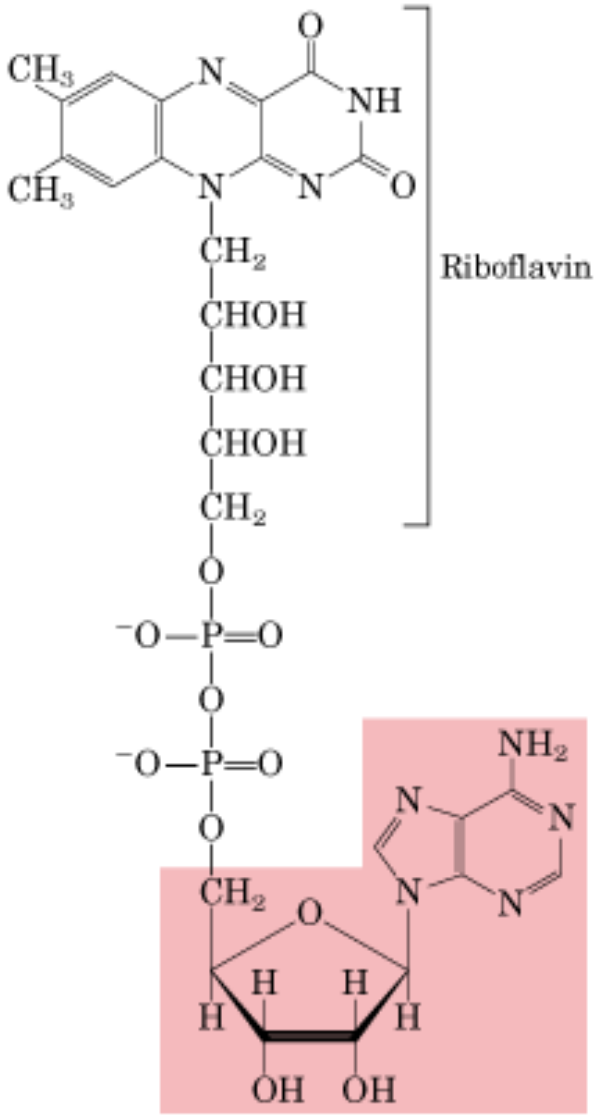
水溶性维生素： B 族、硫辛酸和维生素 C

B_1	硫胺素	谷物种子



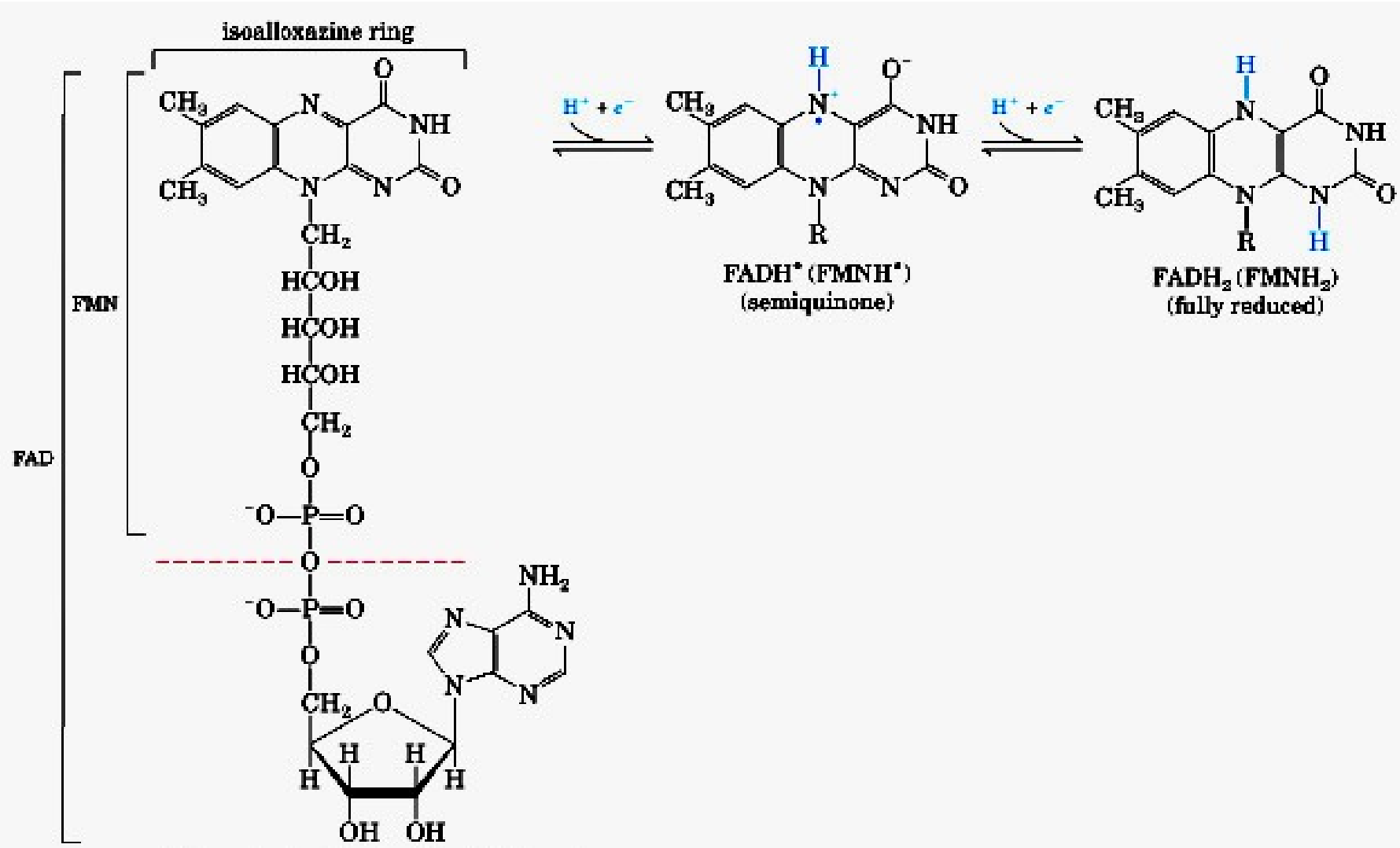
Thiamine pyrophosphate (TPP)

(a)

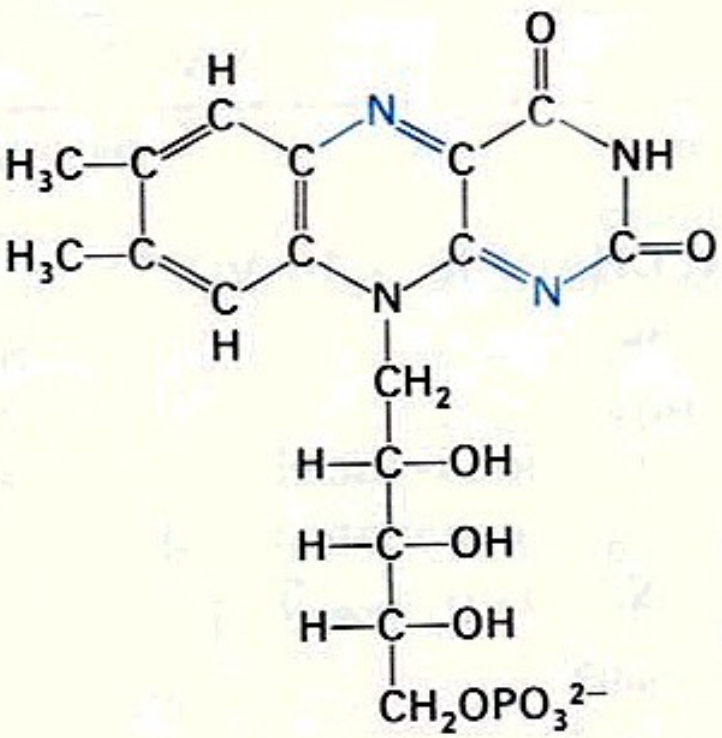


Flavin adenine dinucleotide (FAD)

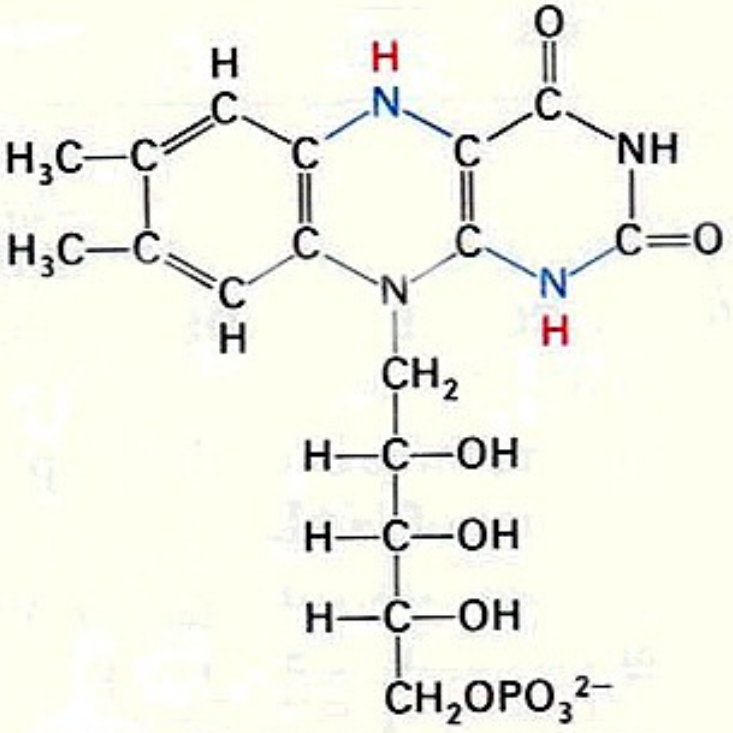
B_2	核黄素	小麦、青菜、蛋黄、肝脏



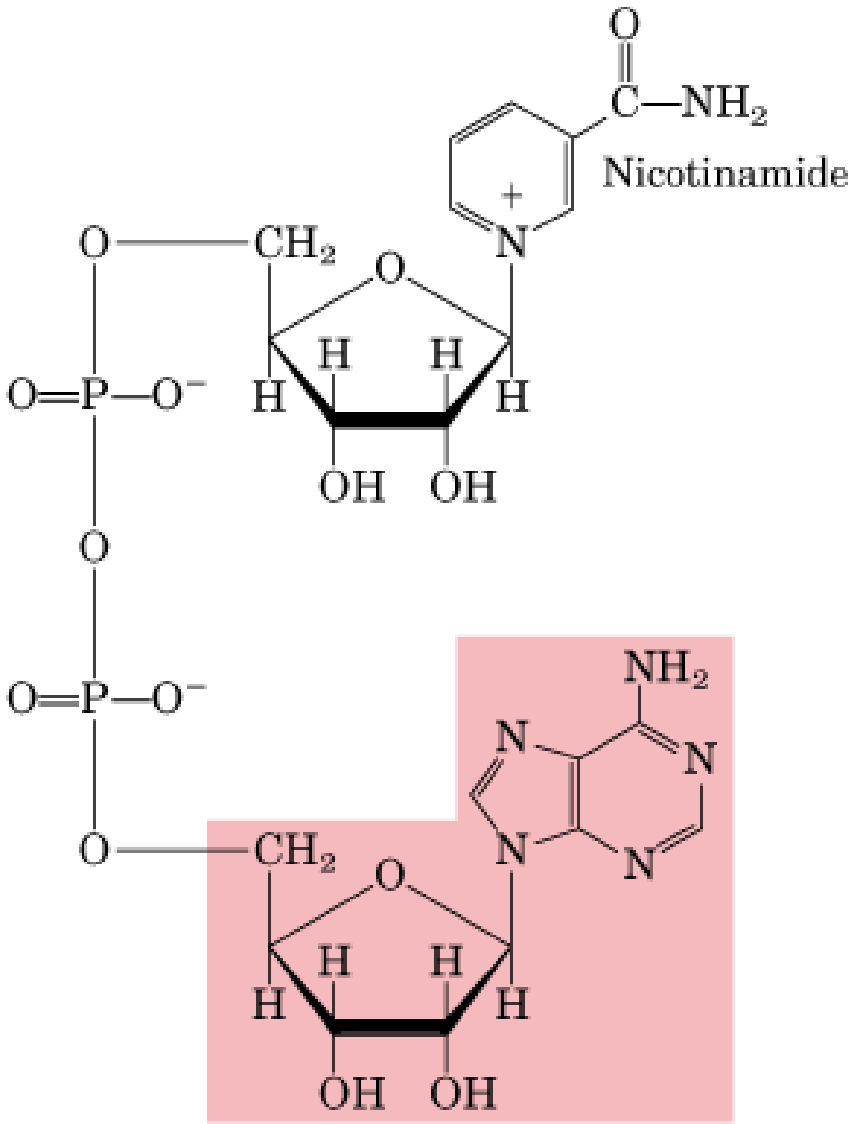
Flavin adenine dinucleotide (FAD) and flavin mononucleotide (FMN)


H
O
N
C
NH
H_3C
H_3C
C
C
C
C
C
C
C
C
C
C
C=O
H
N
N
CH_2
H—C—OH
H—C—OH
H—C—OH
$CH_2OPO_3^{2-}$

(FMN)

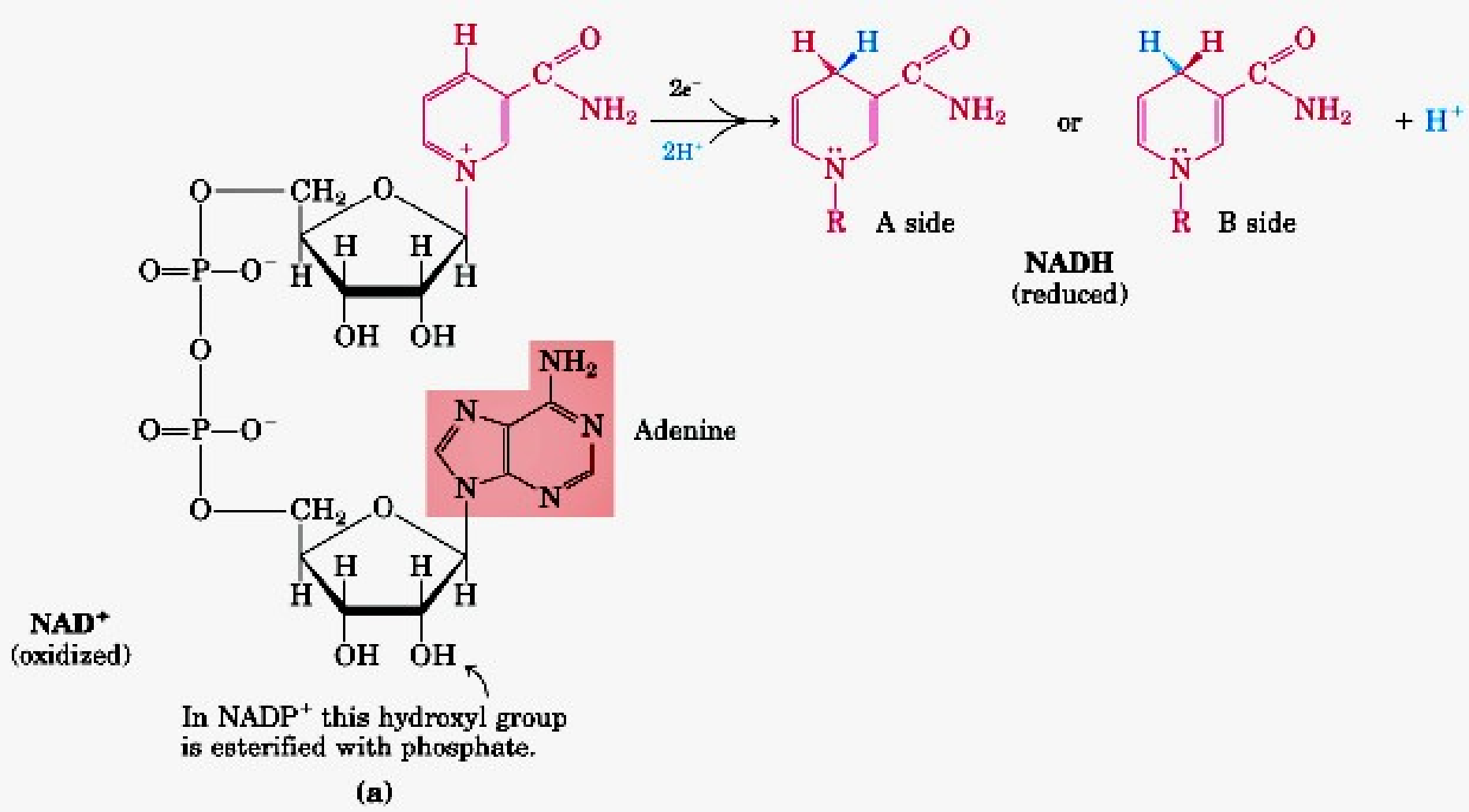

H
H
O
N
C
NH
H_3C
H_3C
C
C
C
C
C
C
C
C
C
C
C=O
H
N
N
H
CH_2
H—C—OH
H—C—OH
H—C—OH
$CH_2OPO_3^{2-}$

(FMNH$_2$)



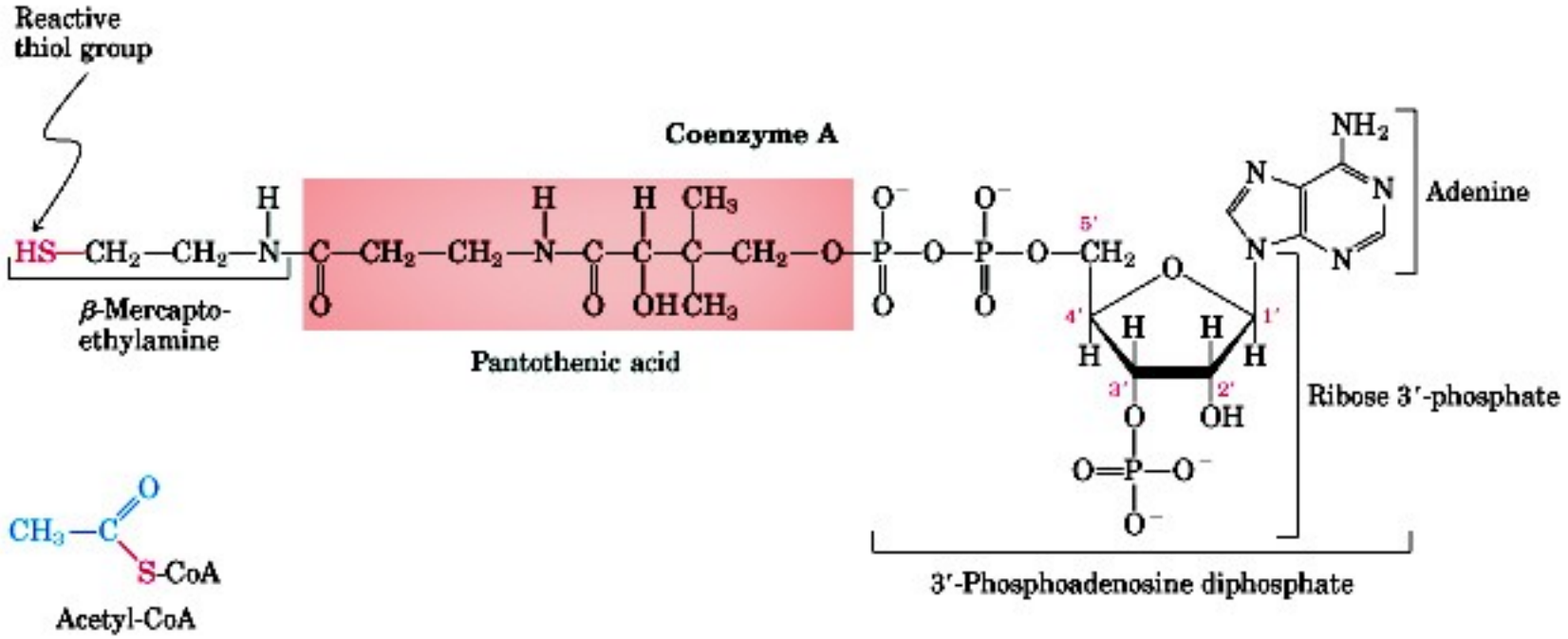
Nicotinamide adenine dinucleotide (NAD^+)

B3	烟酸（烟酰胺）	肉类、谷物、花生



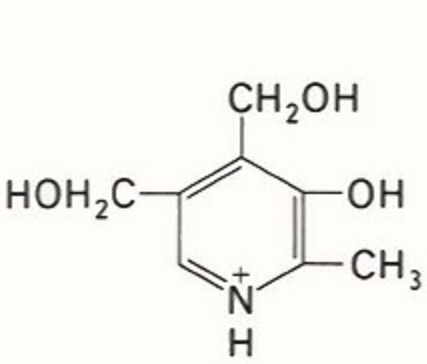
NAD / NADP

辅酶 I / 辅酶 II

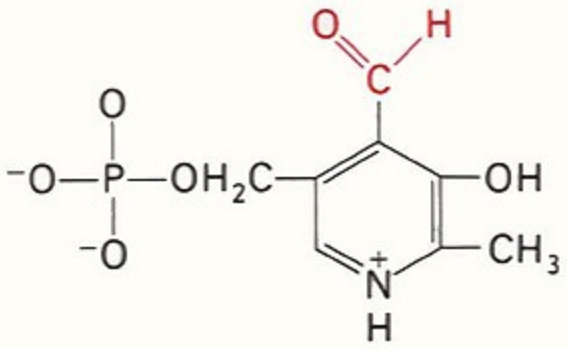


B5 泛酸(遍多酸) | 自然界中广泛 | 未发现典型缺乏病

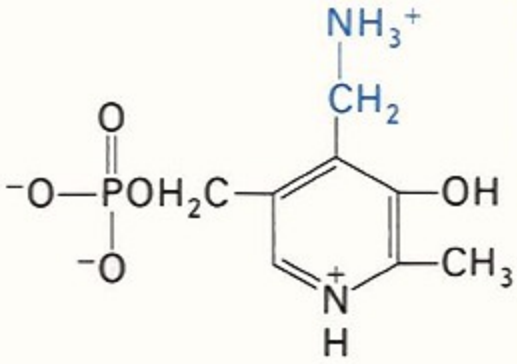
Coenzyme A 辅酶A



吡哆醇

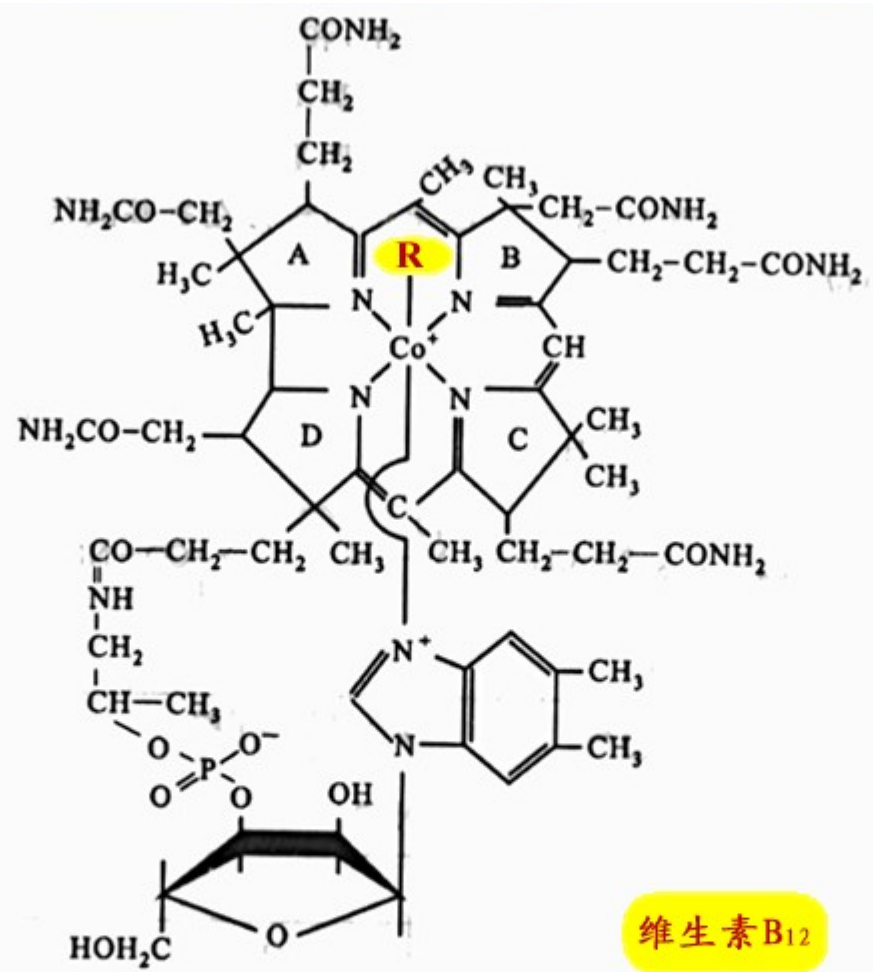


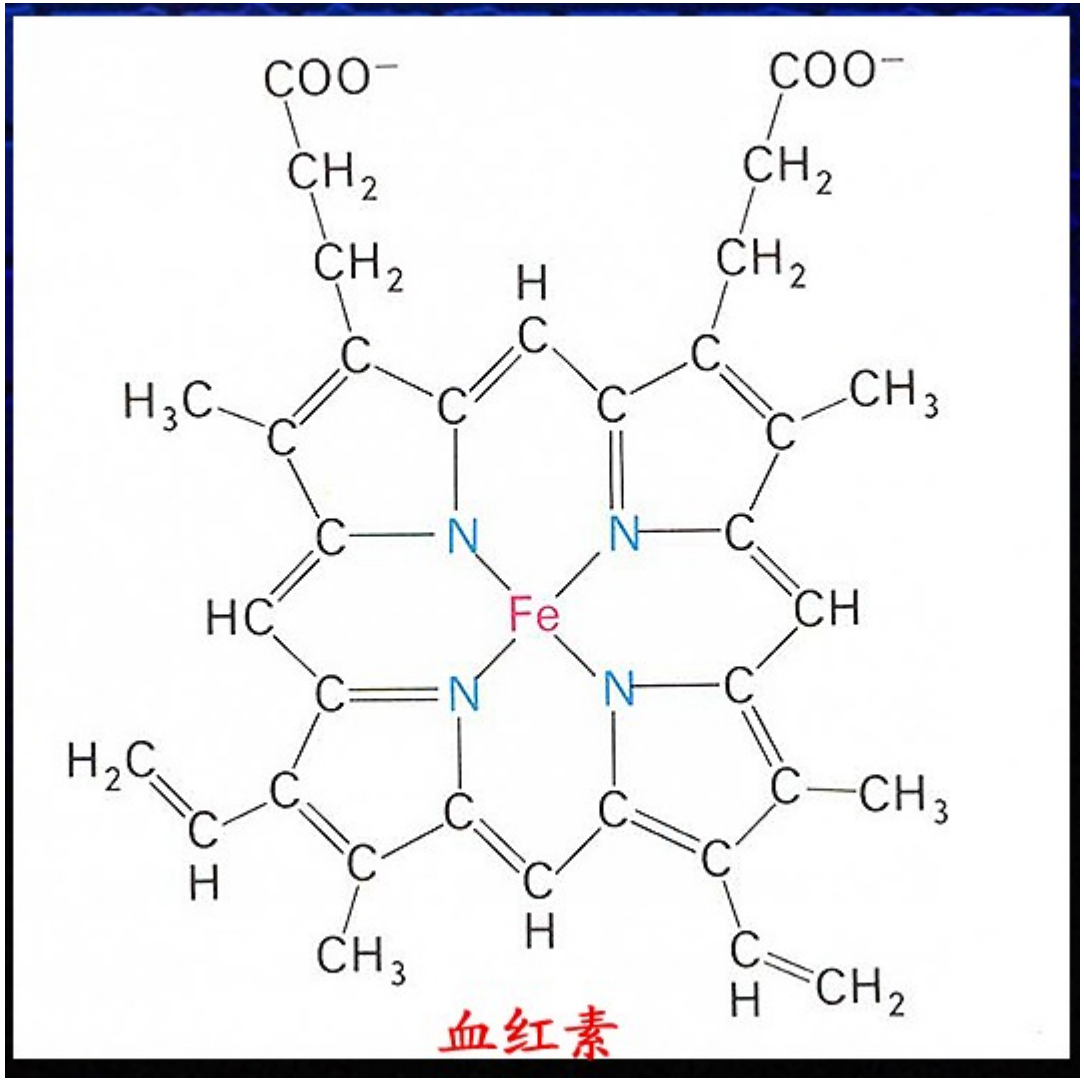
磷酸吡哆醛 PLP



磷酸吡哆胺

B6	吡哆素	动、植物分布广泛


CONH2
CH2
CH2
NH2CO-CH2
H3C
H3C
A
R
B
CH2-CONH2
CH2-CH2-CONH2
N
N
Co+
N
N
CH
NH2CO-CH2
D
C
CH3
CH3
CO-CH2-CH2
CH3
CH3
CH2-CH2-CONH2
NH
CH2
CH-CH3
O
O-
P
O
O
OH
HOH2C
O
N+
N
CH3
CH3
维生素B12
R:
—CN
5′-脱氧腺苷
—OH
—CH3
名称：
氰钴胺素
5′-脱氧腺苷钴胺素
羟钴胺素
甲基钴胺素


COO⁻
CH2
CH2
COO⁻
CH2
CH2
H
H3C
CH3
N
N
Fe
HC
CH
N
N
H2C
CH3
CH3
CH
CH2
血红素

Fe^{2+}
ferrous

table 8–2

Some Coenzymes That Serve as Transient Carriers of Specific Atoms or Functional Groups*

Coenzyme	Examples of chemical groups transferred	Dietary precursor in mammals
Biocytin	CO_2	Biotin
Coenzyme A	Acyl groups	Pantothenic acid and other compounds
5′-Deoxyadenosylcobalamin (coenzyme B_{12})	H atoms and alkyl groups	Vitamin B_{12}
Flavin adenine dinucleotide	Electrons	Riboflavin (vitamin B_2)
Lipoate	Electrons and acyl groups	Not required in diet
Nicotinamide adenine dinucleotide	Hydride ion ($:H^-$)	Nicotinic acid (niacin)
Pyridoxal phosphate	Amino groups	Pyridoxine (vitamin B_6)
Tetrahydrofolate	One-carbon groups	Folate
Thiamine pyrophosphate	Aldehydes	Thiamine (vitamin B_1)

*The structure and mode of action of these coenzymes are described in Part III of this book.

table 8–1

Some Inorganic Elements That Serve as Cofactors for Enzymes

Cu^{2+}	Cytochrome oxidase
Fe^{2+} or Fe^{3+}	Cytochrome oxidase, catalase, peroxidase
K^{+}	Pyruvate kinase
Mg^{2+}	Hexokinase, glucose 6-phosphatase, pyruvate kinase
Mn^{2+}	Arginase, ribonucleotide reductase
Mo	Dinitrogenase
Ni^{2+}	Urease
Se	Glutathione peroxidase
Zn^{2+}	Carbonic anhydrase, alcohol dehydrogenase, carboxypeptidases A and B

3. Enzymes are classified by the reactions they catalyze

3.1 Trivial names are usually given to enzymes.

3.1.1 Many enzymes have been named by adding the suffix "-ase" to the name of their substrate or to a word or phrase describing their activity (type of reaction).

3.2 Enzymes are categorized into six major classes by international agreement.

3.2.1 The six major classes include

Oxidoreductases: catalyzing oxidation-reduction reactions.

Transferases: catalyzing the transfer of a molecular group from one molecule to another.

Hydrolases: catalyzing the cleavage by the introduction of water.

Lyases(such as synthase, hydratase): catalyzing reactions involving removal of a group to form a double bond or addition of groups to double bonds.

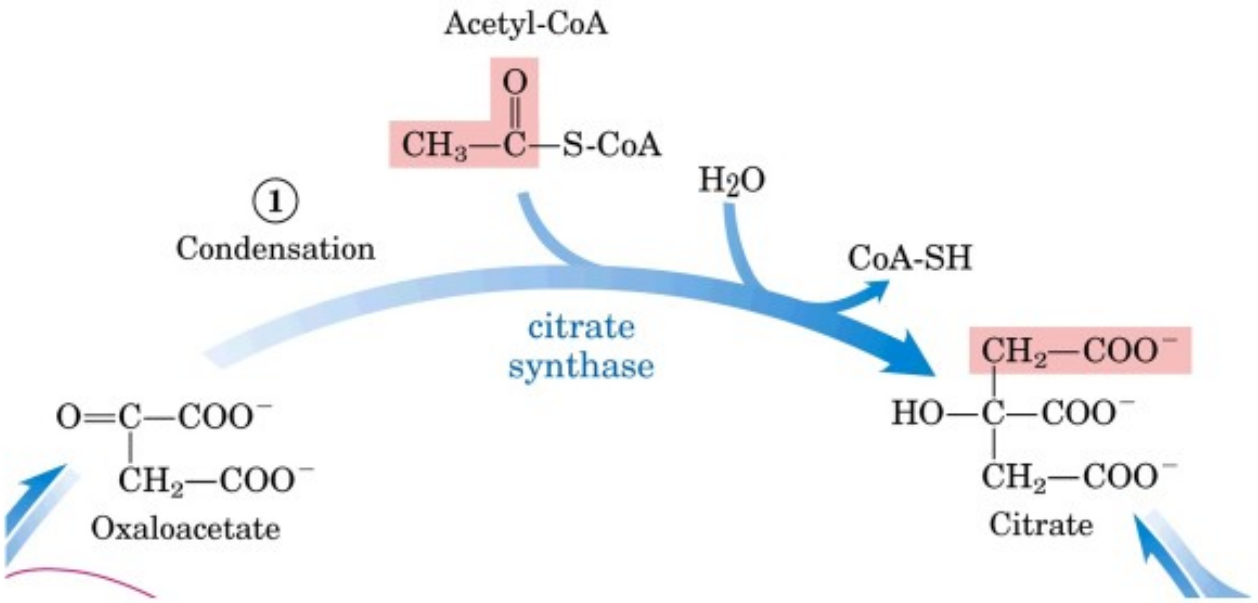
Isomerases: catalyzing reactions involving intramolecular rearrangements.

Ligases (such as synthetases): catalyzing reactions joining together two molecules.

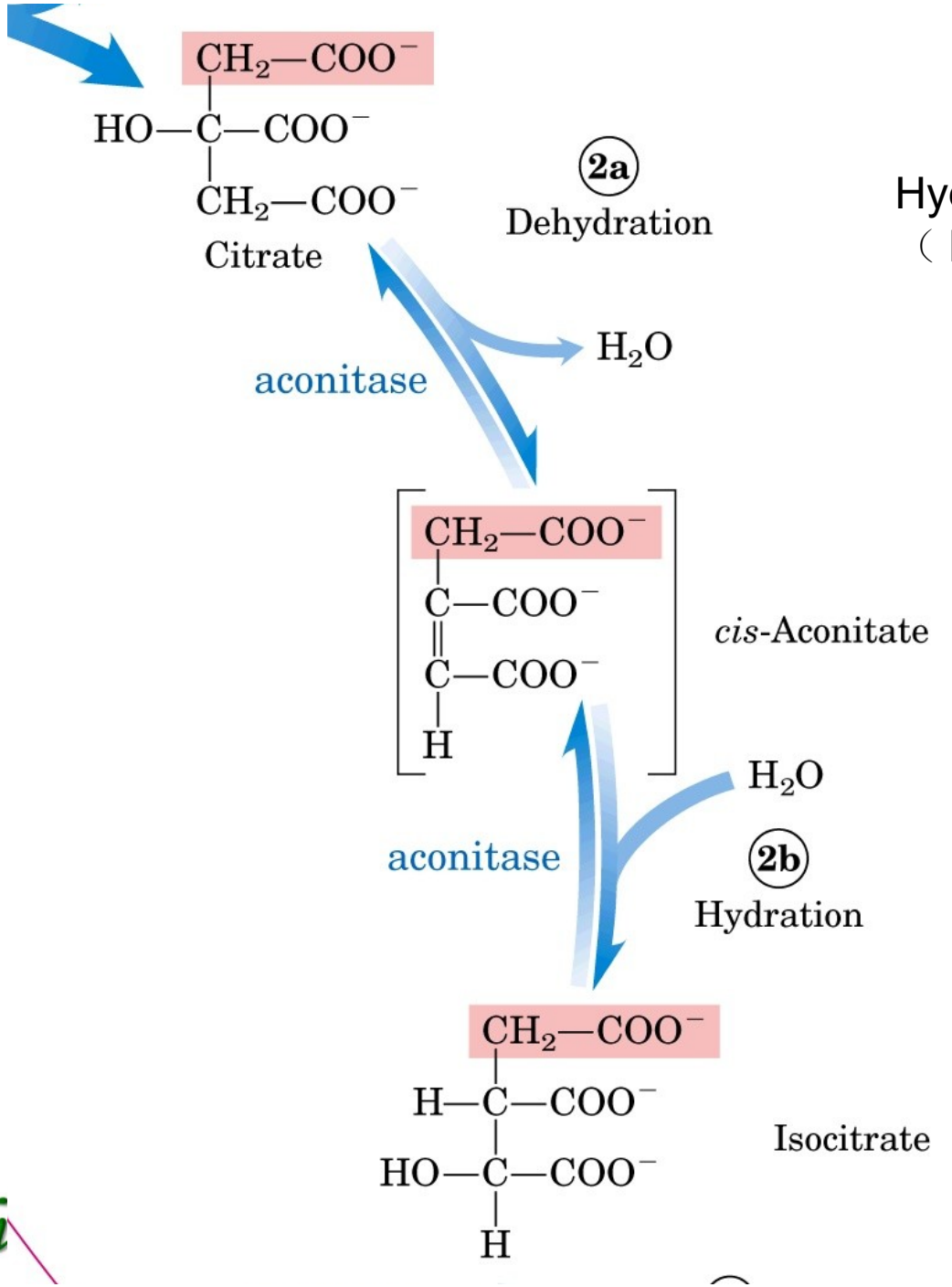
3.2.1 酶的分类

- 1. 氧化还原酶类 $AH_2 + B \rightleftharpoons A + BH_2$
- 2. 转移酶类 $AR + B \rightleftharpoons A + BR$
- 3. 水解酶类 $AB + H_2O \rightleftharpoons AOH + BH$
- 4. 裂合酶类 $AX \rightleftharpoons A=B + X-Y$
- 5. 异构酶类 $\rightleftharpoons$ A B
- 6. 合成酶类 $A + B + ATP \rightleftharpoons AB + ADP + Pi$

举例 synthase （ lyase ）



Biochemistry



Hydratase
（ lyase ）

aminoacyl-tRNA synthetase

Amino acid + ATP + tRNA → aminoacyl-tRNA + AMP +PPi



table 8–3

International Classification of Enzymes*

No.	Class	Type of reaction catalyzed
1	Oxidoreductases	Transfer of electrons (hydride ions or H atoms)
2	Transferases	Group-transfer reactions
3	Hydrolases	Hydrolysis reactions (transfer of functional groups to water)
4	Lyases	Addition of groups to double bonds, or formation of double bonds by removal of groups
5	Isomerases	Transfer of groups within molecules to yield isomeric forms
6	Ligases	Formation of C—C, C—S, C—O, and C—N bonds by condensation reactions coupled to ATP cleavage

*Most enzymes catalyze the transfer of electrons, atoms, or functional groups. They are therefore classified, given code numbers, and assigned names according to the type of transfer reaction, the group donor, and the group acceptor.

3.3 Each enzyme is given a systematic name which identifies the reaction catalyzed (e.g., hexokinase is named as ATP:glucose phosphotransferase).

3.4 Each enzyme is assigned a four-digit number with the first digit denoting the class it belongs, the other three further clarifications on the reaction catalyzed.

Each enzyme is then uniquely identified by a four-digit classification number. Thus trypsin has the Enzyme Commission (EC)number 3.4.21.4. (e.g. Trypsin, EC 3.4.21.4)

glucokinase

$$\text{ATP} + \text{D-glucose} \longrightarrow \text{ADP} + \text{D-glucose-6-phosphate}$$

ATP: D-glucose phosphotransferase

A phosphate group is transferred from ATP to the C-6-OH group of glucose, so the enzyme is a *transferase.*

E.C.2.7.1.2,

Table 14.1

Systematic Classification of Enzymes According to the Enzyme Commission

E.C. Number	Systematic Name and Subclasses
1	*Oxidoreductases* (oxidation–reduction reactions)
1.1	Acting on CH—OH group of donors
1.1.1	With NAD or NADP as acceptor
1.1.3	With O_2 as acceptor
1.2	Acting on the $\diagdown$C=O group of donors
1.2.3	With O_2 as acceptor
1.3	Acting on the CH—CH group of donors
1.3.1	With NAD or NADP as acceptor
2	*Transferases* (transfer of functional groups)
2.1	Transferring C-1 groups
2.1.1	Methyltransferases
2.1.2	Hydroxymethyltransferases and formyltransferases
2.1.3	Carboxyltransferases and carbamoyltransferases
2.2	Transferring aldehydic or ketonic residues
2.3	Acyltransferases
2.4	Glycosyltransferases
2.6	Transferring N-containing groups
2.6.1	Aminotransferases
2.7	Transferring P-containing groups
2.7.1	With an alcohol group as acceptor

3	*Hydrolases* (hydrolysis reactions)
3.1	Cleaving ester linkage
3.1.1	Carboxylic ester hydrolases
3.1.3	Phosphoric monoester hydrolases
3.1.4	Phosphoric diester hydrolases
4	*Lyases* (addition to double bonds)
4.1	C=C lyases
4.1.1	Carboxy lyases
4.1.2	Aldehyde lyases
4.2	C=O lyases
4.2.1	Hydrolases
4.3	C=N lyases
4.3.1	Ammonia-lyases
5	*Isomerases* (isomerization reactions)
5.1	Racemases and epimerases
5.1.3	Acting on carbohydrates
5.2	Cis-trans isomerases
6	*Ligases* (formation of bonds with ATP cleavage)
6.1	Forming C—O bonds
6.1.1	Amino acid–RNA ligases
6.2	Forming C—S bonds
6.3	Forming C—N bonds
6.4	Forming C—C bonds
6.4.1	Carboxylases

4. Enzymes, like all other catalysts, does not affect reaction equilibria, only accelerate reactions.

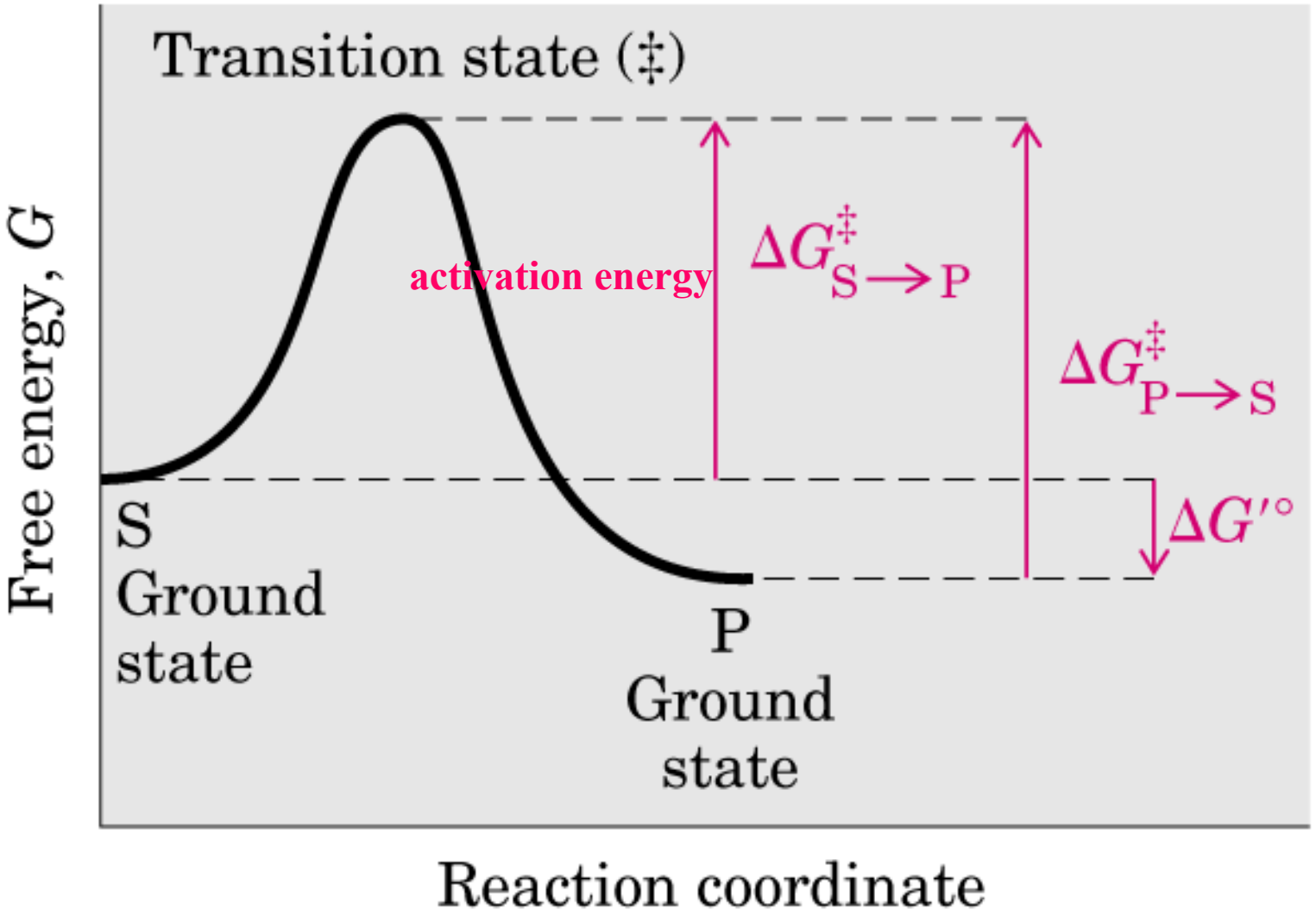
4.1 Equilibrium constant (K_{eq}') of a reaction is related to the free energy difference between the ground states of the substrates and products ($\Delta G^{o'}$)

P $\rightleftharpoons$ **S** $\qquad K'_{eq} = \dfrac{[P]}{[S]}$

$\Delta G^{o'} = -RT\ln K_{eq}'$

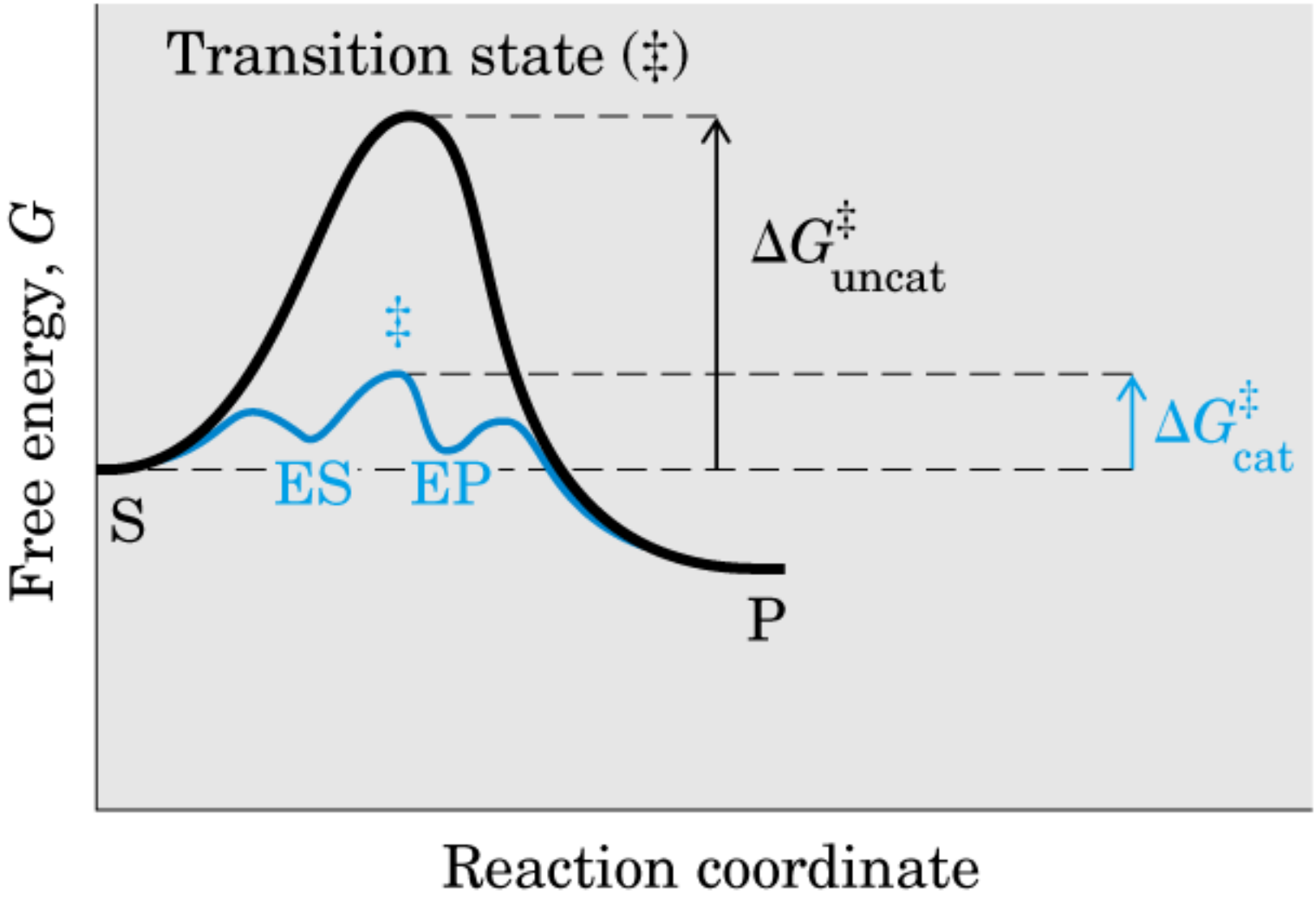
Enzyme catalysis does not affect $\Delta G^{o'}$, thus not K_{eq}'.

4.2 The rate constant of a reaction (*k*) is related to the free energy difference between the transition state and the ground state of the substrate ($\Delta G^{\ddagger}$)



For a biochemical reaction to proceed, the energy barrier needed to transform the substrate molecules into the transition state has to be overcome. The difference in free energy between the substrate and the transition state is termed the Gibbs free energy of activation ($\Delta G^{\ddagger}$)

$\Delta G^{\ddagger}$------ activation energy


Transition state (‡)
$\Delta G^{\ddagger}_{\text{uncat}}$
‡
$\Delta G^{\ddagger}_{\text{cat}}$
ES
EP
S
P
Free energy, G
Reaction coordinate

4.2.1 Transition state is a fleeting molecular moment (not a chemical species with any significant stability) that has the highest free energy during a reaction.

4.2.2 An enzyme increases the rate constant of a reaction (k) by lowering its $\Delta G^{\ddagger}$.

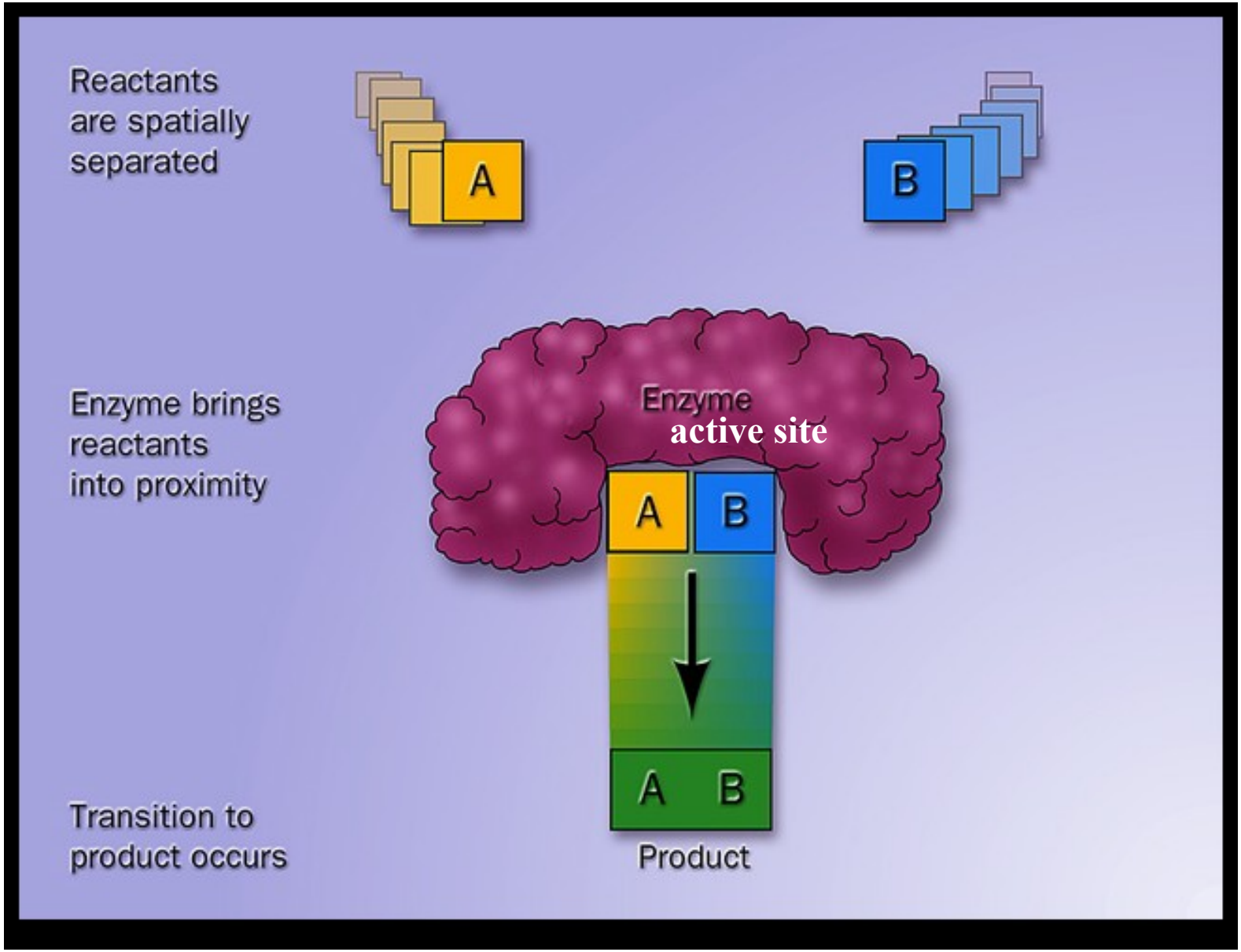
4.2.3 The combination of a substrate and an enzyme creates a new reaction pathway whose transition state energy is lower than that of the reaction in the absence of enzyme.

5. Formation of an enzyme-substrate complex is the first step in enzyme catalysis.

5.1 Substrates are bound to a specific region of an enzyme called the active site.

5.1.1 Much of the catalytic power of enzymes comes from their bringing substrates together in favorable orientations in enzyme-substrate (ES) complexes.

5.1.2 Most enzymes are highly selective in their binding of substrates.


Reactants are spatially separated
A
B
Enzyme brings reactants into proximity
Enzyme
active site
A
B
Transition to product occurs
A
B
Product

5.1.3 Common Features of the Active Sites

1. *The active site takes up a relatively small part of the total volume of an enzyme.*

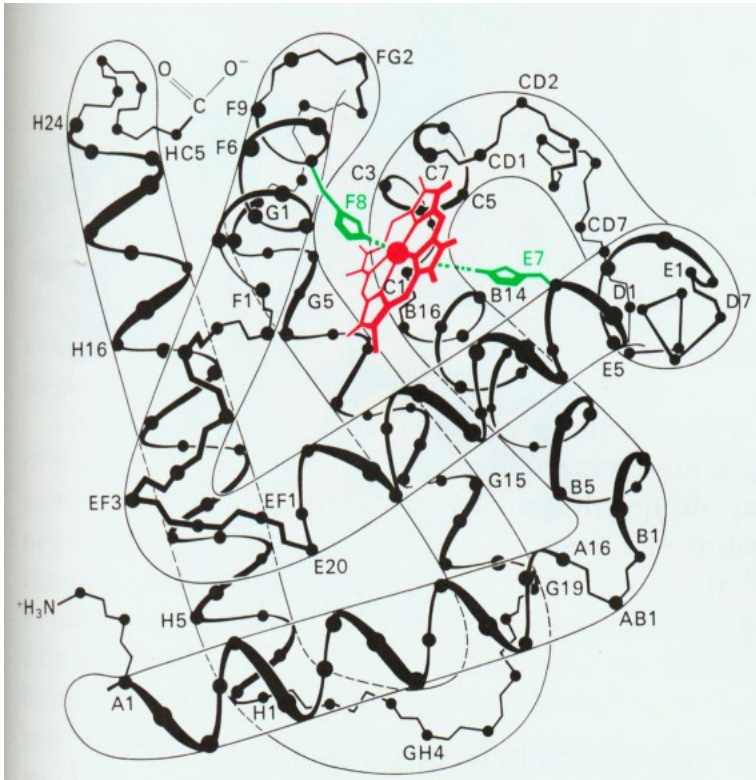


Figure 7-4
Model of myoglobin at high resolution. Only the α carbon atoms are shown. The heme group is shown in red, and two key histidine residues in green. [After R.E. Dickerson. In *The Proteins*, vol. 2, 2nd ed., H. Neurath, ed. (Academic Press, 1964), p. 634.]

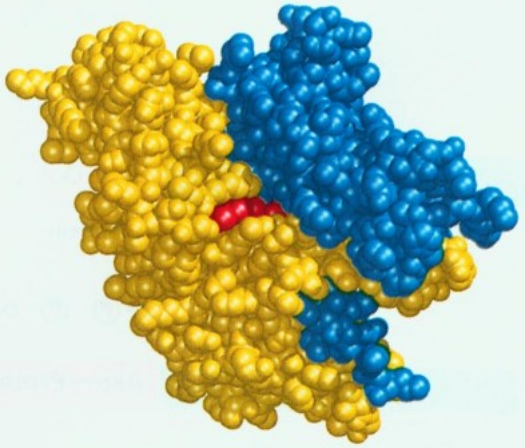


Figure 35-10
Structure of an ATPase fragment derived from hsc70, a cytosolic heat-shock protein. ADP (red) is bound in the cleft between the two domains (yellow and blue). [Drawn from 1ats.pdb. K.M. Flaherty, C. DeLuca-Flaherty, and D.B. McKay. *Nature* 346(1990):623.]

2. *The active site is a three-dimensional entity* formed by groups that come from different parts of the linear amino acid sequence

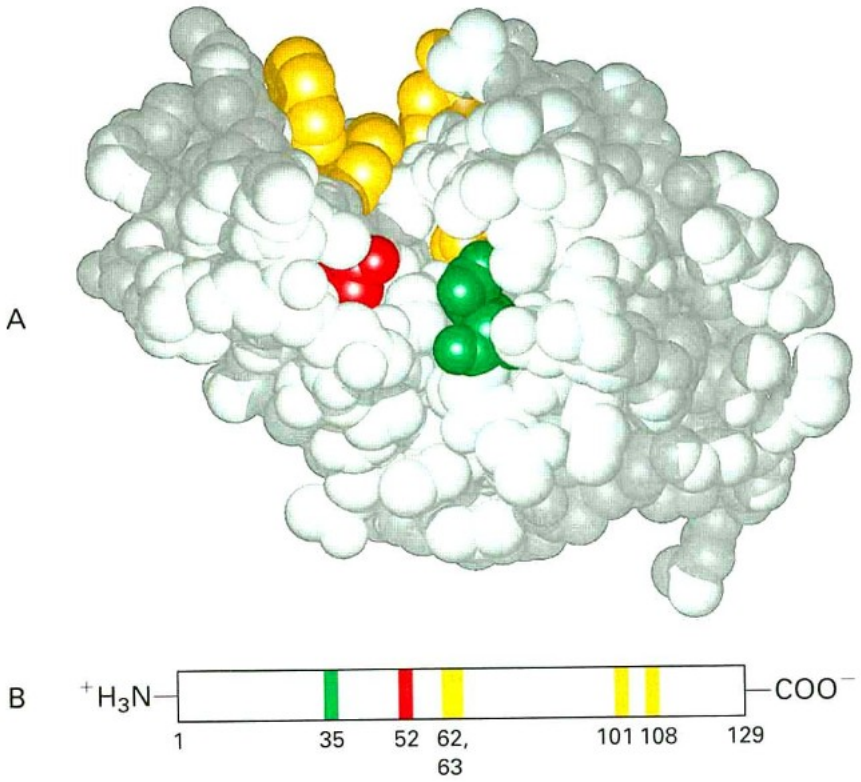


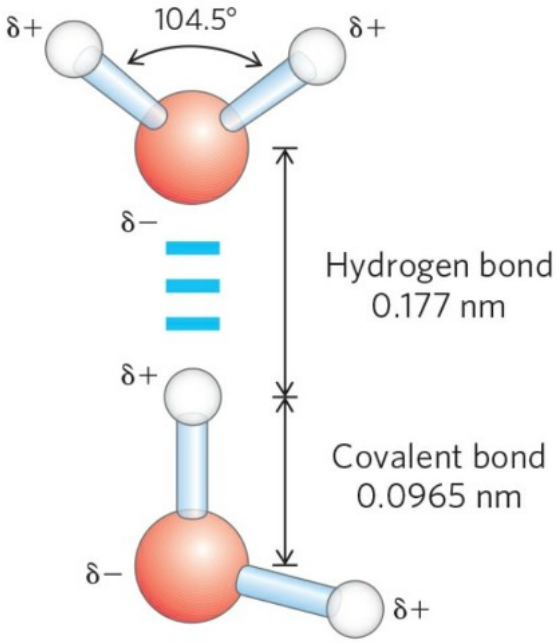
Figure 8-11
(A) Space-filling model of lysozyme. Two catalytically critical residues are shown in red and green. Several other residues in the active site are shown in yellow. (B) Schematic diagram of lysozyme showing that the active site is formed by residues that come from different segments of the polypeptide chain. [(A) Drawn from 6lyz. R. Diamond. *J. Mol. Biol.* 82(1974):371.]

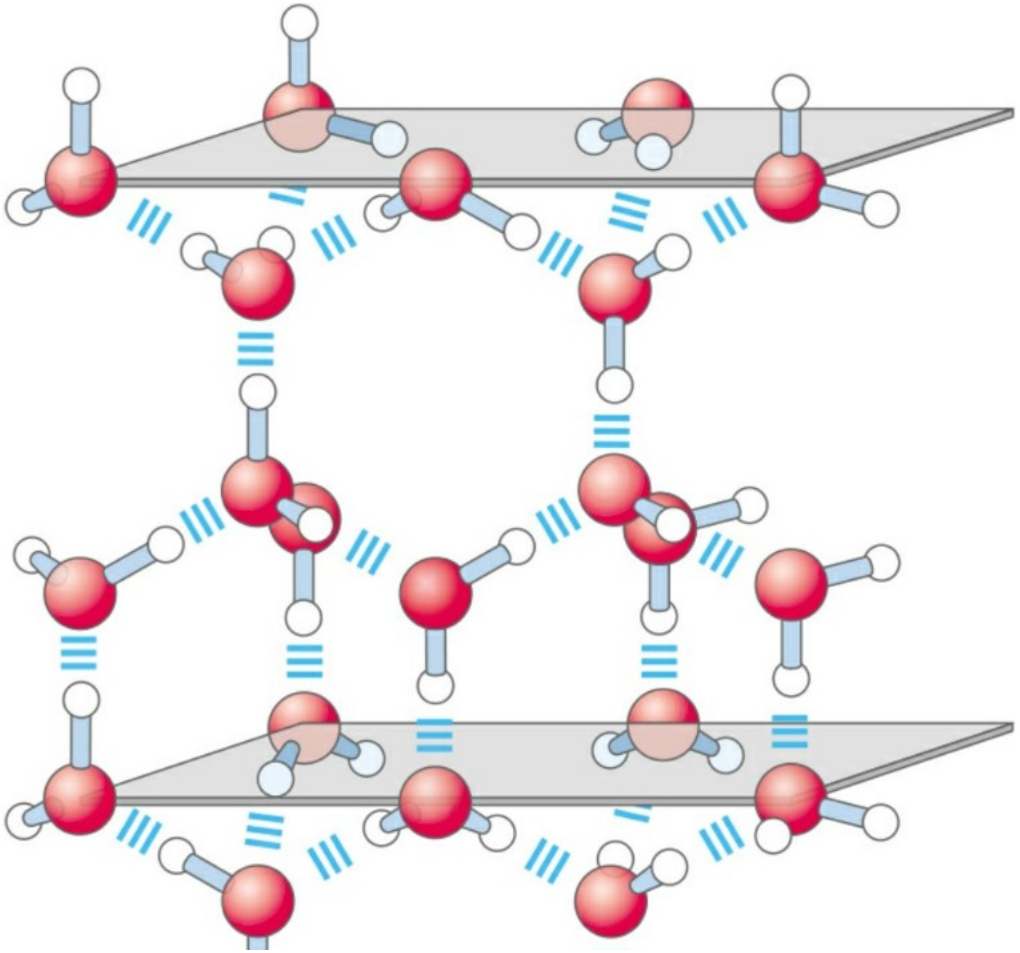
3. *Substrates are bound to enzymes by multiple weak attractions.*

5.1.4 What are the types of the multiple weak interactions?

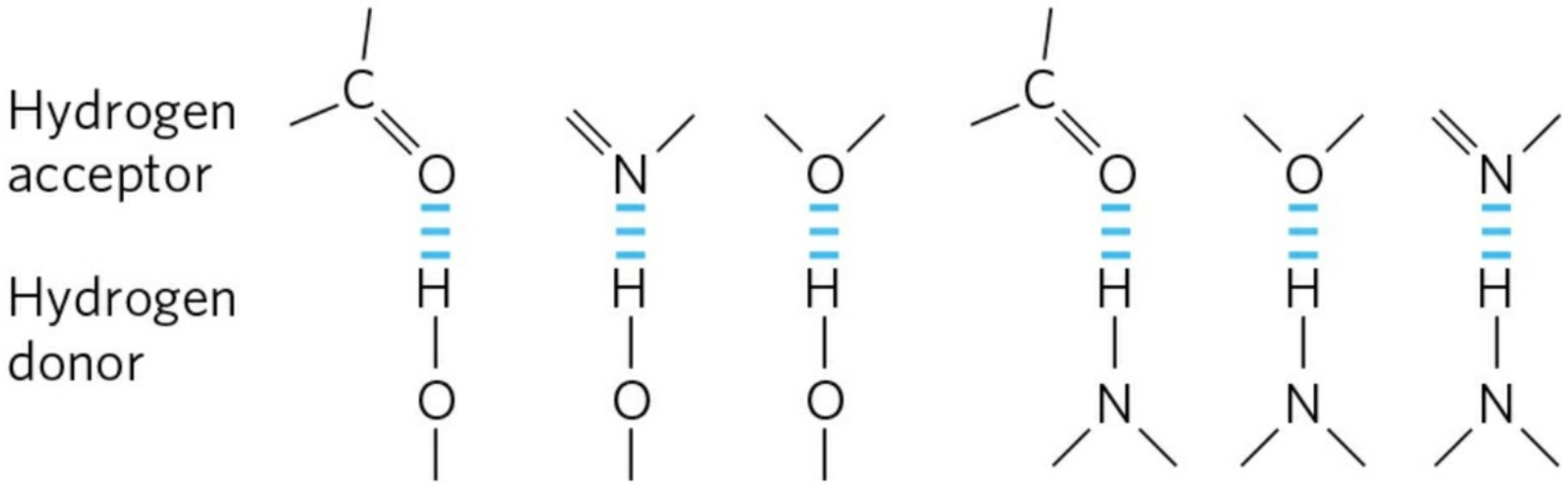
tions of biomolecules are mediated by[1] electrostatic bonds,[2] hydrogen bonds;[3] van der Waals forces, and[4] hydrophobic interactions. Van der Waals forces become significant in binding only when numerous substrate atoms can simultaneously come close to many enzyme atoms. Hence, the enzyme and substrate should have complementary shapes. The directional character of hydrogen bonds between enzyme and substrate often enforces a high degree of specificity.

Note: "electrostatic bonds" should be called as "electrostatic interactions"





Hydrogen bonding in ice



Common hydrogen bonds in biological systems

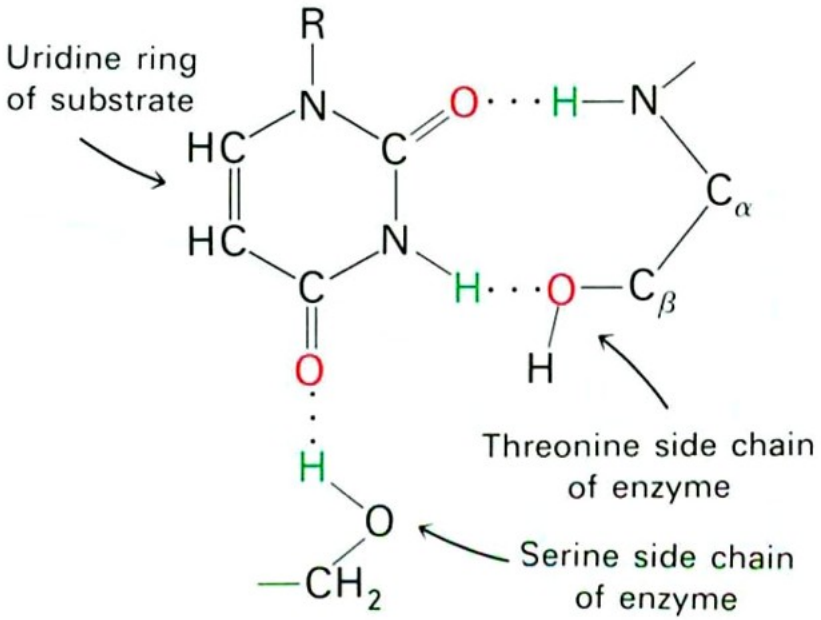


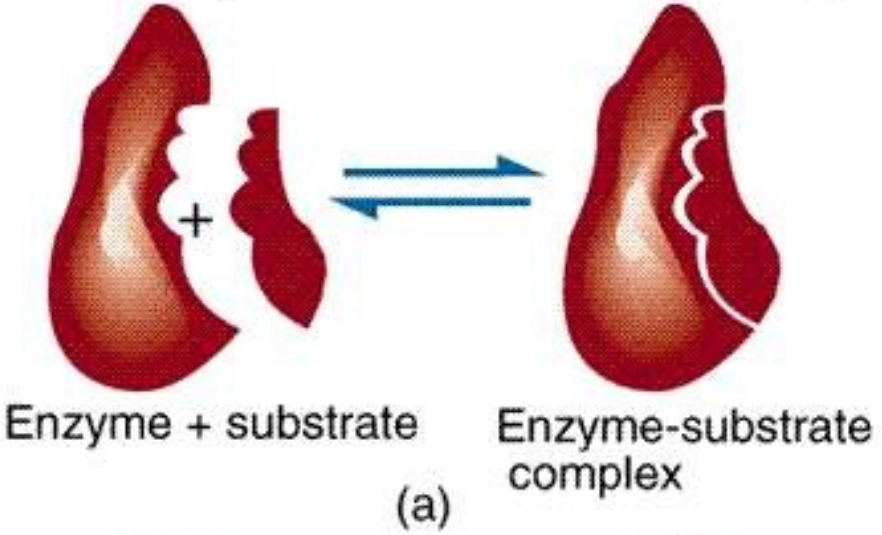
Figure 8-12
Hydrogen bond interactions in the binding of a uridine substrate to ribonuclease. [After F.M. Richards, H.W. Wyckoff, and N. Allewell. In *The Neurosciences: Second Study Program,* F.O. Schmidt, ed. (Rockefeller University Press, 1970), p. 970.]

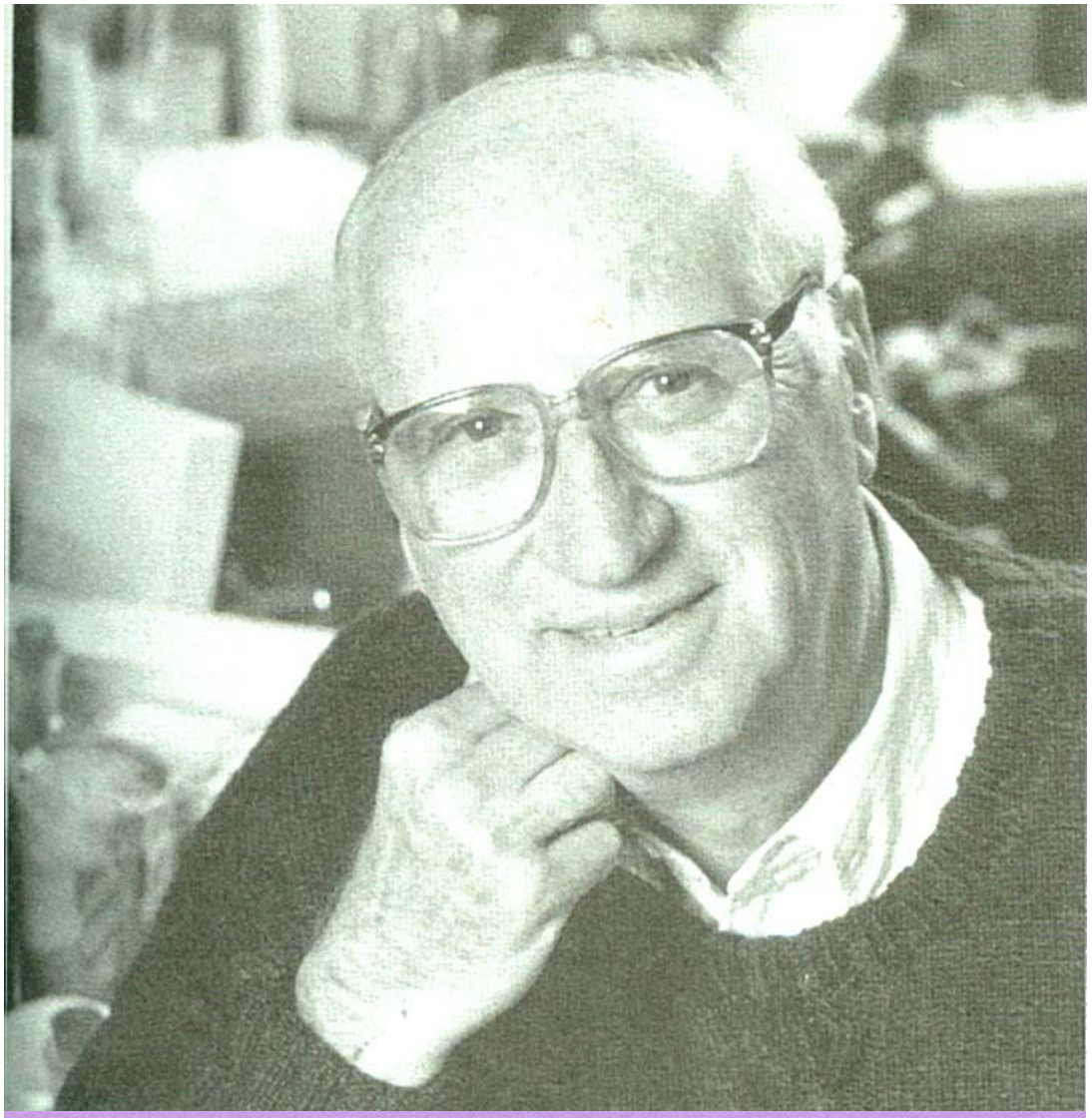
5.1.5 The active sites of some unbound enzymes are complementary in shape to those of their substrates (the lock-and-key metaphor, Emil Fisher).

5.1.6 In many enzymes, the active sites have shapes complementary to those of their substrates only after the substrates are bound (the induced-fit model, Daniel Koshland).

Biochemistry by McKee/McKee, ©1996 McGraw-Hill Higher Education

Models of enzyme-substrate binding. M Figure 6.2





Daniel E. Koshland, Jr.

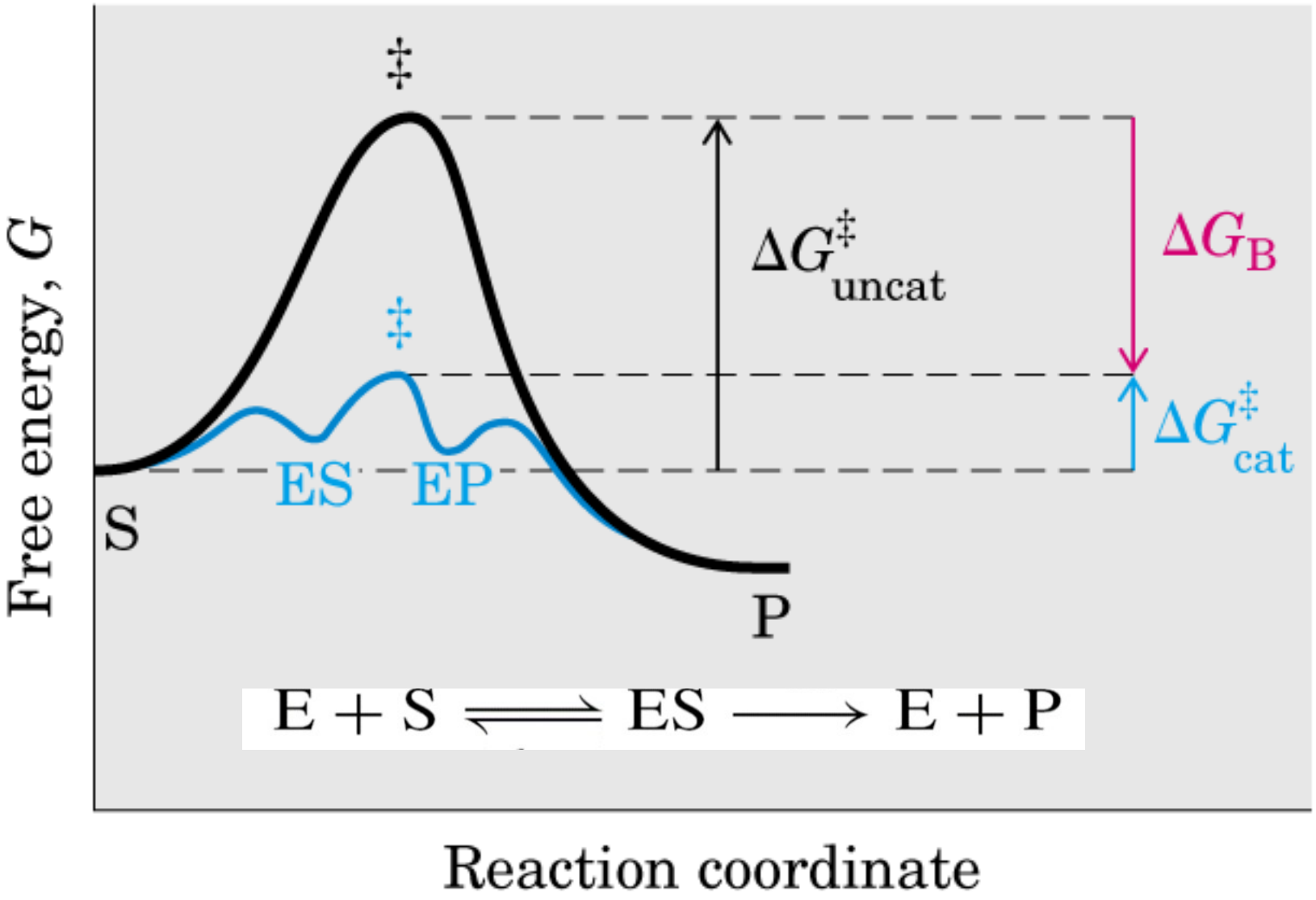
Professor in the Graduate School
(Division of Biochemistry & Molecular Biology)

Biochem

5.2 The existence of ES complexes has been shown in a variety of ways.

5.2.1 The saturation effect: at a constant concentration of an enzyme, the reaction rate increases with increasing substrate concentrations until a Vmax is reached.

5.2.2 ES complexes have been directly observed by electron microscopy and X-ray crystallography.

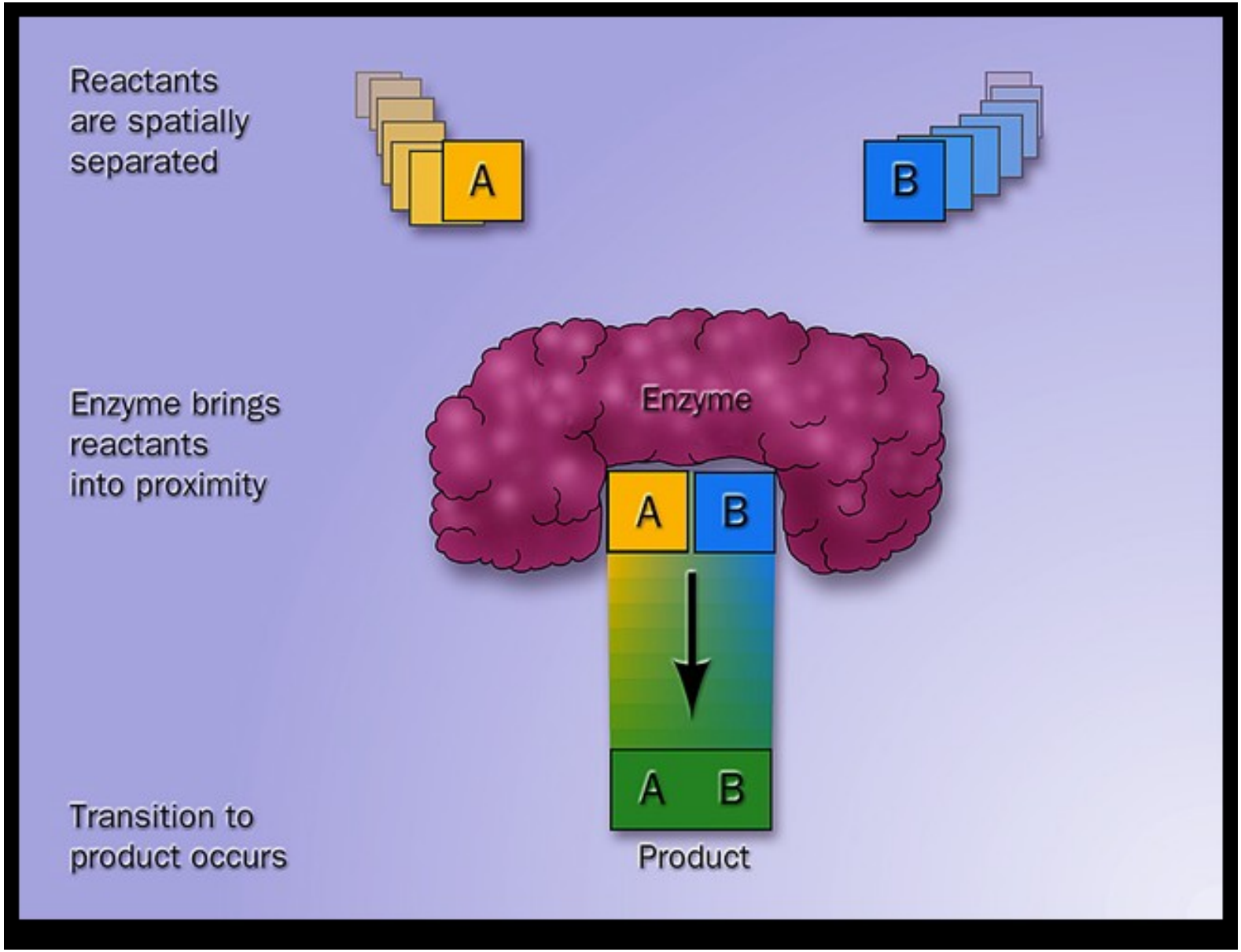

Free energy, G
‡
‡
ΔG‡uncat
ΔGB
ΔG‡cat
ES
EP
S
P
E + S ⇌ ES ⟶ E + P
Reaction coordinate

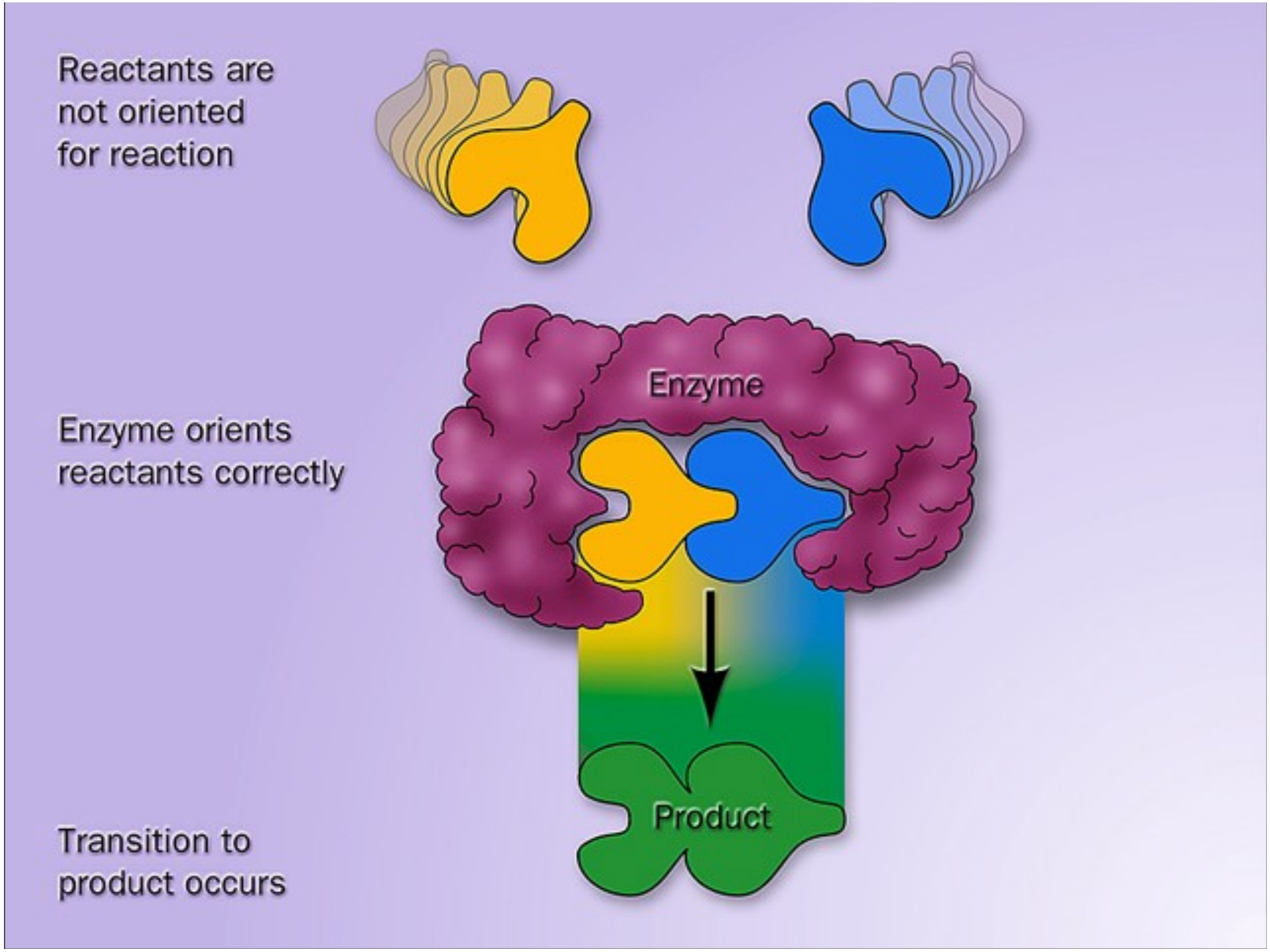
6. Various factors leading to rate enhancements are proximity and orientation effects, strain or distortion acid-base catalysis, covalent catalysis, and changes in environment.

6.1 Proximity and orientation effect

An enzyme could increase the rate of a reaction involving more than one substrate by binding the substrates at adjacent sites and therefore bringing them into close proximity with each other.

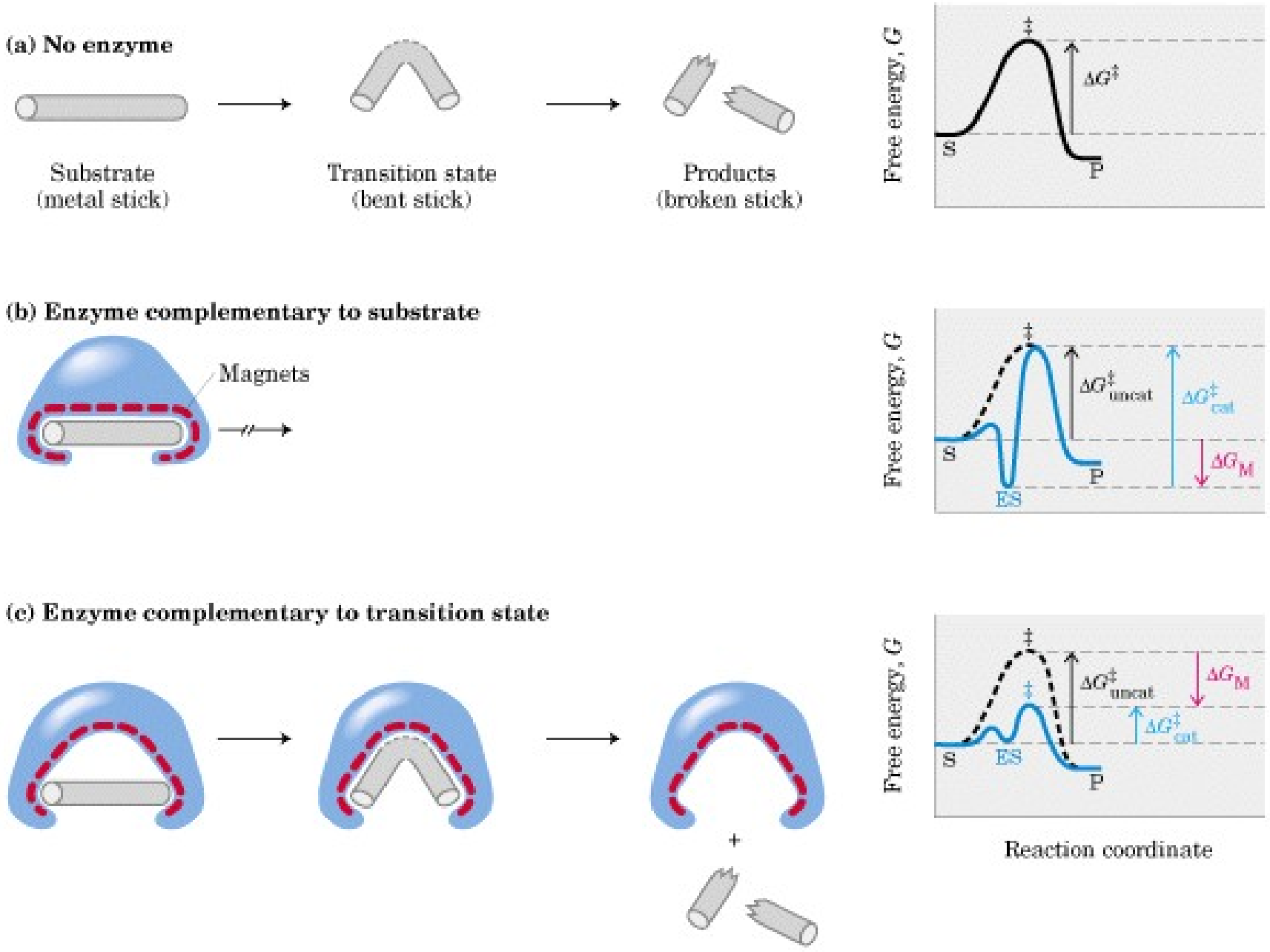
The orientation of the reacting molecules with respect to each other can greatly influence the rate of reaction.


Reactants are spatially separated
A
B
Enzyme brings reactants into proximity
Enzyme
A
B
Transition to product occurs
A B
Product


Reactants are not oriented for reaction
Enzyme orients reactants correctly
Transition to product occurs
Enzyme
Product

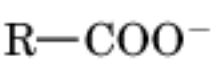
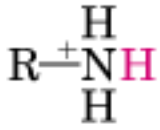
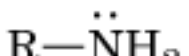
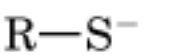
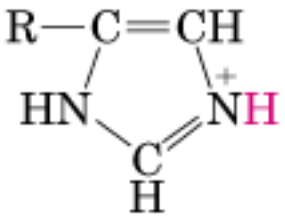
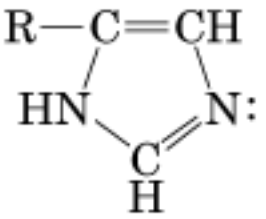
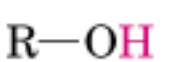
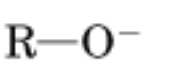
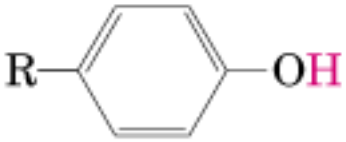
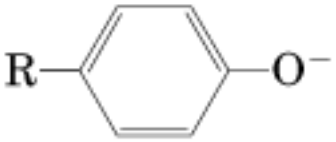
6.2 Strain or distortion

The bonds in a substrate may be distorted on binding to the appropriate enzyme. This would speed up the reaction if the distortion (or strain) lowered the free energy of activation by making the geometry and electronic structure of the substrate.

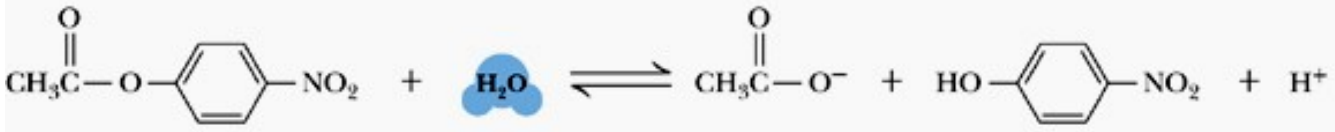

(a) No enzyme
Substrate
(metal stick)
Transition state
(bent stick)
Products
(broken stick)
Free energy, G
‡
ΔG‡
S
P
(b) Enzyme complementary to substrate
Magnets
Free energy, G
‡
ΔG‡uncat
ΔG‡cat
ΔGM
S
P
ES
(c) Enzyme complementary to transition state
+
Free energy, G
‡
ΔG‡uncat
ΔGM
‡
ΔG‡cat
S
ES
P
Reaction coordinate

6.3 Acid-base catalysis

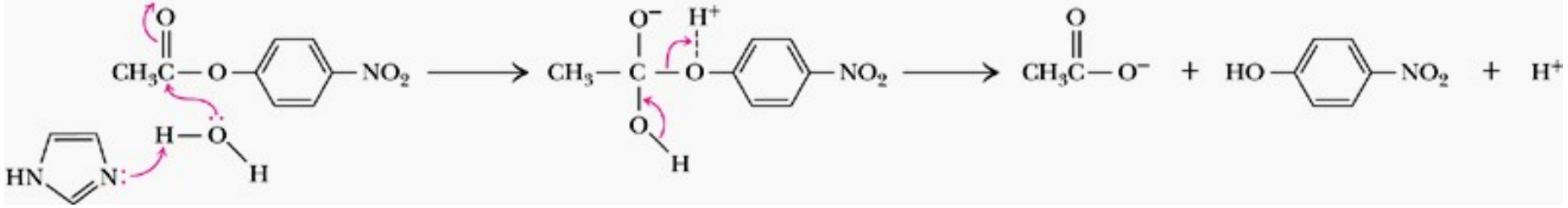
Since enzymes contain a number of amino acid side-chains that are capable of acting as proton donors or acceptors, it is reasonable to suppose that acid-base catalysis would be important in enzyme-catalysed reaction.

Amino acid residues	General acid form (proton donor)	General base form (proton acceptor)
Glu, Asp		
Lys, Arg		
Cys		
His		
Ser		
Tyr		

Reaction



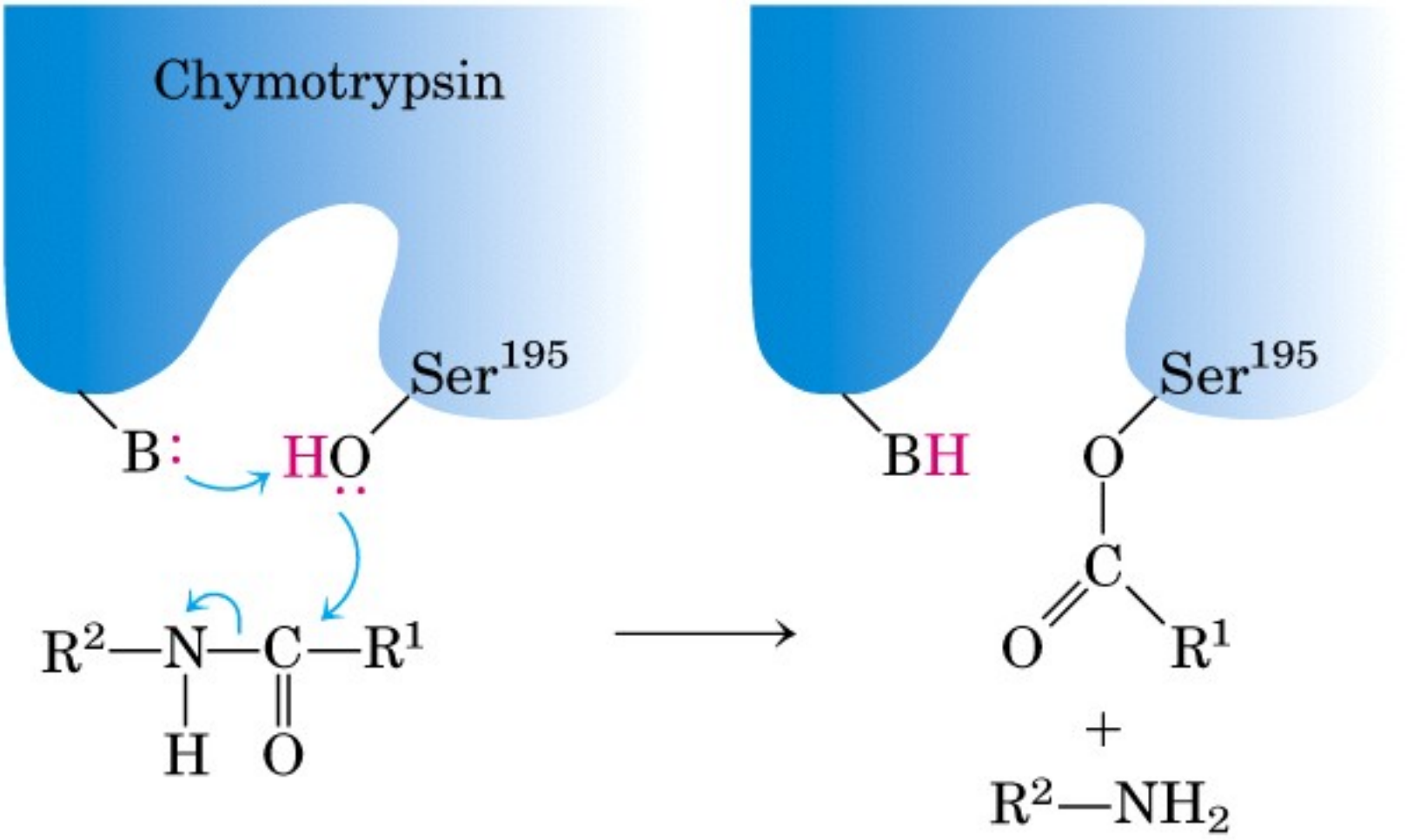
Mechanism

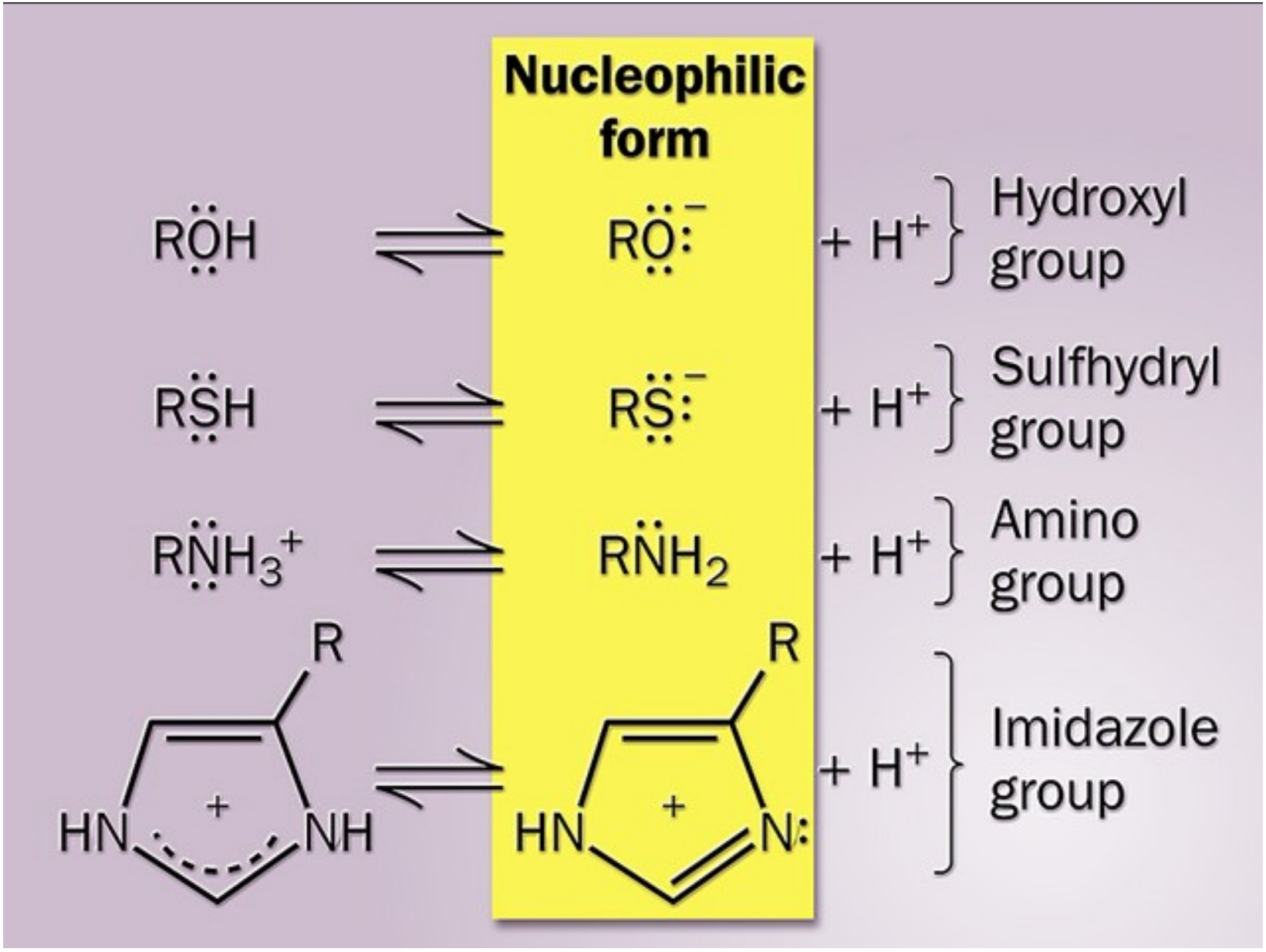


6.4 Covalent catalysis (intermediate formation)

In this type catalysis, a transient covalent bond is formed between the enzyme and the substrate.

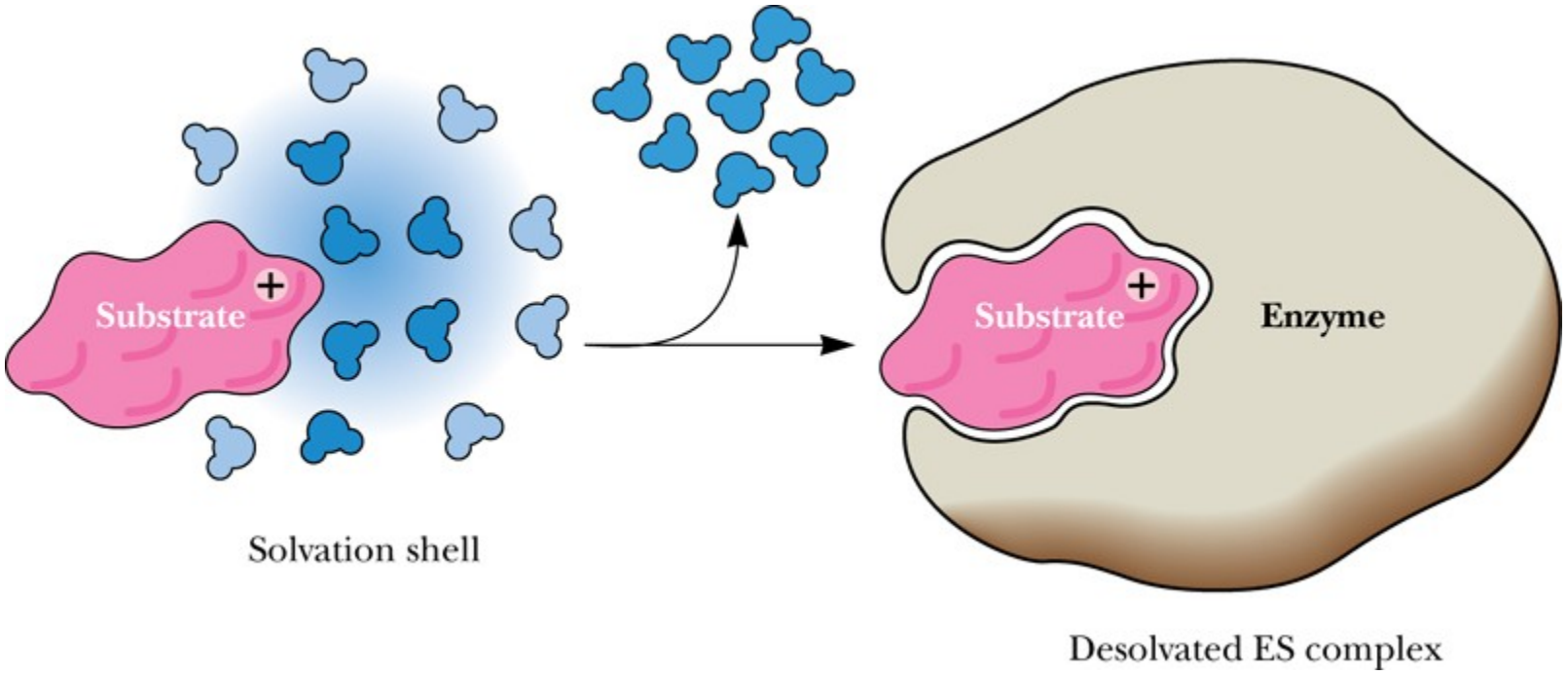
Reactions can be speed up by the formation of intermediates, provided that such intermediates are both rapidly formed and rapidly broken down.

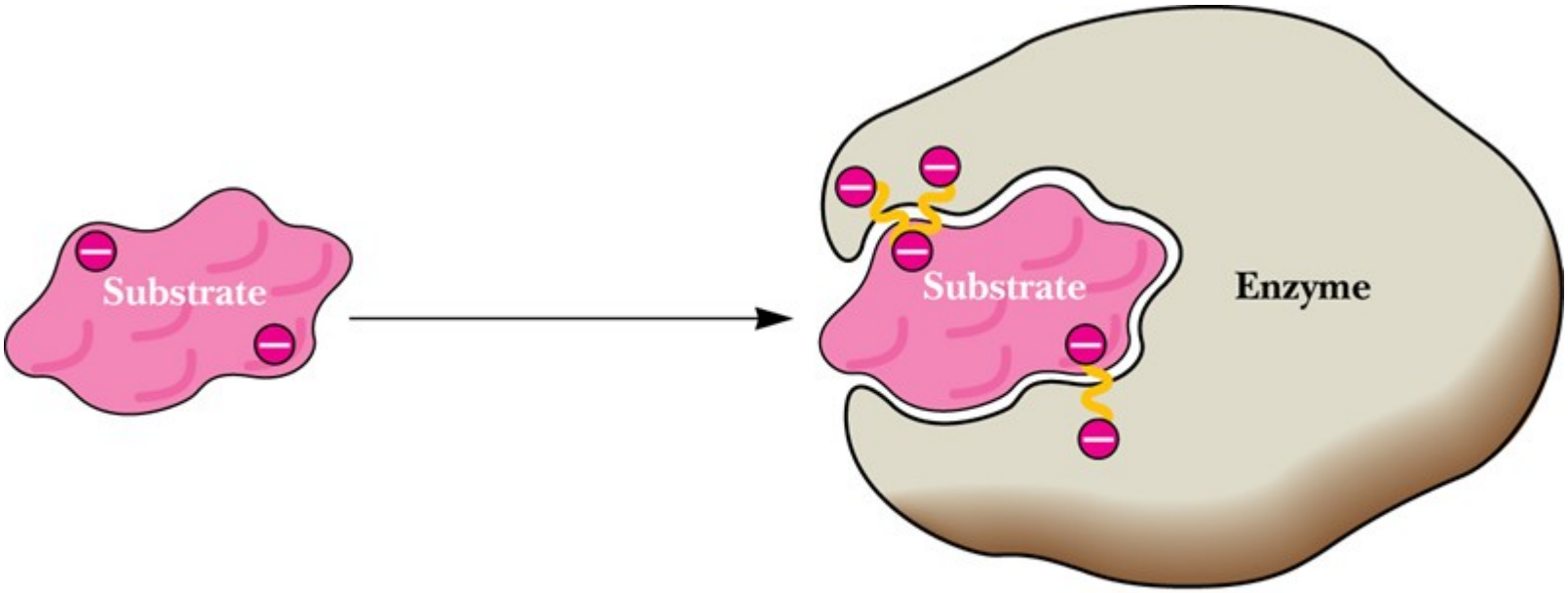

Chymotrypsin
Ser195
B:
HO
R2—N—C—R1
H O
BH
O
C
O
R1
+
R2—NH2


Nucleophilic form
RÖH
RÖ:⁻
+ H⁺
Hydroxyl group
RS̈H
RS̈:⁻
+ H⁺
Sulfhydryl group
RN̈H₃⁺
RN̈H₂
+ H⁺
Amino group
R
HN
NH
+
R
HN
N:
+
+ H⁺
Imidazole group

6.5 Changes in environment

X-ray crystallographic studies show that enzymes are capable of providing unusual environments for reactions. For instance, in the case of lysozyme, the cleft in the molecule that forms the substrate-binding site is lined by a number of non-polar amino acid side-chains; these clearly provide an environment markedly different from that of the solvent water.


Substrate
+
Solvation shell
Substrate
+
Enzyme
Desolvated ES complex

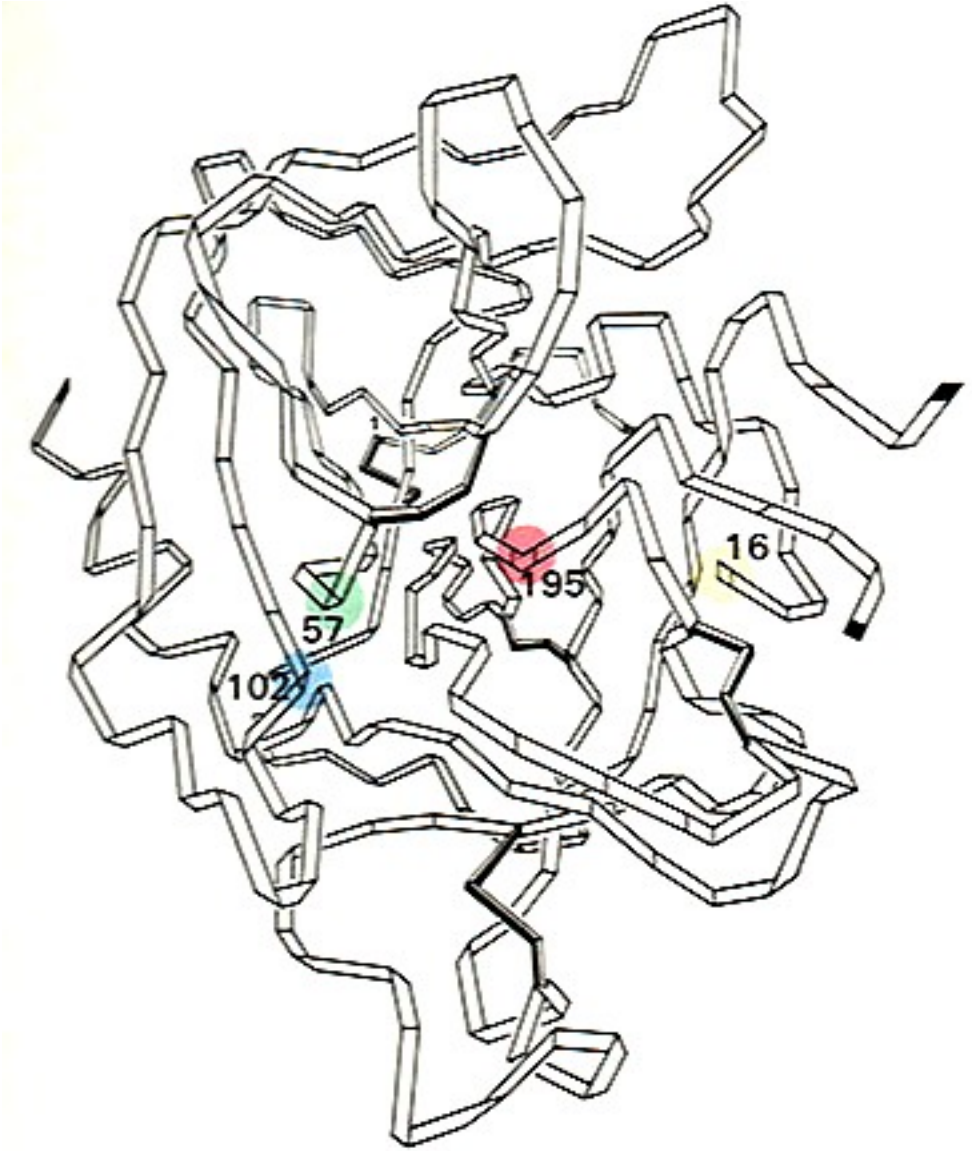


Electrostatic destabilization
in ES complex

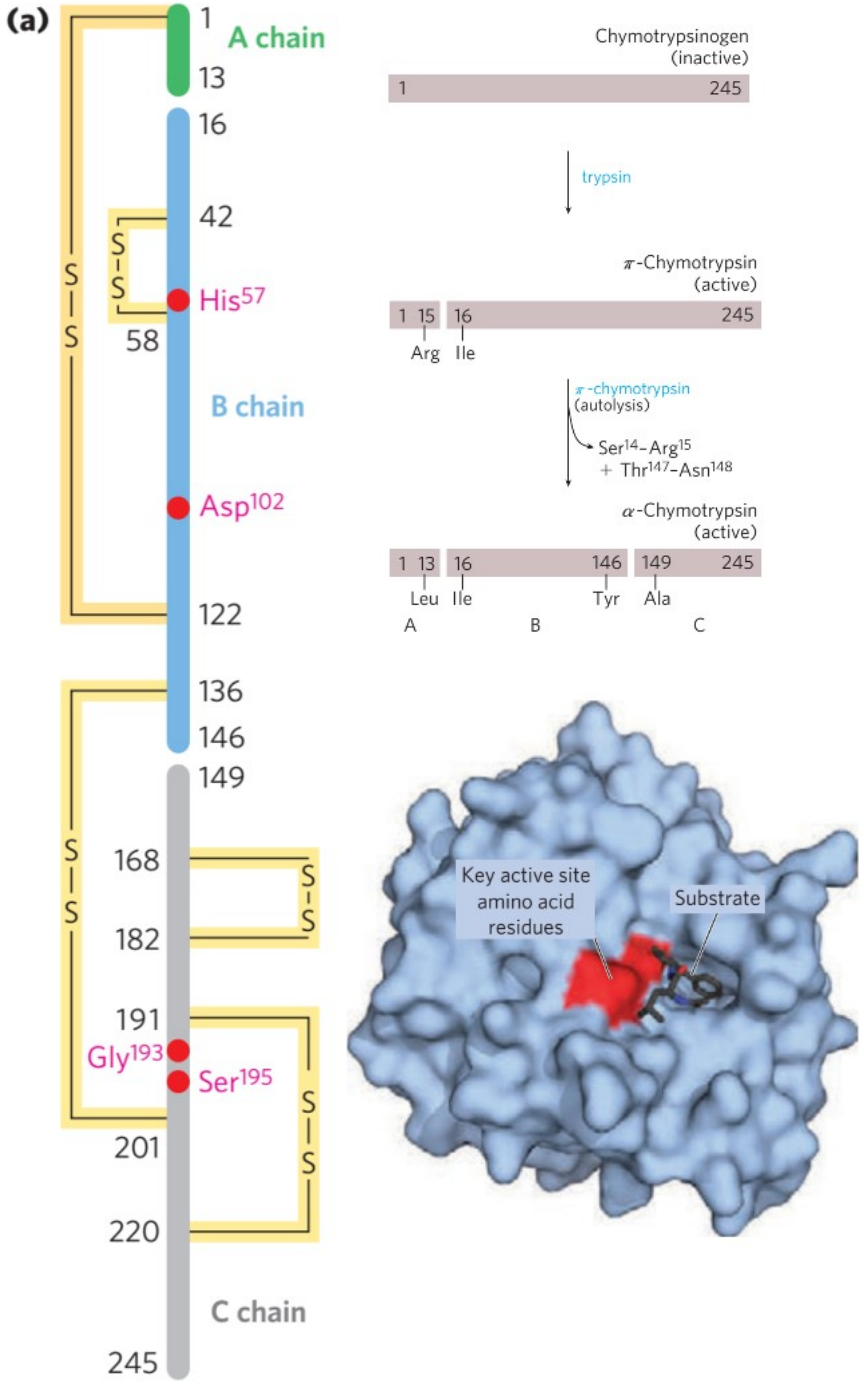
An enzyme may use a combination of several catalytic strategies to bring about a rate enhancement.

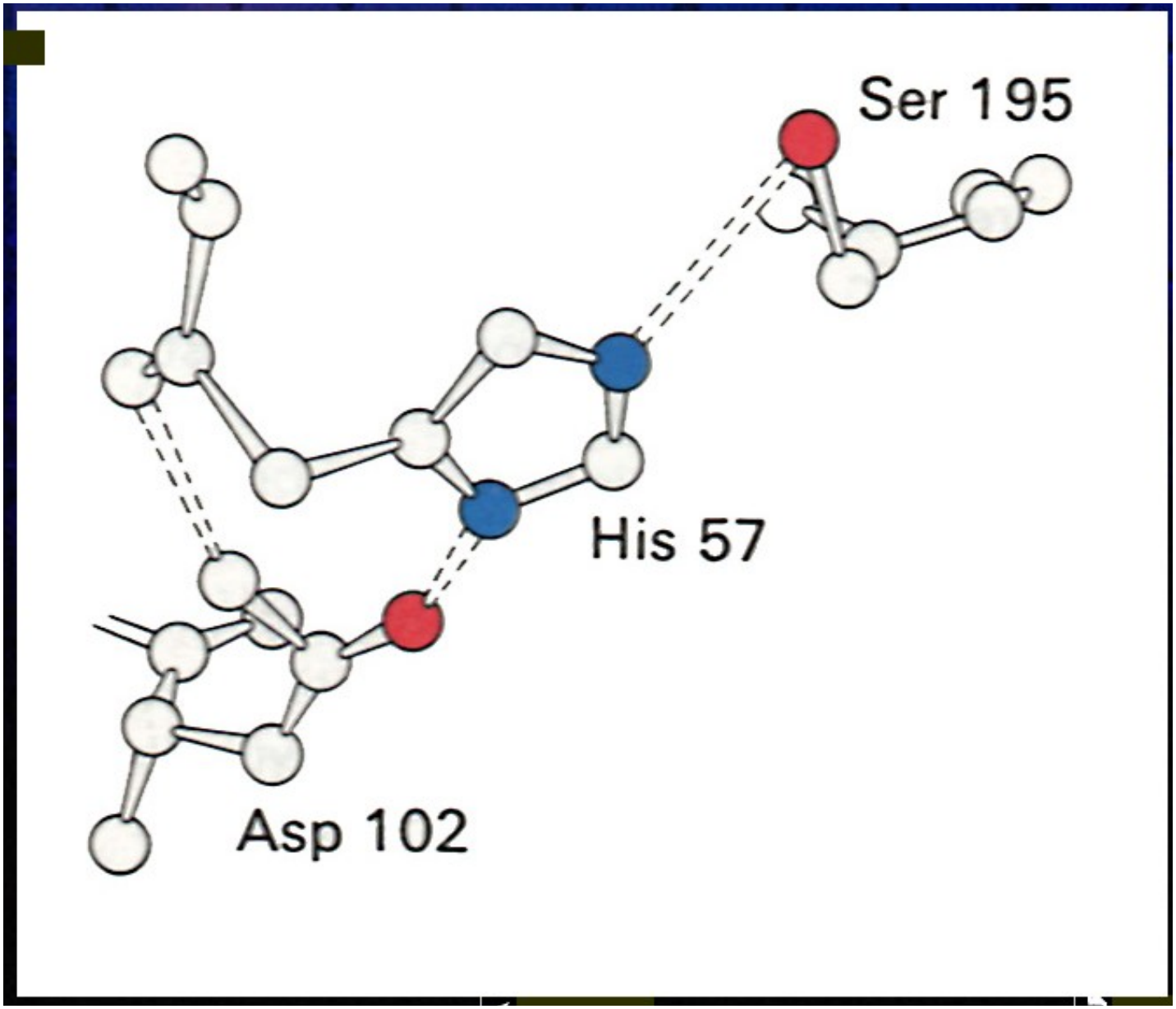
Chymotrypsin uses both covalent catalysis and general acid-base catalysis.

作用机理实例——胰凝乳蛋白酶

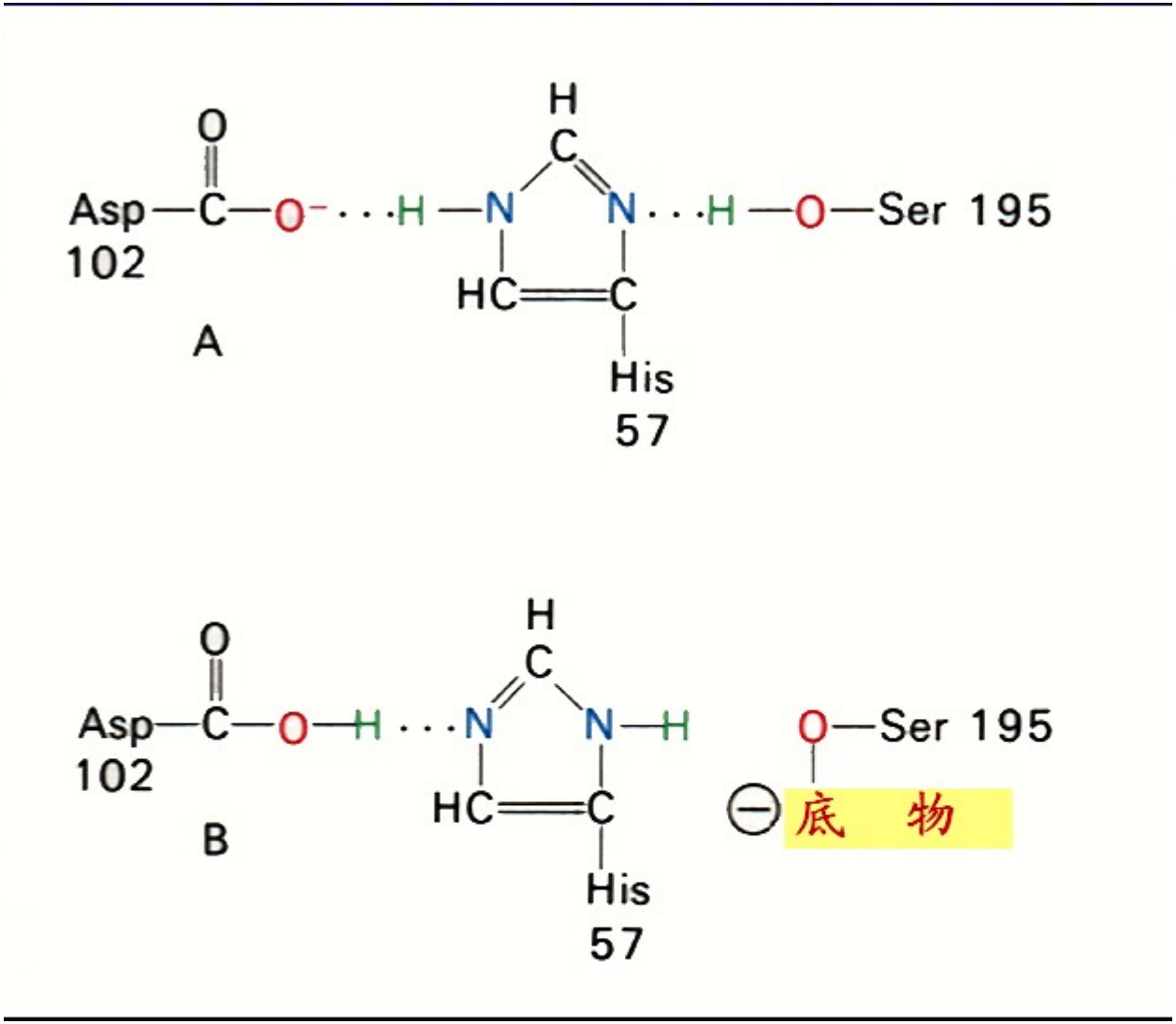


A Chymotrypsin

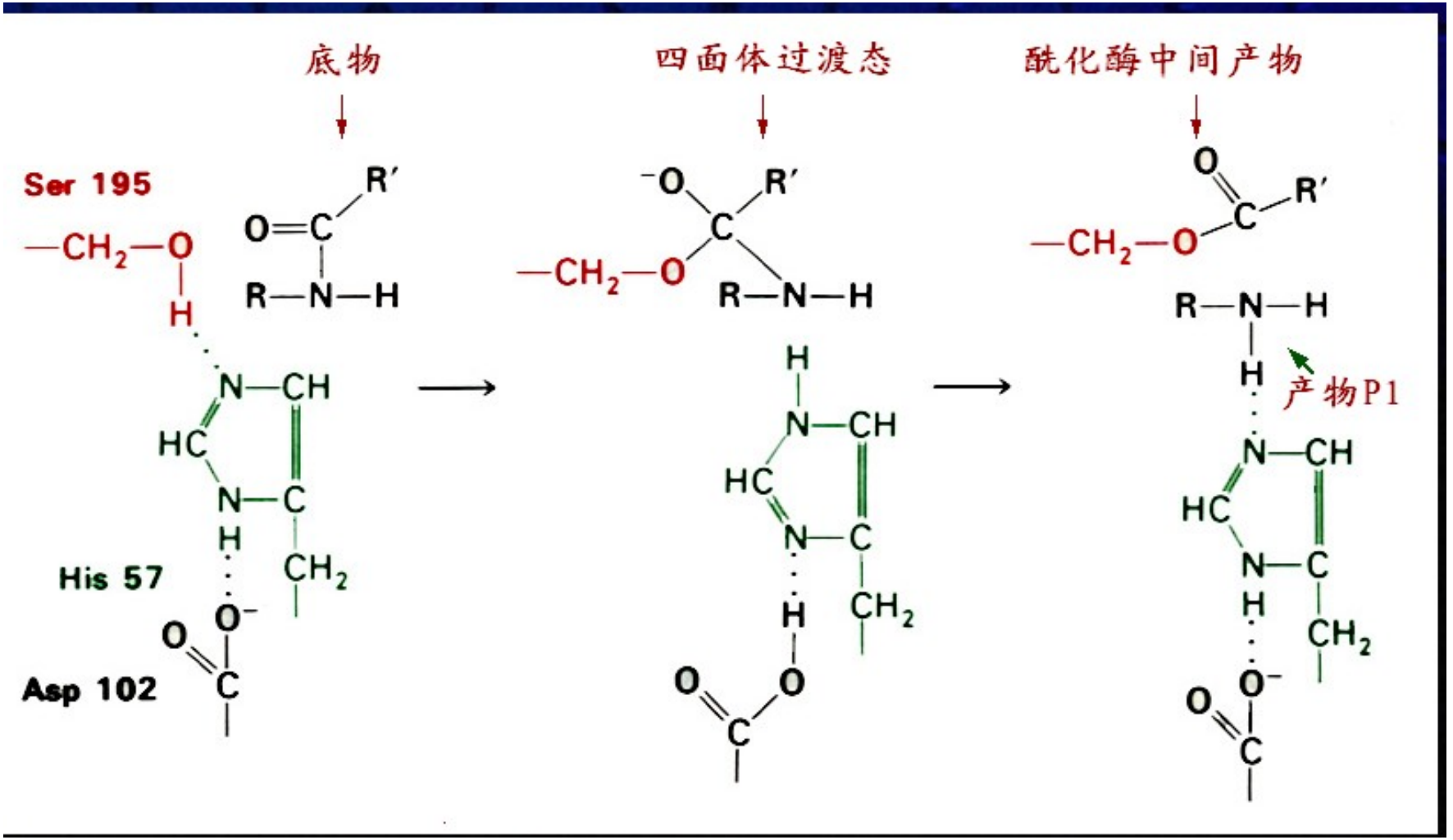

(a)
1
A chain
13
16
42
S
S
S
S
His57
58
B chain
Asp102
122
136
146
149
S
S
168
S
S
182
191
Gly193
Ser195
S
S
201
220
C chain
245
Chymotrypsinogen
(inactive)
1
245
trypsin
π-Chymotrypsin
(active)
1 15
16
245
Arg
Ile
π-chymotrypsin
(autolysis)
Ser14–Arg15
+ Thr147–Asn148
α-Chymotrypsin
(active)
1 13
16
146
149
245
Leu
Ile
Tyr
Ala
A
B
C
Key active site
amino acid
residues
Substrate


Ser 195
His 57
Asp 102

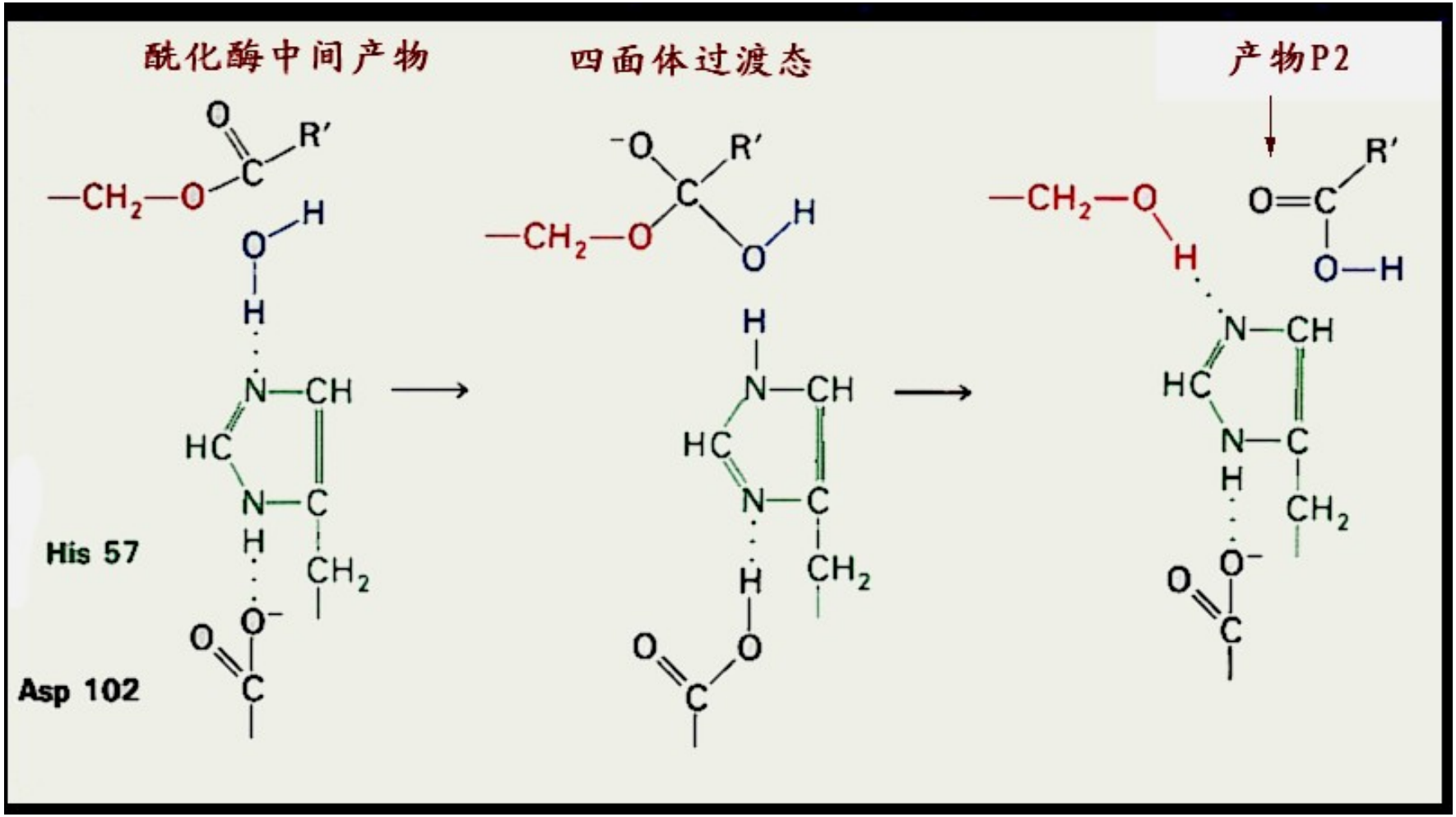
1．通过电荷转移，使 Ser 有更强的亲核力

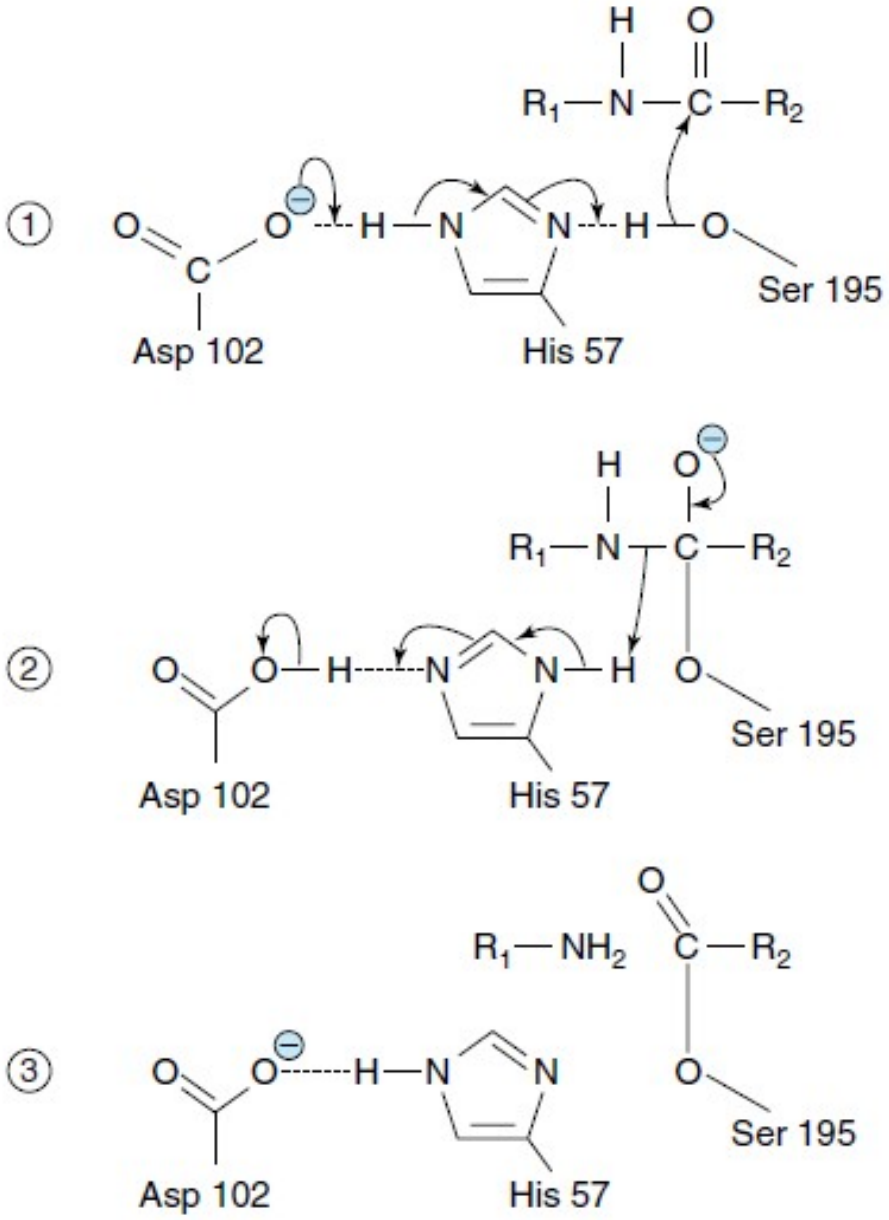


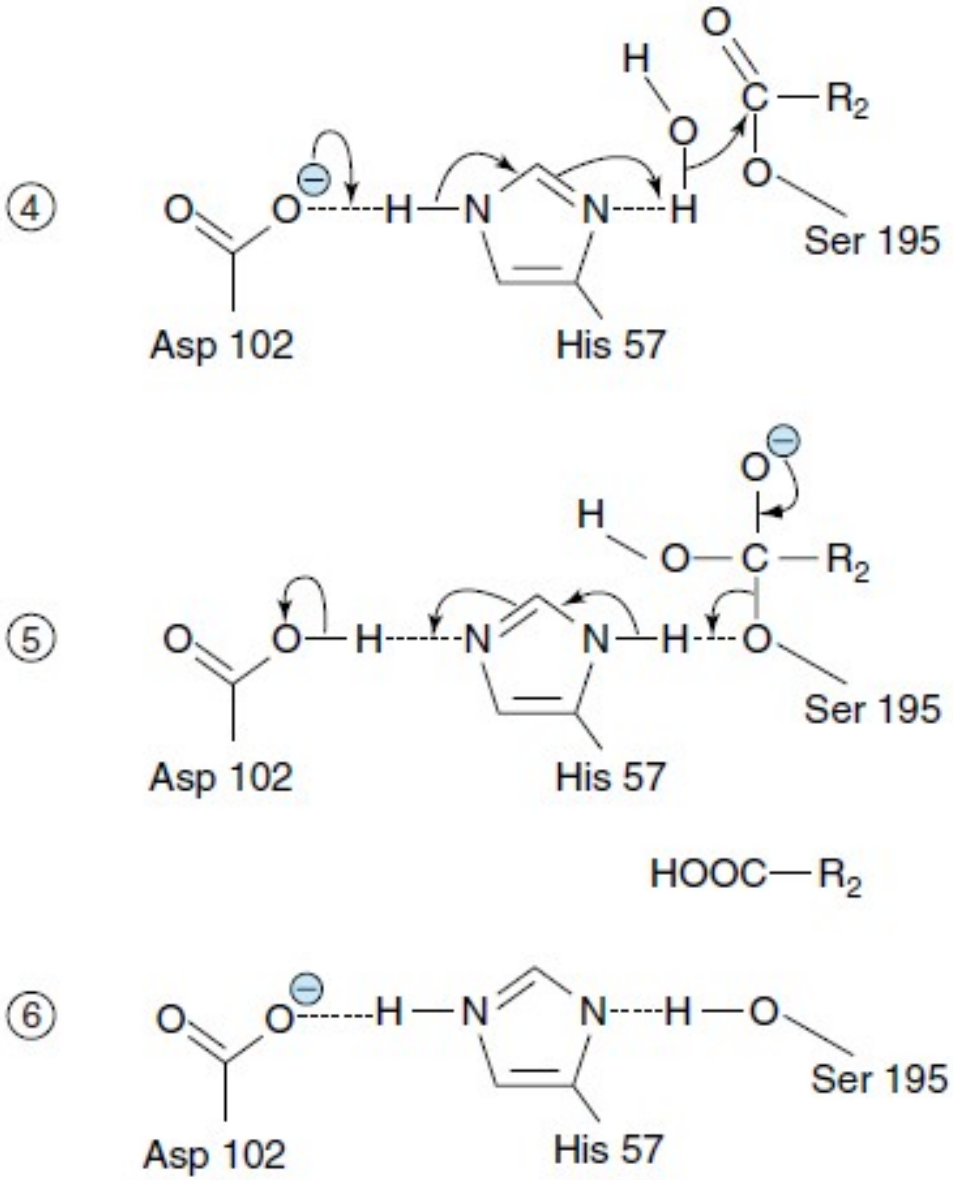
2．形成酰化酶，放出产物 P1



3．脱酰（水解），形成产物 P2




①
H
O
R1—N—C—R2
O
O
H
N
N
H
O
C
Ser 195
Asp 102
His 57
②
H
O
R1—N—C—R2
O
O
H
N
N
H
O
Ser 195
Asp 102
His 57
③
O
R1—NH2
C—R2
O
O
H
N
N
Ser 195
Asp 102
His 57


④
O
H
C
R2
O
O
O
H
N
N
H
Ser 195
Asp 102
His 57
⑤
O
H
O
C
R2
O
O
H
N
N
H
O
Ser 195
Asp 102
His 57
HOOC—R2
⑥
O
O
H
N
N
H
O
Ser 195
Asp 102
His 57

C3 Enzyme Kinetics

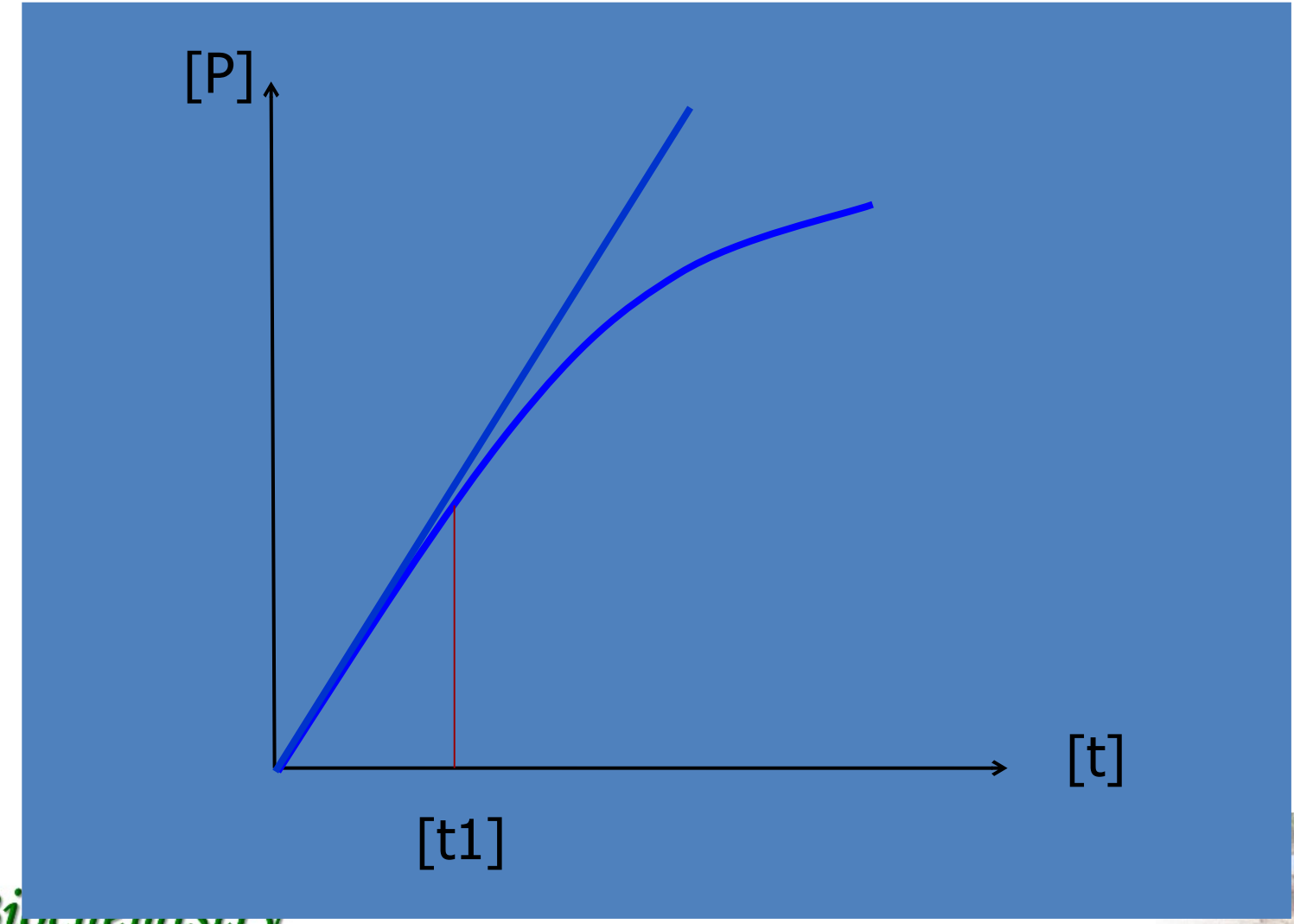
1. Michaelis-Menten equation reflects the kinetic behavior of many enzymes

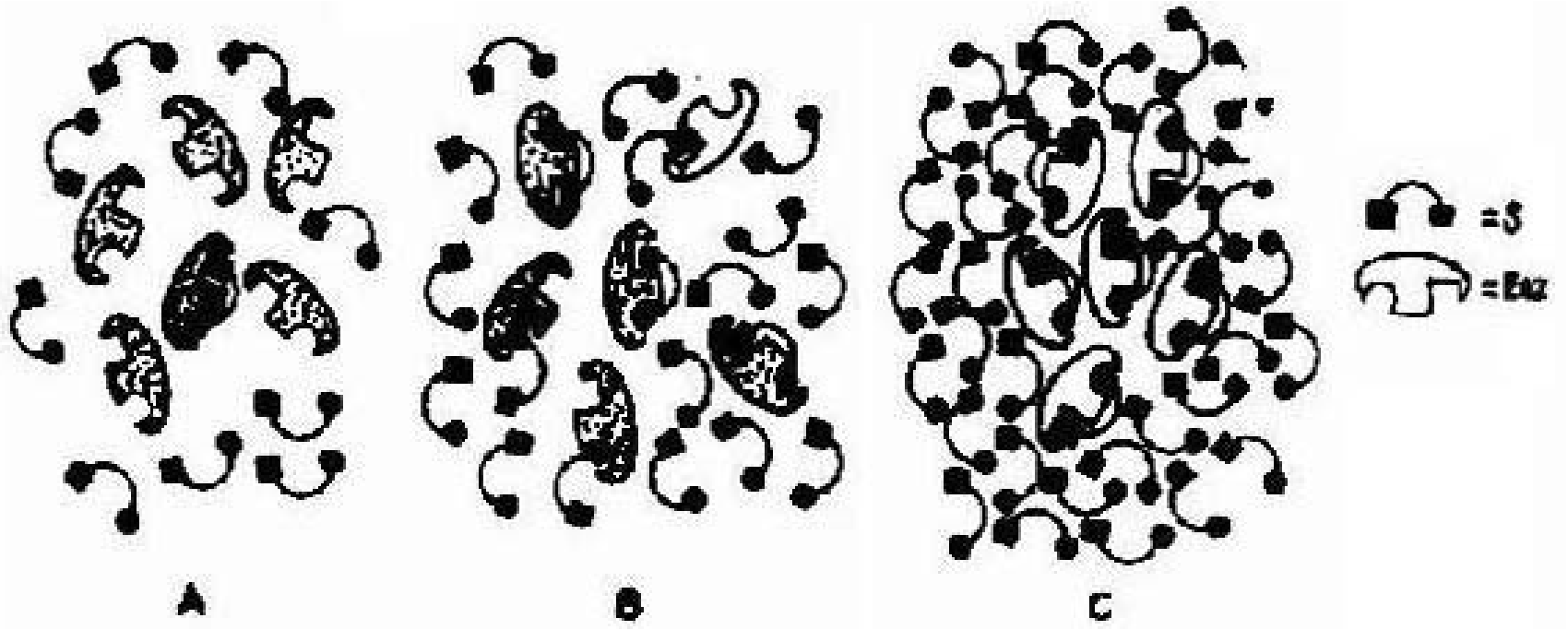
1.1 Saturation effect was observed in enzyme catalysis when plotting the initial velocity (V_o) against the substrate concentration ([S]).

1.2 The ES complex was proposed to be a necessary step in enzyme catalysis based on this kinetic pattern.

酶促反应速度的测量

（酶促反应初始速率 initial rate or initial velocity ）

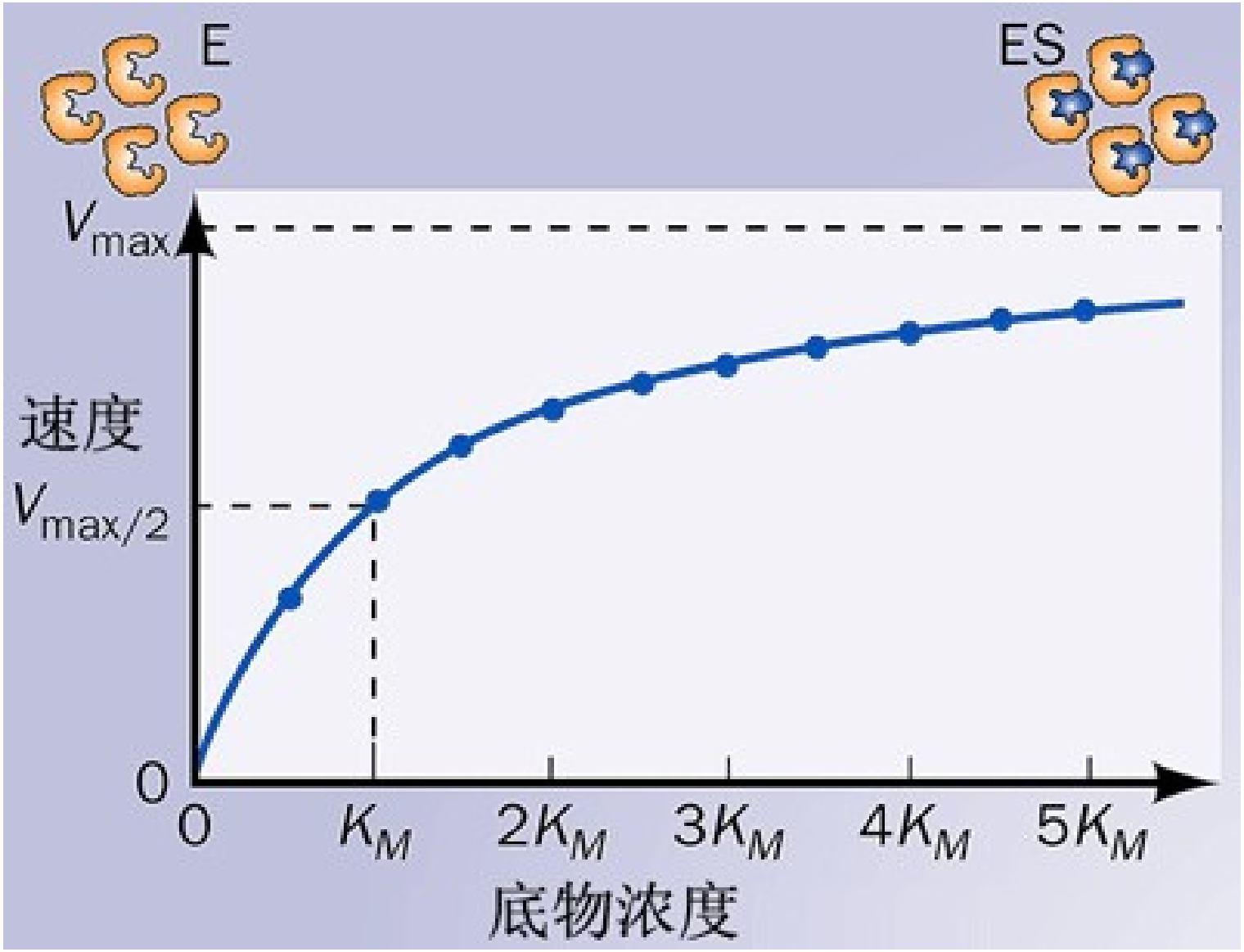




Saturation effect V_{max}

TABLE 3.1 Velocity as a Function of Substrate Concentration for a Putative Enzyme

Substrate Concentration (m*M*)	Velocity (nmol L^{-1} min^{-1})				
	a	b	c	d	e
0	0	0	0	0	0
8.33	13.8	11.5	10	12.6	15
10	16	14.5	17	10	21
12.5	19	16	21	13	23
16.7	23.6	21.4	26	19.5	27
20	26.7	22	28	20	29
25	40	38.6	42.5	39	41
33.3	36.3	41	35	37	40
40	40	39	42	37.6	43
50	44.4	38.6	47	36	50
60	48	47	49	45	51.2
80	50	48.4	52.6	46.3	54.6
100	70	65	75	62.5	76
150	60	59.5	63.8	57.3	65.8
200	66.7	62.5	70	61	72

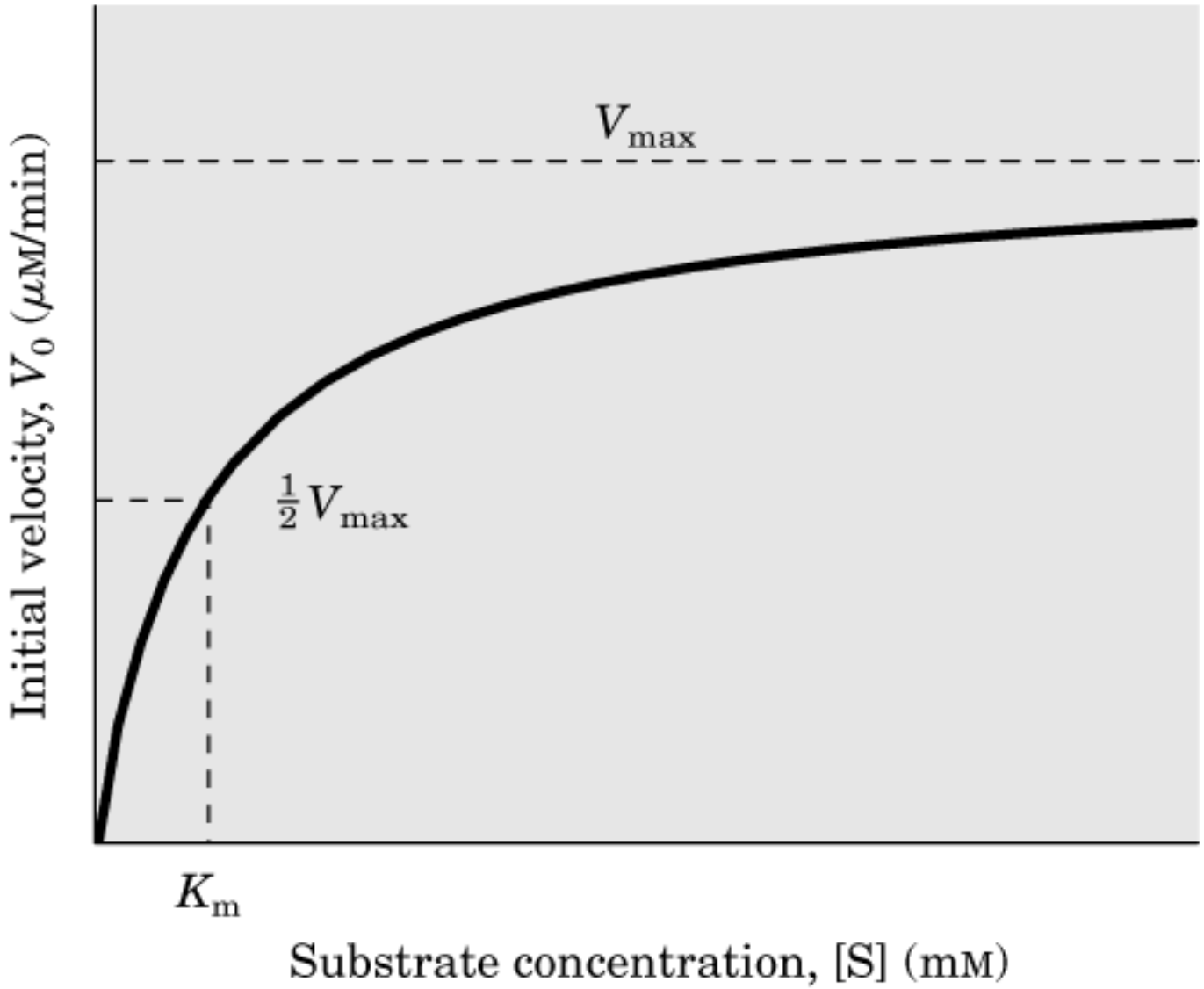

E
ES
V_{max}
速度
$V_{max/2}$
0
0
K_M
$2K_M$
$3K_M$
$4K_M$
$5K_M$
底物浓度



Leonor Michaelis
1875–1949



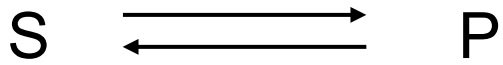
Maud Menten
1879–1960


V_{max}
$\frac{1}{2}V_{max}$
K_m
Initial velocity, V_0 (μM/min)
Substrate concentration, [S] (mM)

$$V_0 = \frac{V_{\max}[\mathrm{S}]}{K_\mathrm{m} + [\mathrm{S}]}$$

1.3 The Michaelis-Menten equation was established to account for the observed relationship between Vo and [S].

1.3.1 A single-substrate, single-product reaction is considered for simplicity.



1.3.2 At early times in the reaction, the concentration of the product ([P]) is and the overall reaction can be written as (no reverse reaction)

$$\mathrm{E} + \mathrm{S} \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} \mathrm{ES} \xrightarrow{k_{\mathrm{cat}}} \mathrm{E} + \mathrm{P}$$

1.3.3 It is hypothesized that the rate-limiting step in enzymatic reactions is the breakdown of the ES complex to form the product and the free enzyme:

$$v = k_{\mathrm{cat}}[\mathrm{ES}]$$

$V=K_2[ES]$

1.3.4

the rate of ES formation = the rate of ES breakdown

the rate of ES formation = $k_1([Et]-[ES])[S]$,

the rate of ES breakdown = $k_{-1}[ES]+k_2[ES]$.

1.3.5 The free enzyme concentration

$$[E]=[E_t]-[ES]$$

($[E_t]$ is the total enzyme concentration).

1.3.6 The amount of substrate bound by the enzyme at any given time is negligible compared with the total [S], because [S] is usually far greater than $[E_t]$ ([S]>>[Et], still true for low [S]); therefore, the free substrate concentration is [S].

1.3.7 $K_m = (k_{-1}+k_2)/k_1$, is the Michaelis-Menten constant.

1.3.8 米氏方程推导过程

the rate of ES formation = $k_1([Et]-[ES])[S]$,

the rate of ES breakdown = $k_{-1}[ES]+k_2[ES]$.

$$k_1([Et]-[ES])[S]= k_{-1}[ES]+k_2[ES]$$

$$([Et]-[ES])[S]=(k_{-1}+k_2)/ k_1\ [ES]$$

Michaelis constant
米氏常数

$$K_m = \frac{k_2 + k_{-1}}{k_1}$$

$$K_m[ES] = [E_t][S] - [ES][S]$$

$$v = k_2[ES]$$

$$v = \frac{k_2[E_t][S]}{K_m + [S]} = \frac{V_{\max}[S]}{K_m + [S]}$$

This is the Michaelis-Menten equation.

The equation fits the observed curve very well.

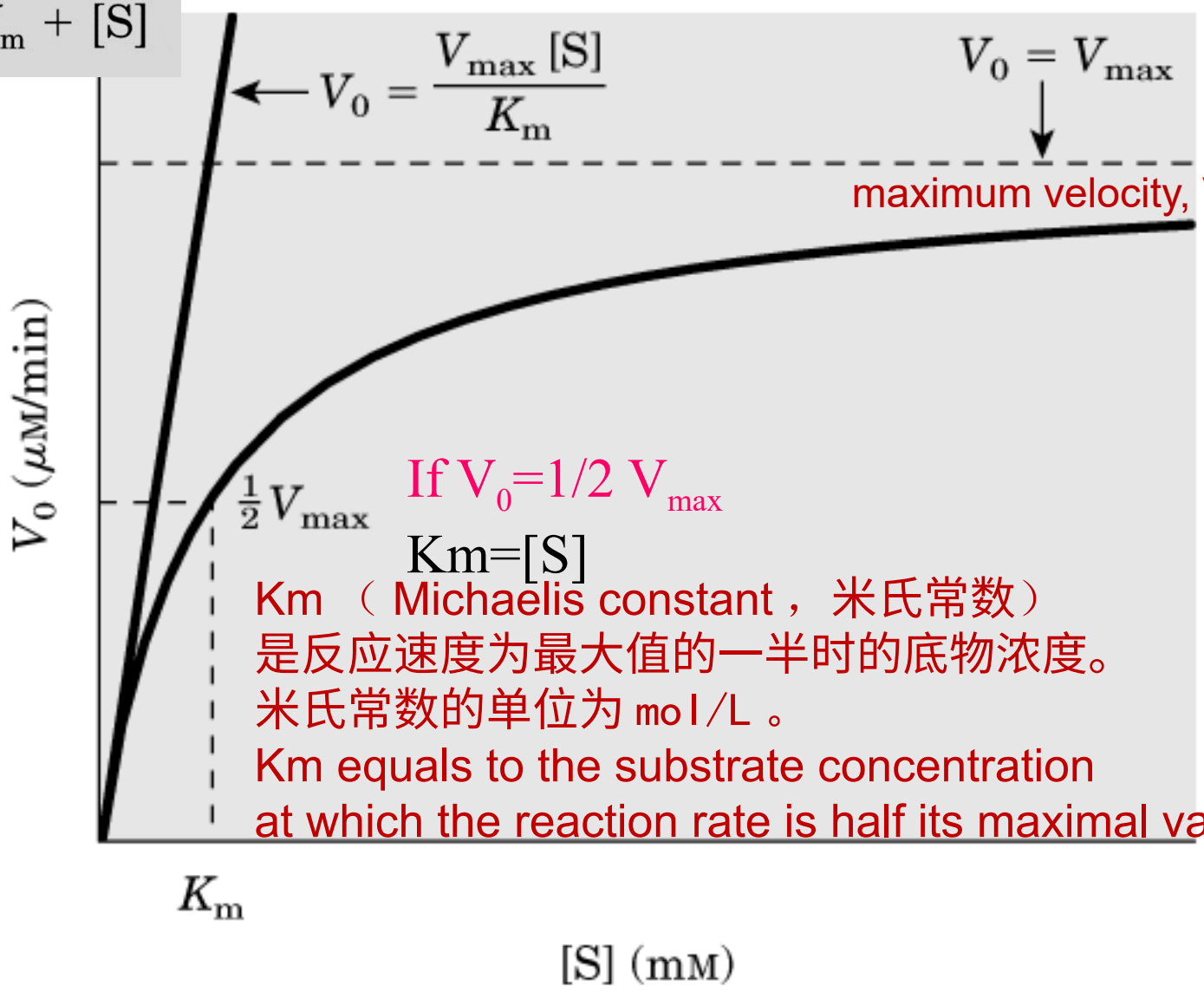
When [S] is very low (<<Km), Vo=Vmax[S]/Km

when [S] is very high (>>Km), Vo=Vmax

$$V_0 = \frac{V_{max}[S]}{K_m + [S]}$$

If [S]<<Km,

If [S]>>Km,



If V_0=1/2 V_{max}

Km=[S]

Km （ Michaelis constant ， 米氏常数）
是反应速度为最大值的一半时的底物浓度。
米氏常数的单位为 mol/L 。
Km equals to the substrate concentration
at which the reaction rate is half its maximal value

米氏常数 $K\mu$ 的意义

- **重要特征物理常数，与酶浓度无关。不同的酶具有不同 Kμ 值，** $K\mu=10^{-3}-10^{-5}$ mol/L.
- **Kμ 值是在固定的底物，一定的温度和 πH ，一定的缓冲体系中测得的，不同条件下具有不同的 Kμ 值。**
- **Kμ 值可表示酶与底物之间的亲和力： Kμ 值大表示亲和程度小，酶的催化活性低 ; Kμ 值小表示亲和程度大 , 酶的催化活性高**
- **同一种酶有几种底物就有几个 Kμ 值，其中 Kμ 值最小的底物一般称为该酶的最适底物或天然底物**



1.4 Km equals to the substrate concentration at which the reaction rate is half its maximal value (Vo=Vmax/2).

Km and Vmax can be determined by varying the substrate concentrations--The Michaelis-Menten equation can be transformed by taking double reciprocal of both sides into the following form:

$$\frac{1}{V_o} = \frac{K_m}{V_{max}} \cdot \frac{1}{[S]} + \frac{1}{V_{max}}$$

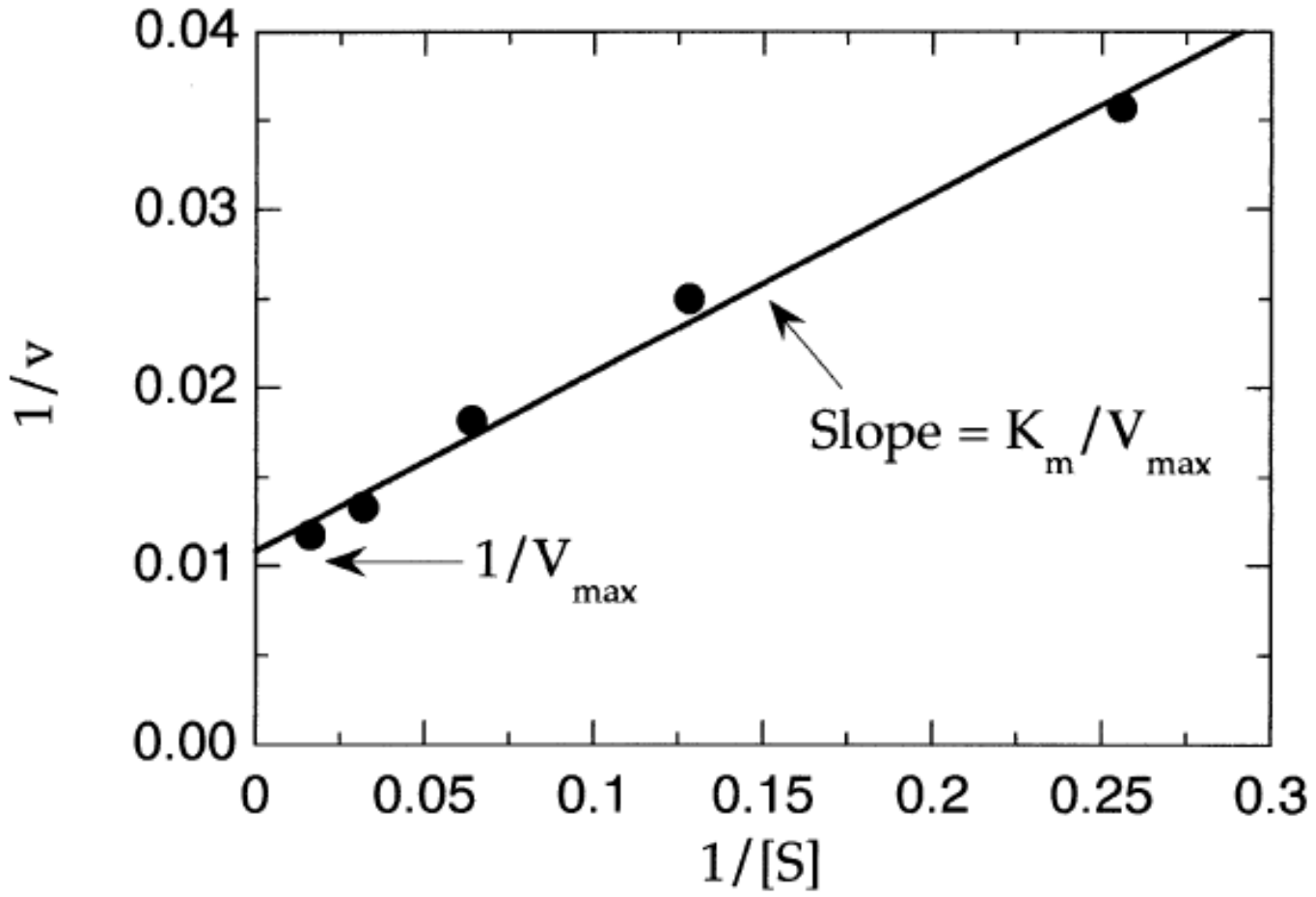
1.4.1 A plot of 1/Vo versus 1/[S] yields a straight line, The double reciprocal plot is called the Lineweaver-Burk plot.

$$\frac{1}{V_0} = \frac{K_m}{V_{max}[S]} + \frac{1}{V_{max}}$$

Table 5.1 Initial velocity (with random error added) as a function of substrate concentration for a model enzymatic reaction

[S](μM)[a]	v(μM product formed s^{-1})	$1/v$(μM^{-1}·s)	$1/[S]$(μM^{-1})
0.98	10	0.100	1.024
1.95	12	0.083	0.512
3.91	28	0.036	0.256
7.81	40	0.025	0.128
15.63	55	0.018	0.064
31.25	75	0.013	0.032
62.50	85	0.012	0.016
125.00	90	0.011	0.008
250.00	97	0.010	0.004

[a]Substrate concentrations reflect a twofold serial dilution starting with an initial solution that provides 250 μM substrate to the final assay reaction mixture.


0.04
0.03
0.02
0.01
0.00
1/v
0
0.05
0.1
0.15
0.2
0.25
0.3
1/[S]
Slope = K_m/V_max
1/V_max

$$\frac{1}{V_0} = \frac{K_m}{V_{max}[S]} + \frac{1}{V_{max}}$$

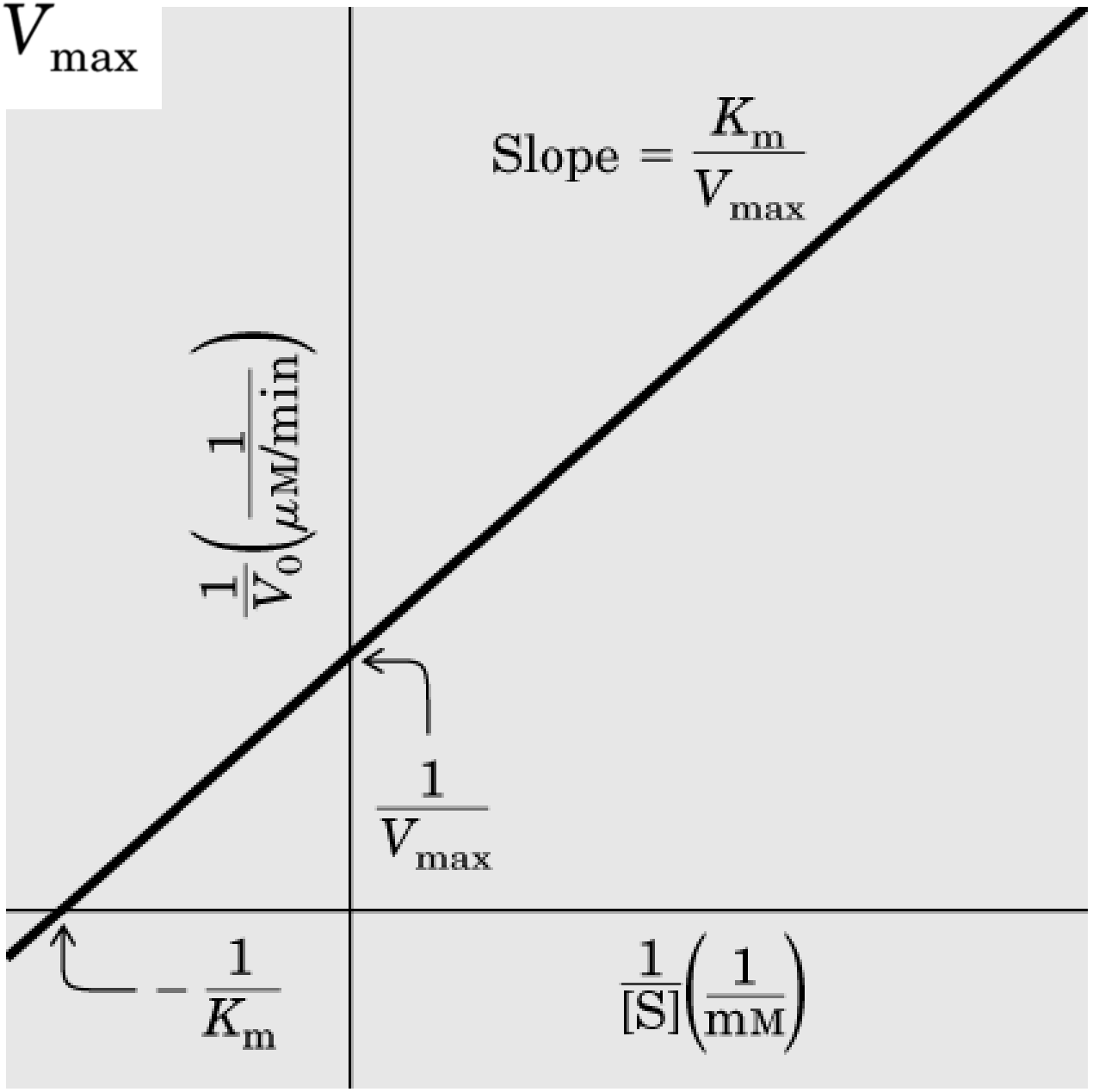


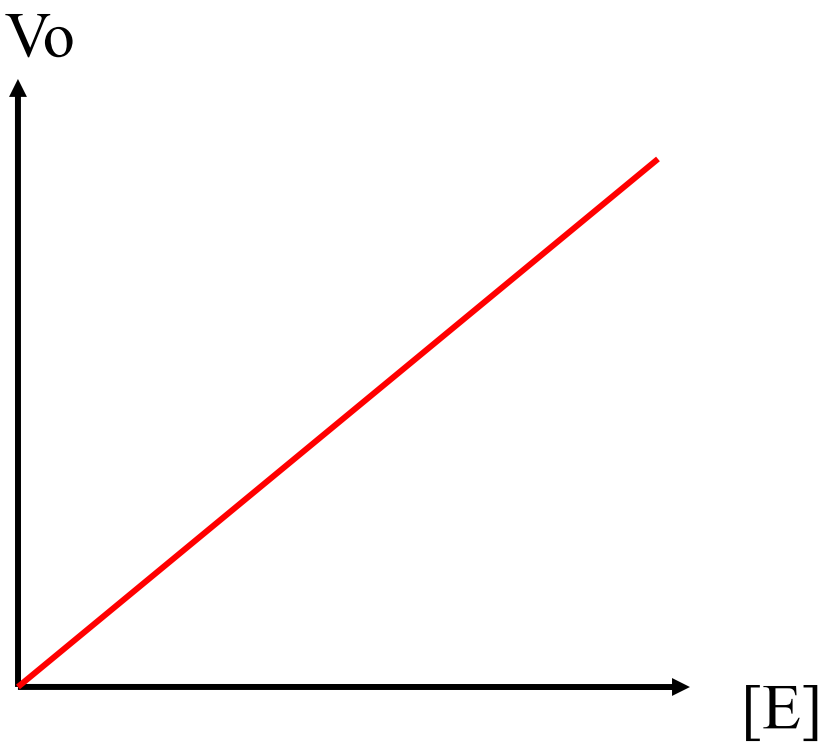
table 8–6

K_m for Some Enzymes and Substrates

Enzyme	Substrate	K_m (mM)
Catalase	H_2O_2	25
Hexokinase (brain)	ATP	0.4
	D-Glucose	0.05
	D-Fructose	1.5
Carbonic anhydrase	HCO_3^-	26
Chymotrypsin	Glycyltyrosinylglycine	108
	N-Benzoyltyrosinamide	2.5
β-Galactosidase	D-Lactose	4.0
Threonine dehydratase	L-Threonine	5.0

K_m 是酶的特征常数， $1/K_m$ 近似表示酶与底物的亲和力

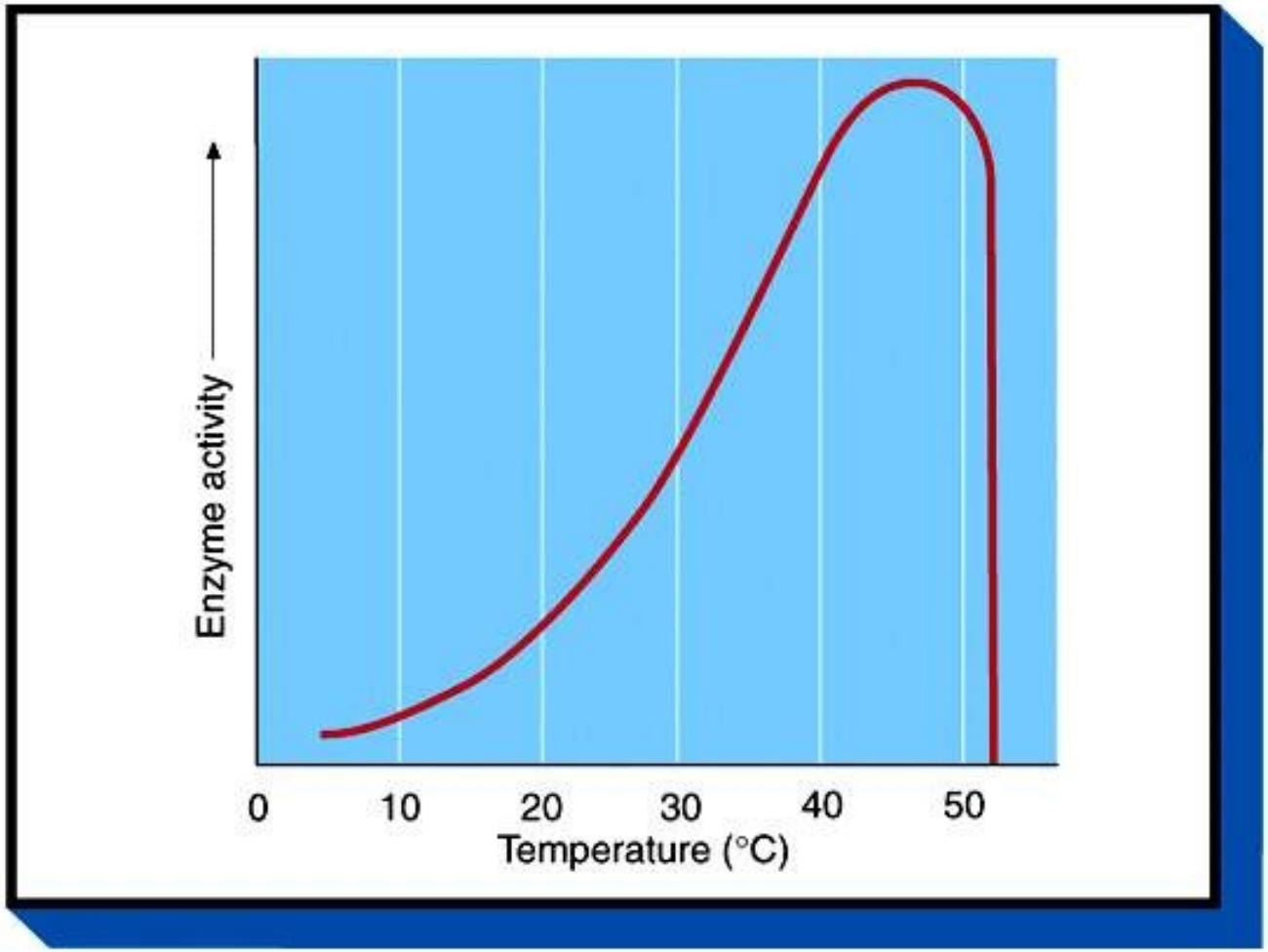
2. When [S] is saturating, a doubling of the enzyme concentration leads to a doubling of Vo.



3. Effects of temperature on the rate

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Effect of temperature on the activity of a typical enzyme. M Figure 6.19



4. Enzyme activity is affected by pH

4.1 Each enzyme has an optimal pH or pH range (where the enzyme has maximal activity).

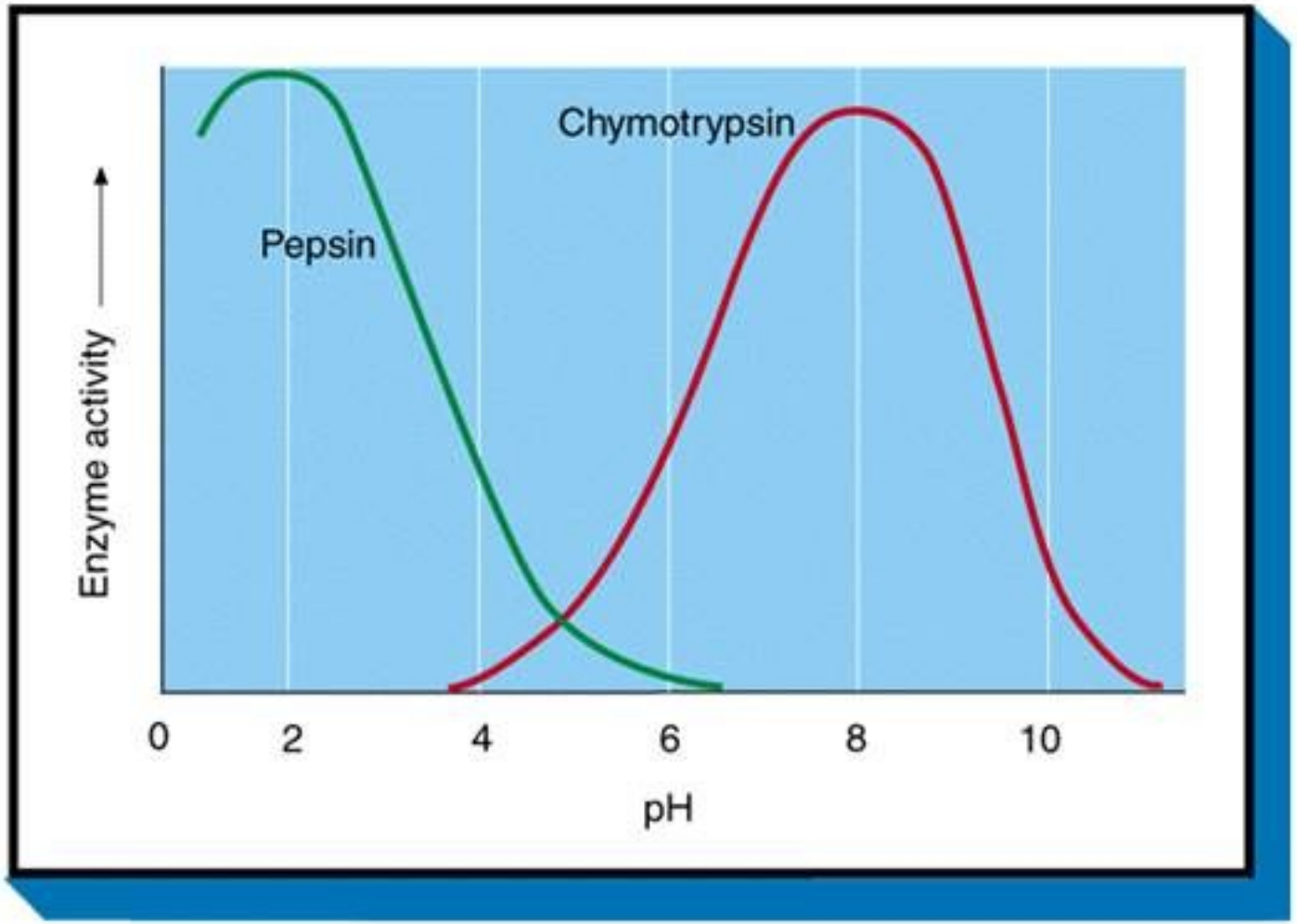
4.1.1 Requirements for the catalytic groups in the active site in appropriate ionization state is a common reason for this phenomenon.

4.1.2 Change of ionization state of surface groups (which may affect the protein structure) sometimes is responsible for this phenomenon.

4.1.3 In rare cases, it is the change of ionization state of substrate that is responsible for this phenomenon.

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The effect of pH on the catalytic activity of two enzymes. M Figure 6.20



4.2 The pH range over which activity changes can provide a clue to what amino acid residues are involved.

4.2.1 This has to be treated with great caution, because in a closely packed environment of a protein, the pK values of amino acid side chains can change significantly (e.g., a nearby positive charge will increase the pK value of a Lys residue, and a negative one will decrease the pK!).

5. Enzyme assays (quantification of enzyme (activity))

- **An enzyme assay measures the conversion of substrate to product, under conditions of cofactors, pH and temperature at which the enzyme is optimally active.**
- **High substrate concentrations are used so that the initial reaction rate is proportional to the enzyme concentration.**
- **Enzyme activity is commonly expressed by the initial rate (Vo). The units of Vo are μmol min^{-1}**
- **1 U=1 μmol min^{-1}**
- **1 kat = 1 mol s^{-1} 1 kat = 6 x 10^7 U**

Σπεχιφιχ αχτιωιτψ (μmol min^{-1} $μγ^{-1}$) 比活力 : Τηε νυμβερ οφ μιχρομολεσ (μολ) οφ α συβστρατε τρανσφορμεδ βψ αν ενζψμε πρεπαρατιον περ μινυτε περ μιλλιγραμ οφ προτειν ατ 25°C; α μεασυρε οφ **ενζψμε πυριτψ**.

Σπεχιφιχιτψ: Τηε αβιλιτψ οφ αν ενζψμε ορ ρεχεπτορ το δισχριμινατε αμονγ χομπετινγ συβστρατεσ ορ λιγανδσ.

- 酶偶联法 **Linked enzyme assays**

$$\Sigma \xrightarrow{E1} \Pi 1 \xrightarrow{E2} \Pi 2$$

条件： S ⟶ P1 反应很慢， P1 ⟶ P2 反应很快

偶联工具酶 E2 纯度高，加入酶量过量

P2 有光吸收或荧光变化



Isoenzymes

- Isoenzymes are different forms of an enzyme which catalyze the same reaction, but which exhibit different physical or kinetic properties. The isoenzymes of Lactate DeHydrogenase (LDH) can be separated electrophoretically and can be used clinically to diagnose a myocardial infarction.



同工酶（isozyme）

概念：催化同一化学反应，但其酶蛋白本身的组成、分子结构有所不同的一组酶，一般由２个或以上的亚基构成。

同工酶技术的应用，与酶制剂应用不同，不是将酶单纯的作为催化剂，追求酶的高效而专一的催化功能，它更多是把同工酶作为一组分子标记物，利用酶灵敏而专一的检测之便，显示生物物种或个别体基因型的共性和个性。

命名

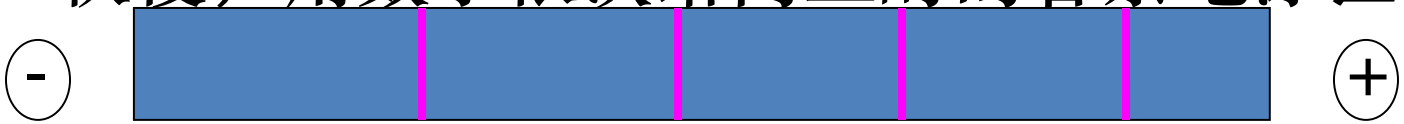
1. 组织命名：酶名称前加组织名称

肝（L） 肌（M） 肾（K） 心（H） 脑（B)

LDH M 型 H 型

2. 数字命名

1976 年 IUB：在规定条件下，按向阳极移动速度快慢，用数字依次给同工酶的各条电泳区带编号。



4 3 2 1

3. A、B、C 命名法

不同亚基用 A、B、C 表示

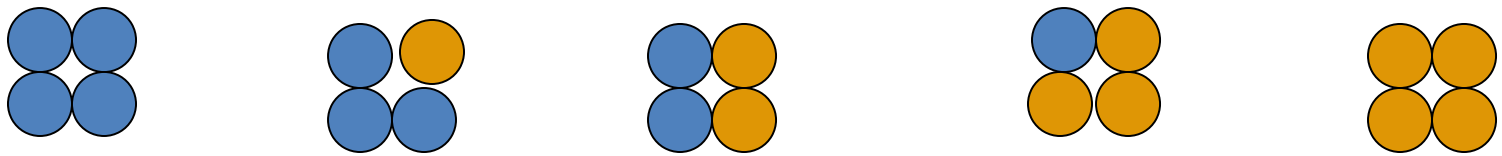
LDH 亚基分为 M 型和 H 型，4 个亚基组成

M 型（肌型）——含碱性氨基酸

H 型（心型）——含酸性氨基酸

LDH 的 5 种组分：

LDH_1	LDH_2	LDH_3	LDH_4	LDH_5
H_4	H_3M	H_2M_2	HM_3	M_4



应用：

1. 同工酶与遗传的关系

基因变异引起同工酶谱的变化

如： LDH 的基因型：　　A 基因　　B 基因

如果等位基因 A 和 A 中的一个基因发生了变异，变成 A* ，

这时由 A 、 A* 、 B 三种基因编码，分别产生三种亚基，

A　A*　B

15 种分子形式

LDH_5	LDH_4	LDH_3	LDH_2	LDH_1
A_4	BA_3	B_2A_2	B_3A	B_4
A_3A^*	BA_2A^*	B_2AA^*	B_3A^*	
$A_2A^*_2$	BAA^*_2	$B_2A^*_2$		
AA^*_3	BA^*_3			
A^*_4				

2. 同工酶与个体发育的关系

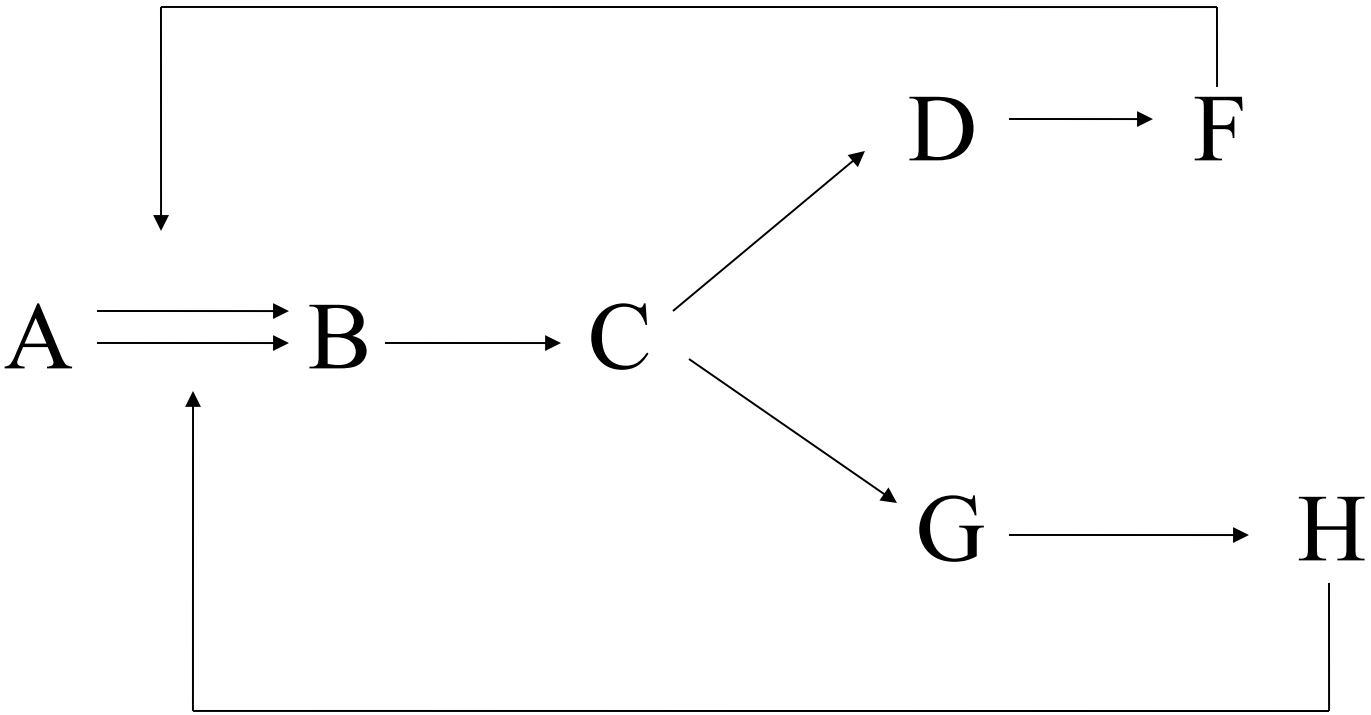
乙醇脱氢酶　　ADH：

发育早期：质片中　ADH_1、ADH_2

稍后：ADH_3、ADH_4

种子萌发的前 10 天　ADH_3，而 ADH_1、ADH_2 ADH_4 仅在萌发的第一天存在，第四天后消失

3. 同工酶与代谢调节的关系



4. 同工酶　用于疾病诊断，灵敏且可靠：

心脏疾病　　LDH_1↗ LDH_2↗ LDH_3

DH_5↘

急性肝炎　　LDH_5↗

肝硬化　　LDH_5　LDH_1　LDH_3↗

原发生肝癌　LDH_3　LDH_4　LDH_5↗，

但 $LDH_5 > LDH_4$

转移性肝癌　LDH　LDH　LDH↗

C4 Enzyme inhibition

- Enzyme inhibition
- Reversible inhibition
- Irreversible inhibition



1. The inhibition of enzymatic activity by specific small molecules and ions is important.

1.1 It serves as a major control mechanism in biological systems.

1.2 Many drugs and toxic agents act by inhibiting enzymes.

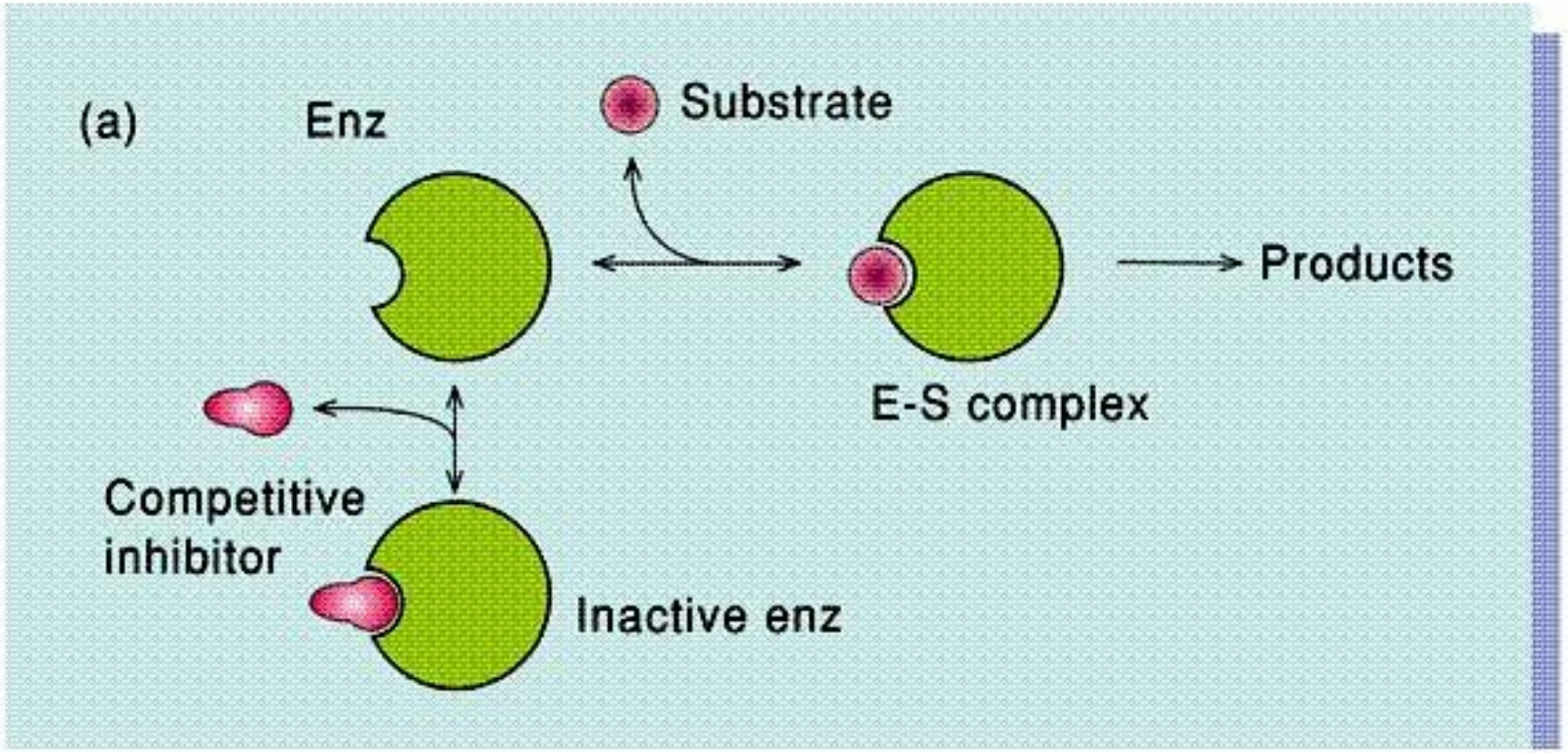
1.3 Inhibition can be a source of insight into the mechanism of enzyme action: residues critical for catalysis can often be identified by using specific (irreversible) inhibitors.

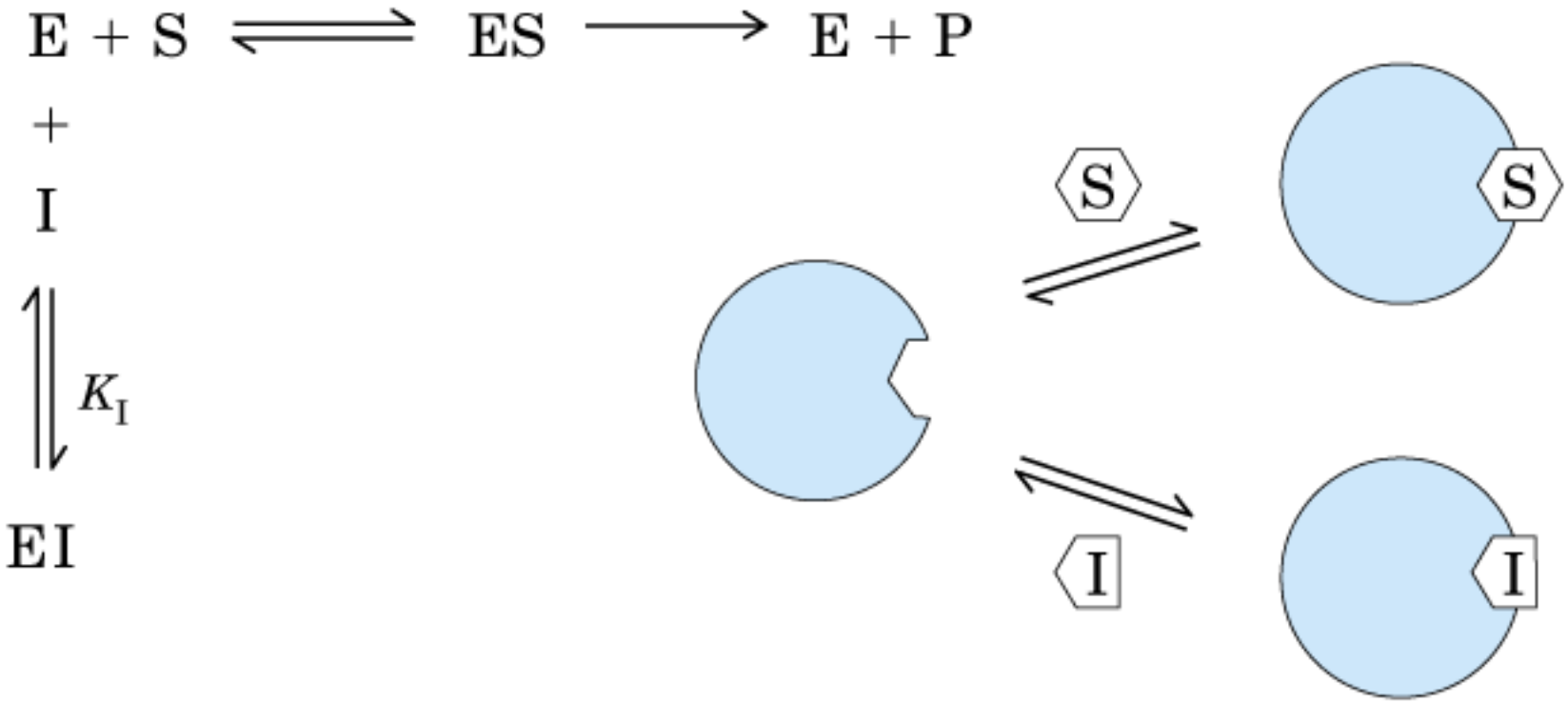
2. Reversible inhibition can usually be divided into different types.

2.1 Reversible inhibitors bind to enzyme noncovalently.

2.2 In competitive inhibition, the inhibitor competes with the substrate for the active site (binding of one prevents binding of the other, forming ES or EI complexes but no ESI complexes.).

2.3 Competitive inhibitors are often compounds that resemble the substrates.


(a)
Enz
Substrate
Products
E-S complex
Competitive
inhibitor
Inactive enz



(a) Competitive inhibition

COMPETITIVE INHIBITION

In this type of reversible inhibition, a compound competes with an enzyme's substrate for binding to the active site,

$$\begin{array}{ccccc} \mathrm{E} + \mathrm{S} & \overset{K_s}{\rightleftharpoons} & \mathrm{ES} & \xrightarrow{k_{\mathrm{cat}}} & \mathrm{E} + \mathrm{P} \\ + & & & & \\ \mathrm{I} & & & & \\ \| & K_i & & & \\ \mathrm{EI} & & & & \end{array} \qquad (4.1)$$

$$v = k_{\text{cat}}\,[\text{ES}]$$

$$K_s = \frac{[\text{E}][\text{S}]}{[\text{ES}]} \qquad K_i = \frac{[\text{E}][\text{I}]}{[\text{EI}]} \tag{4.2}$$

$$[\text{E}_T] = [\text{E}] + [\text{ES}] + [\text{EI}] = [\text{E}] + \frac{[\text{E}][\text{S}]}{K_s} + \frac{[\text{E}][\text{I}]}{K_i}$$

Normalization of the rate equation by total enzyme concentration ($v/[\text{E}_T]$) and rearrangement results in the following expression for the velocity of an enzymatic reaction in the presence of a competitive inhibitor:

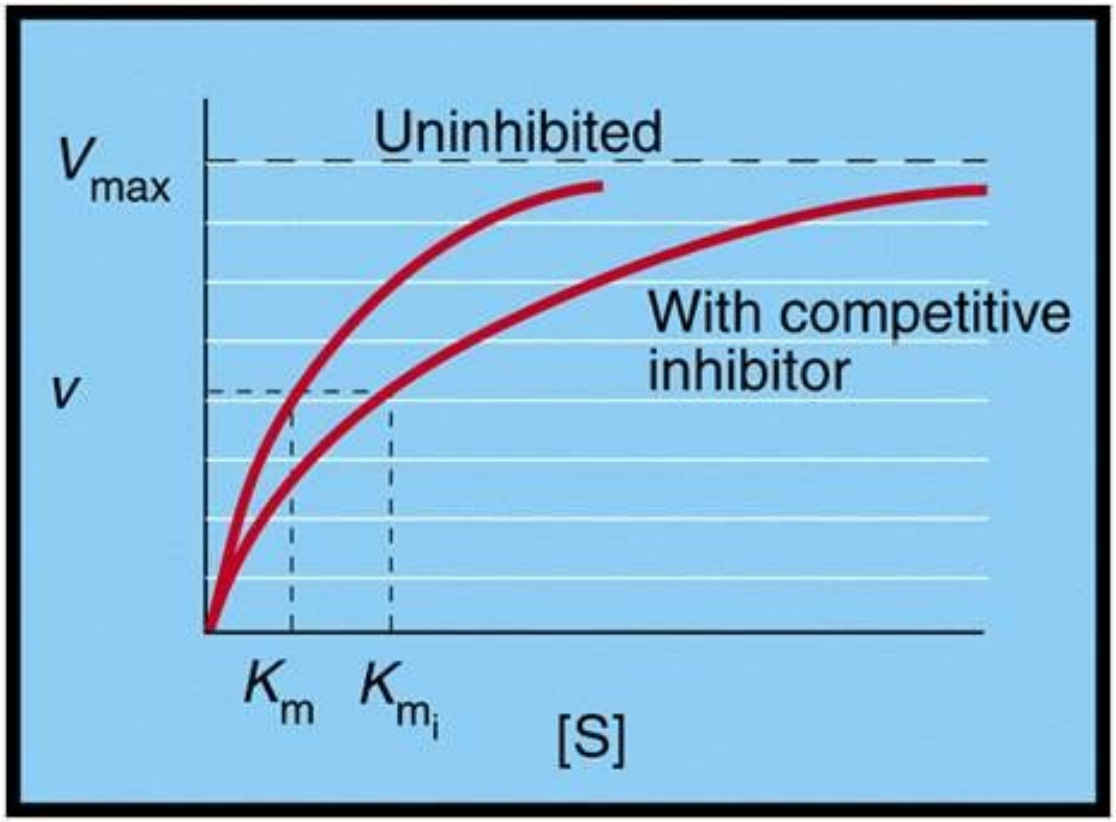
$$v = \frac{V_{\max}[\text{S}]}{K_s^* + [\text{S}]} = \frac{V_{\max}[\text{S}]}{\alpha K_s + [\text{S}]} \tag{4.3}$$

where K_s^* corresponds to the apparent enzyme–substrate dissociation constant in the presence of an inhibitor. In the case of competitive inhibition, $K_s^* = \alpha K_s$, where

$$\alpha = 1 + \frac{[\text{I}]}{K_i} \tag{4.4}$$

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Michaelis-Menten plot of uninhibited enzyme activity versus competitive inhibition. M Figure 6.9



τηε $K_{\mu\iota}$ οβσερωεδ ιν τηε πρεσενχε οφ τηε ινηιβιτορ, ισ οφτεν χαλλεδ τηε "αππαρεντ" K_{μ}

$\alpha = 1 + [I]/K_I \qquad \alpha > 1$

$$\frac{1}{V_0} = \left(\frac{\alpha K_m}{V_{max}}\right)\frac{1}{[S]} + \frac{1}{V_{max}}$$

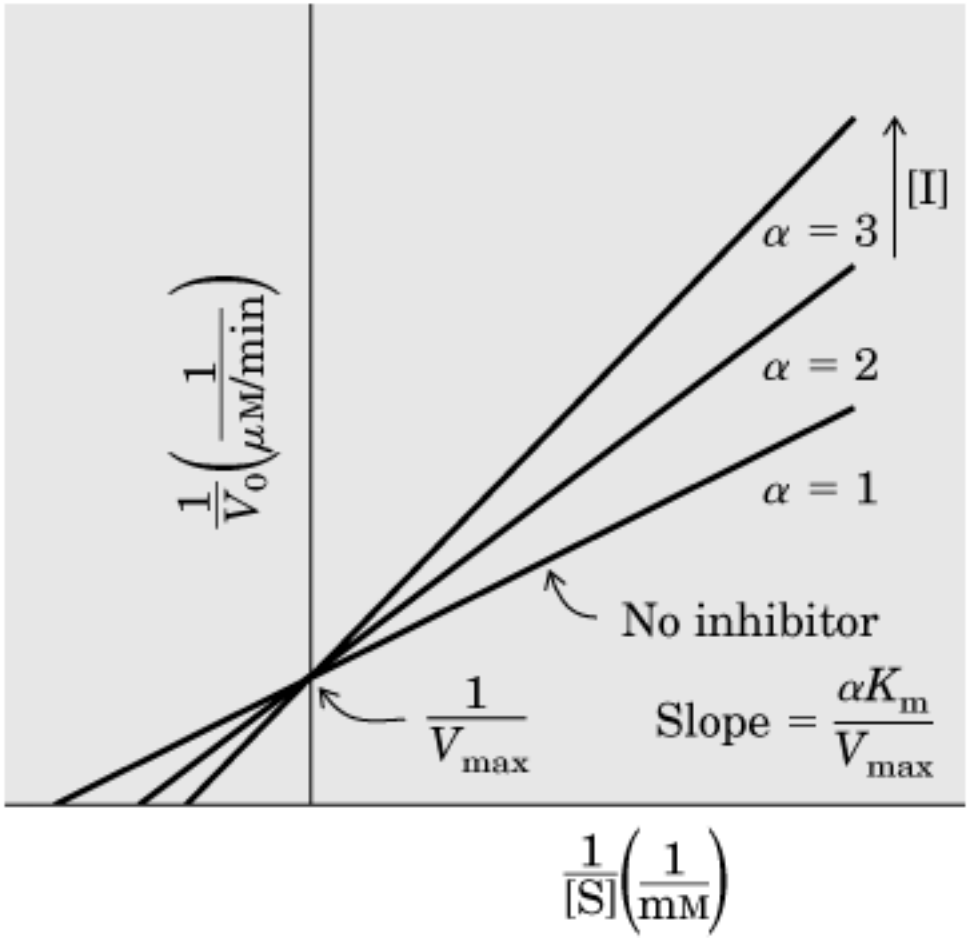


Table 8.1 Hypothetical velocity as a function of substrate concentration at three fixed concentrations of a competitive inhibitor

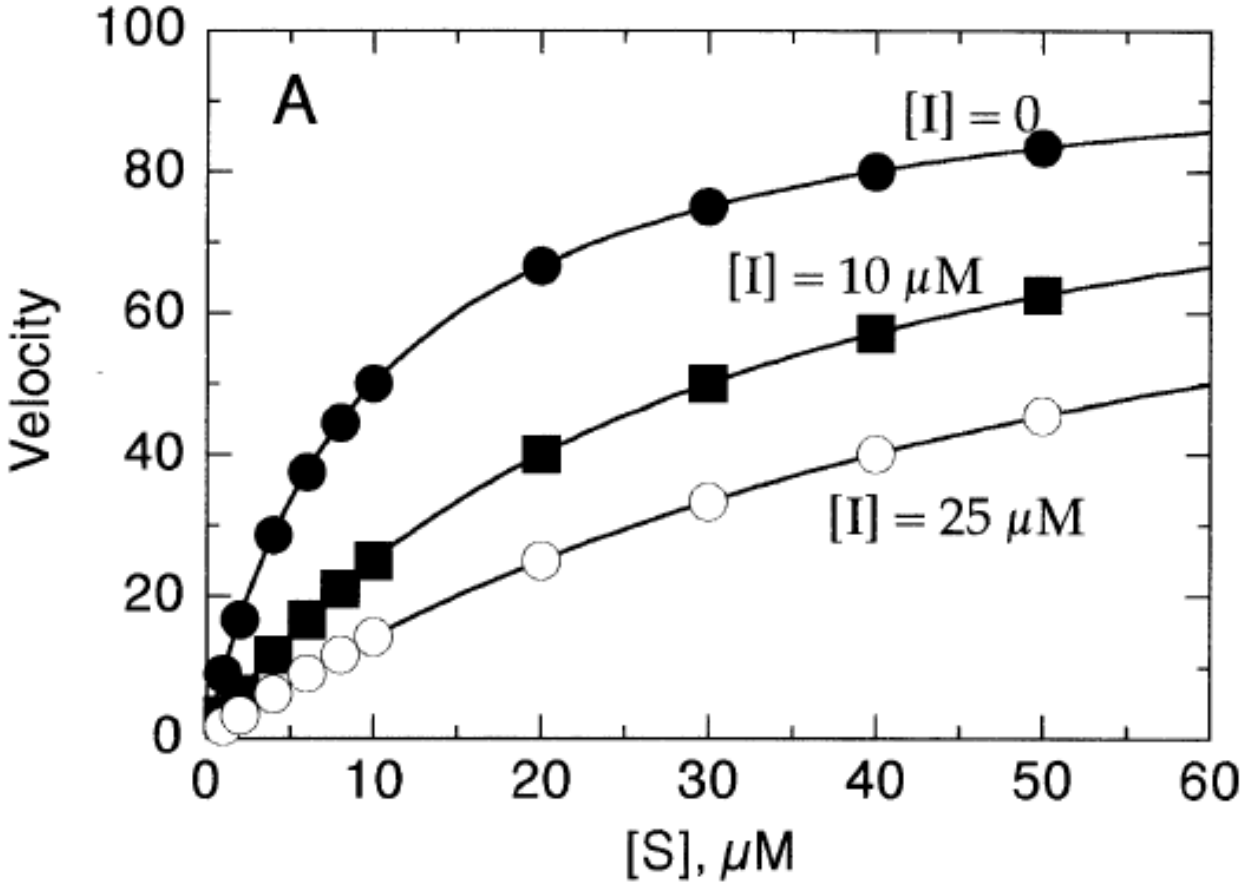
	Velocity (arbitrary units)		
[S] (μM)	[I] = 0	[I] = 10 μM	[I] = 25 μM
1	9.09	3.23	1.69
2	16.67	6.25	3.23
4	28.57	11.77	6.25
6	37.50	16.67	9.09
8	44.44	21.05	11.77
10	50.00	25.00	14.29
20	66.67	40.00	25.00
30	75.00	50.00	33.33
40	80.00	57.14	40.00
50	83.33	62.50	45.46

"αππαρεντ" K_μ
表观 Km

$$[\mathrm{I}] = 0\ \mu\mathrm{M} \qquad V_{\max} = 100, \qquad K_m = 10.00\ \mu\mathrm{M}$$

$$[\mathrm{I}] = 10\ \mu\mathrm{M}, \qquad V_{\max}^{\mathrm{app}} = 100, \qquad K_{\mathrm{m}}^{\mathrm{app}} = 30.00\ \mu\mathrm{M}$$

$$[\mathrm{I}] = 25\ \mu\mathrm{M}, \qquad V_{\max}^{\mathrm{app}} = 100, \qquad K_{\mathrm{m}}^{\mathrm{app}} = 60.00\ \mu\mathrm{M}$$


A
[I] = 0
[I] = 10 μM
[I] = 25 μM
Velocity
[S], μM
0
20
40
60
80
100
0
10
20
30
40
50
60

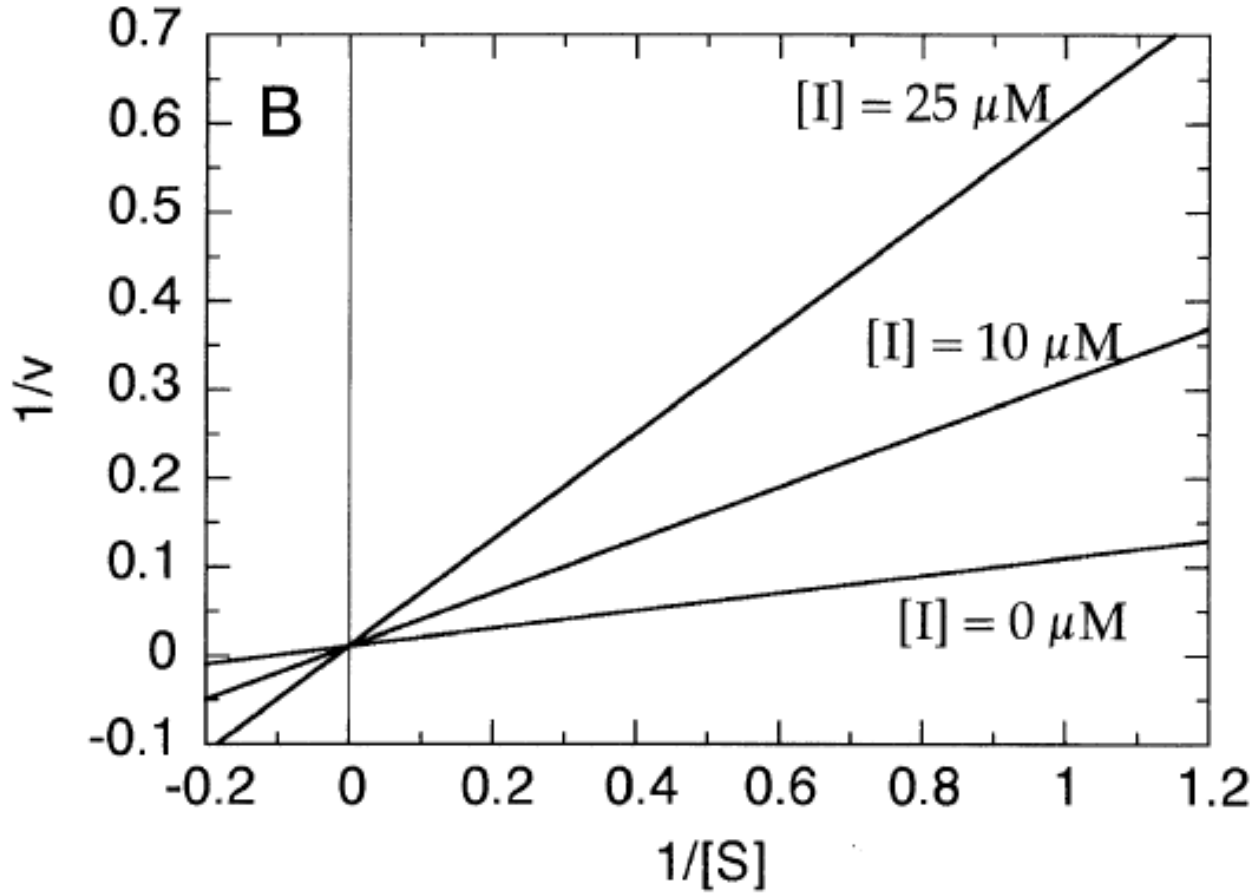
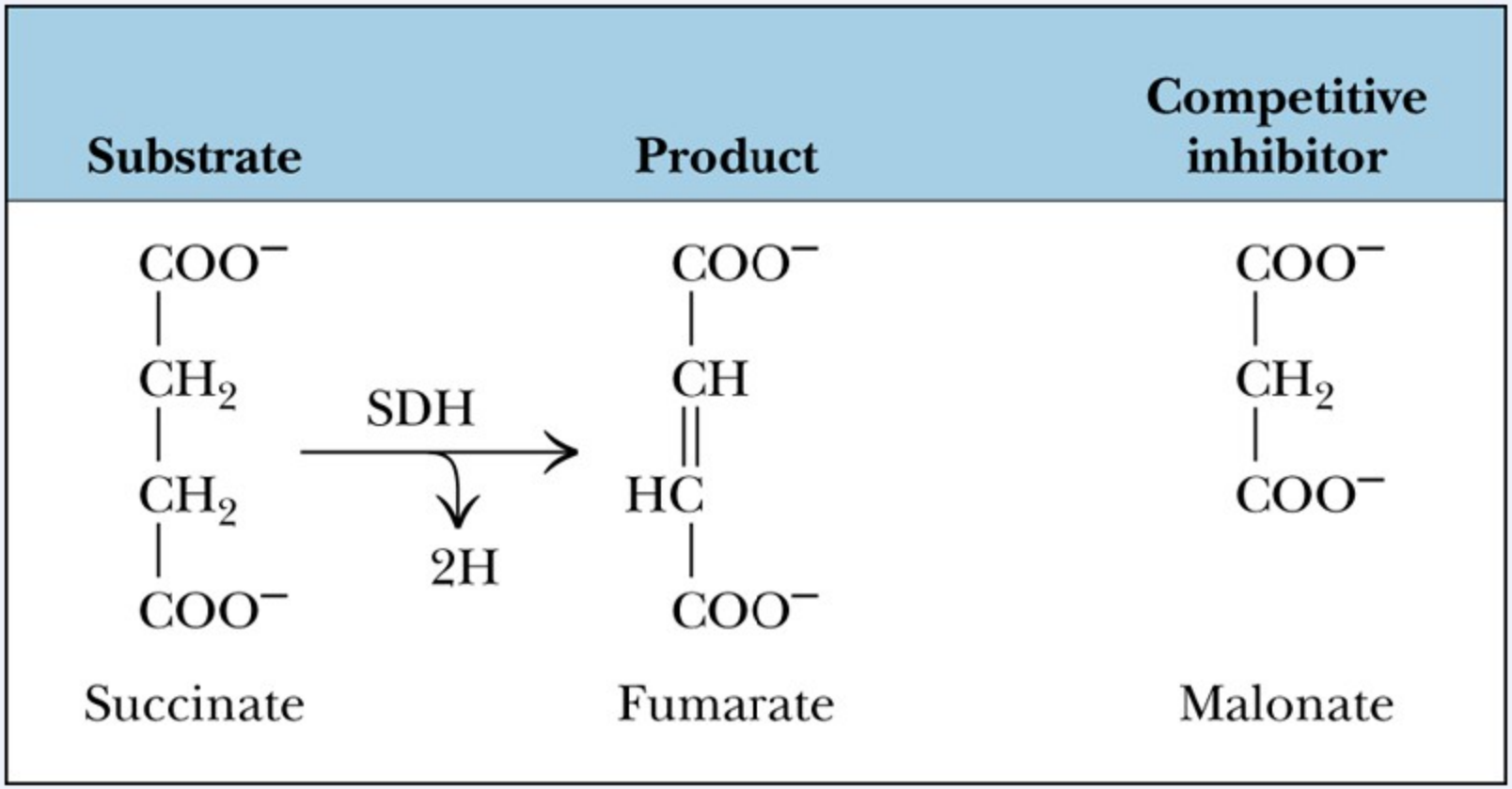


Figure 8.3 Untransformed (A) and double-reciprocal (B) plots for the effects of a competitive inhibitor on the velocity of an enzyme catalyzed reaction. The lines drawn in (B) are obtained by applying Equation 5.24 to the data in (A) and using the apparent values of the kinetic constants in conjunction with Equation 5.34. See text for further details.

Vmax is not affected by the presence of a competitive inhibitor (There is always some high substrate concentration that will replace the inhibitor from the enzyme's active site).

Km is increased due to the presence of a competitive inhibitor. Higher substrate concentration is needed to achieve Vmax/2.



Inhibition of succinate dehydrogenase by malonate

In noncompetitive (or mixed competitive)

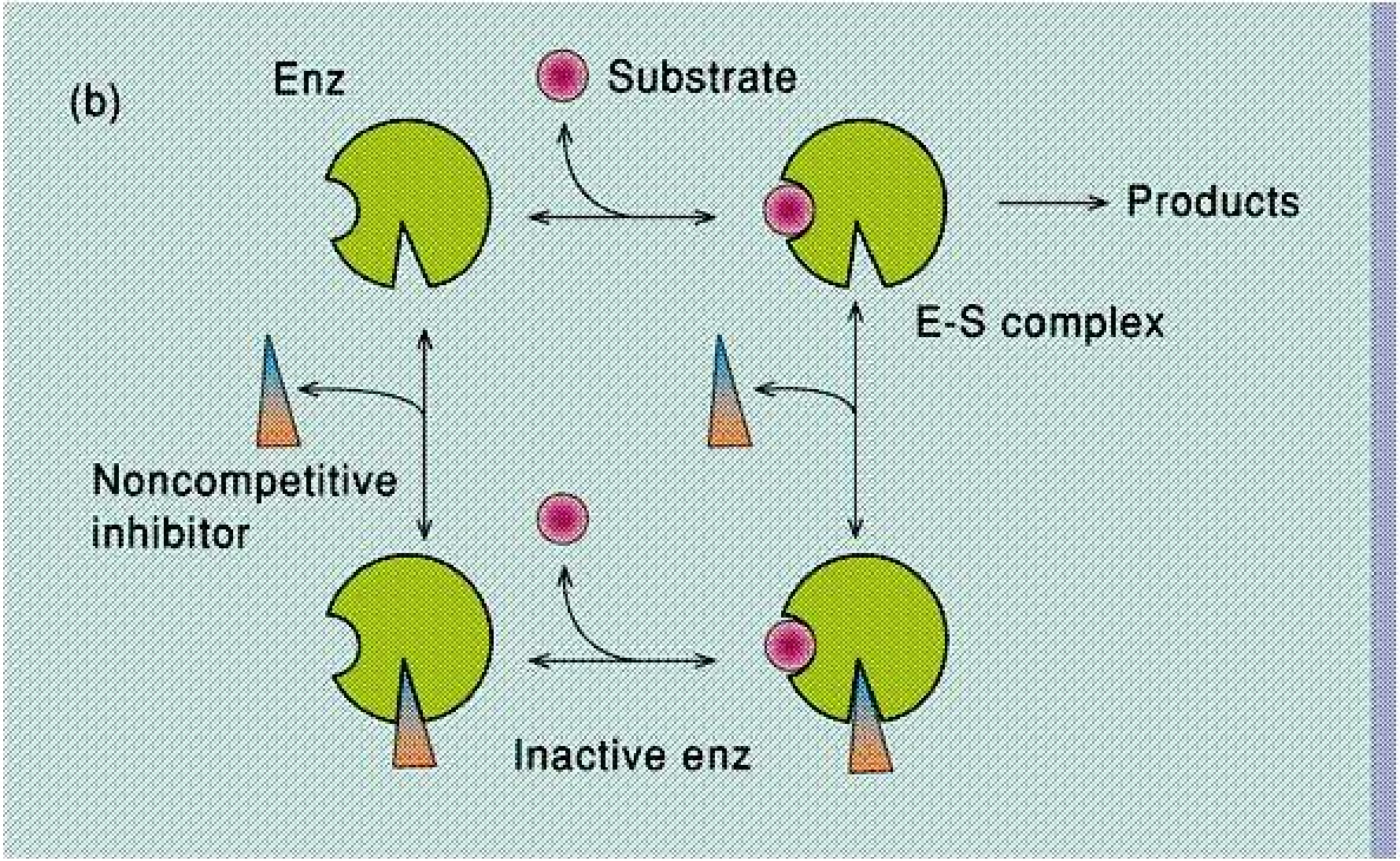
inhibition, the inhibitor binds to a site distinct from that (the active site) which binds the substrate.

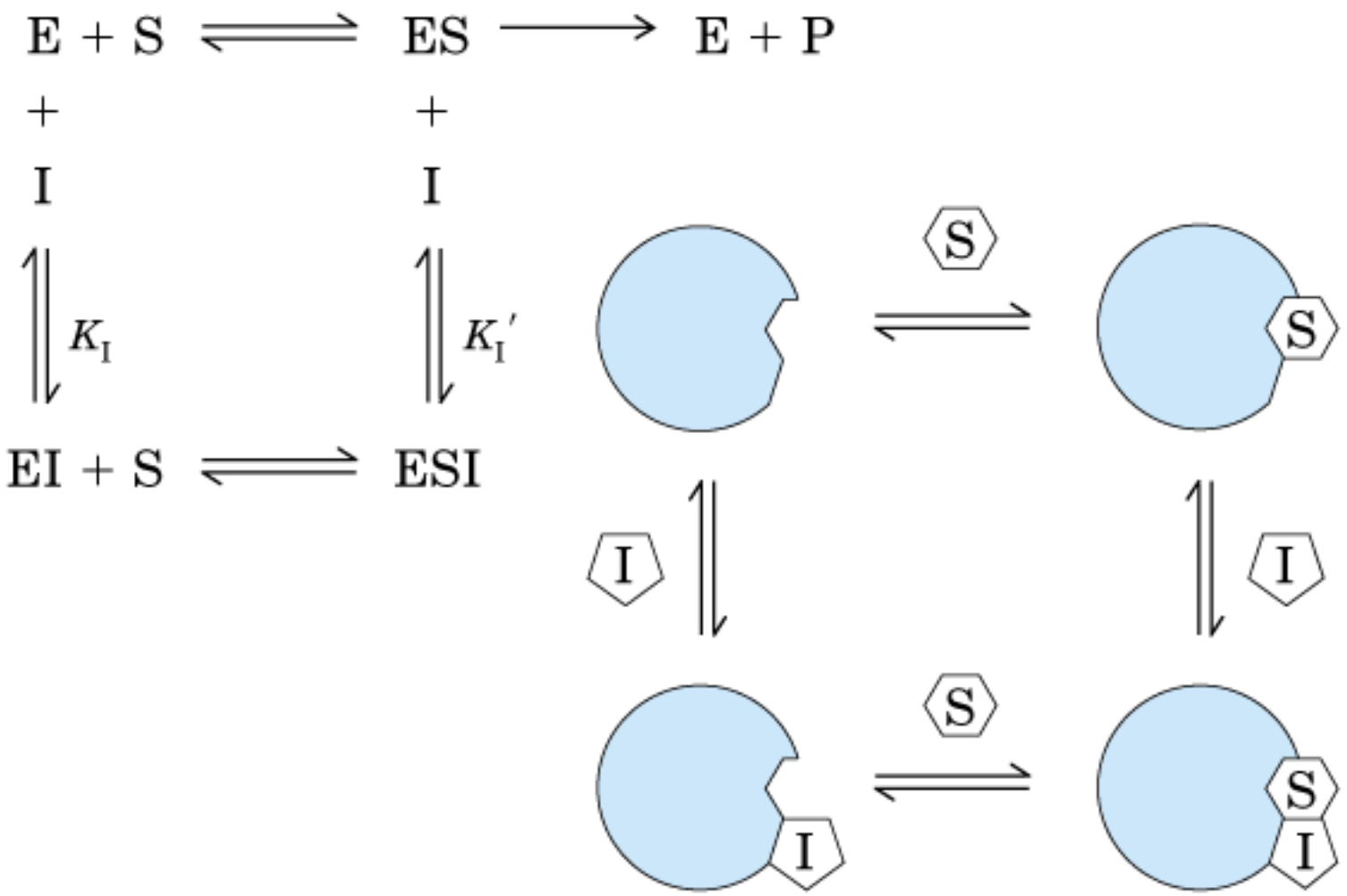
Inhibitor binding does not affect substrate binding and vice versa (i.e., inhibitor can bind to ES complex, substrate can bind to EI complex).

The enzyme is inactivated when inhibitor is bound (whether or not substrate is also present). (e.g.)

The apparent Vmax is lowered (due to the concentration decrease of active enzymes) , apparent Km is increased (mixed) or not changed (noncompetitive).

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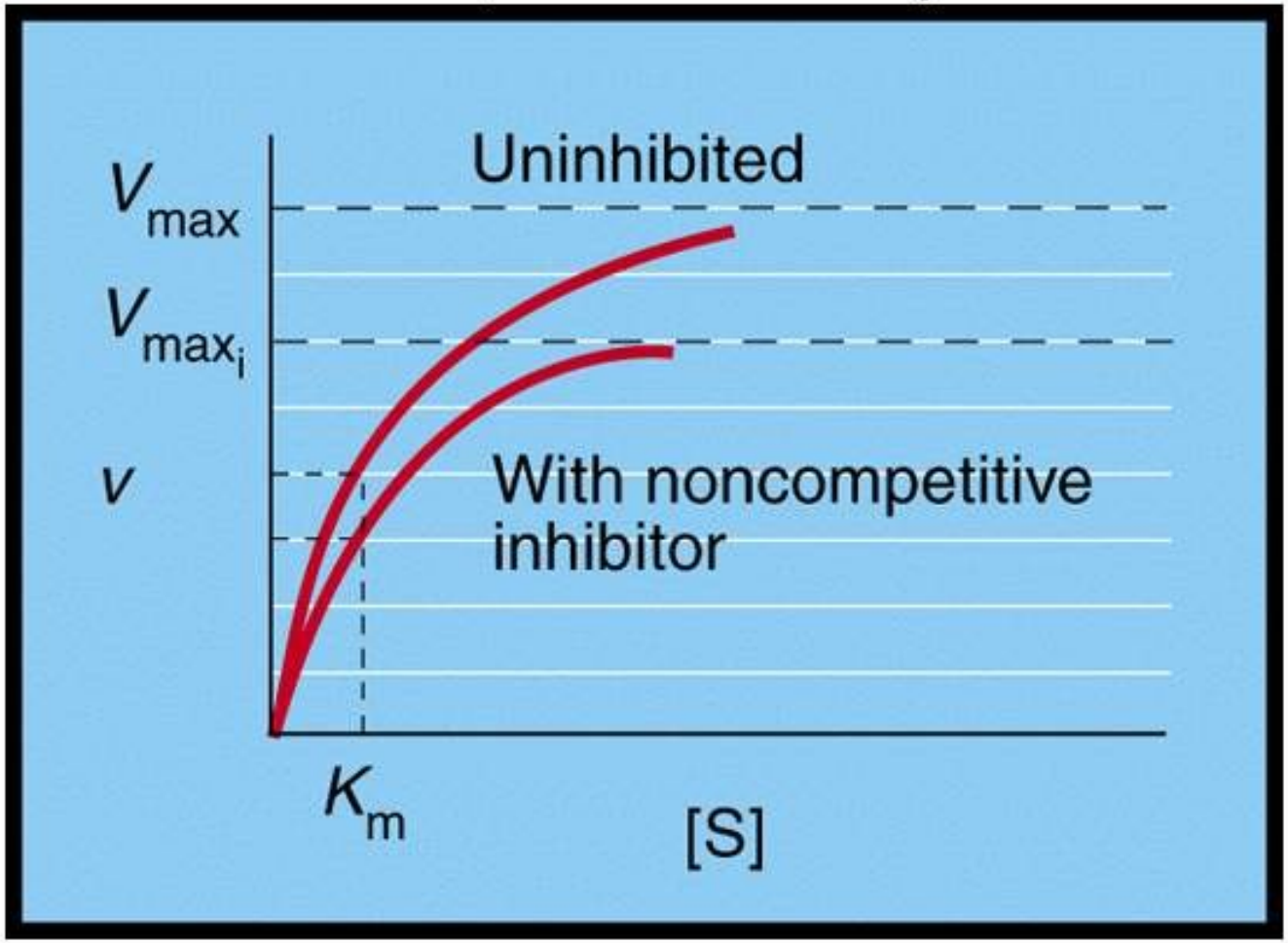
(c) Mixed inhibition

Noncompetitive inhibition

$K_I = K_I'$

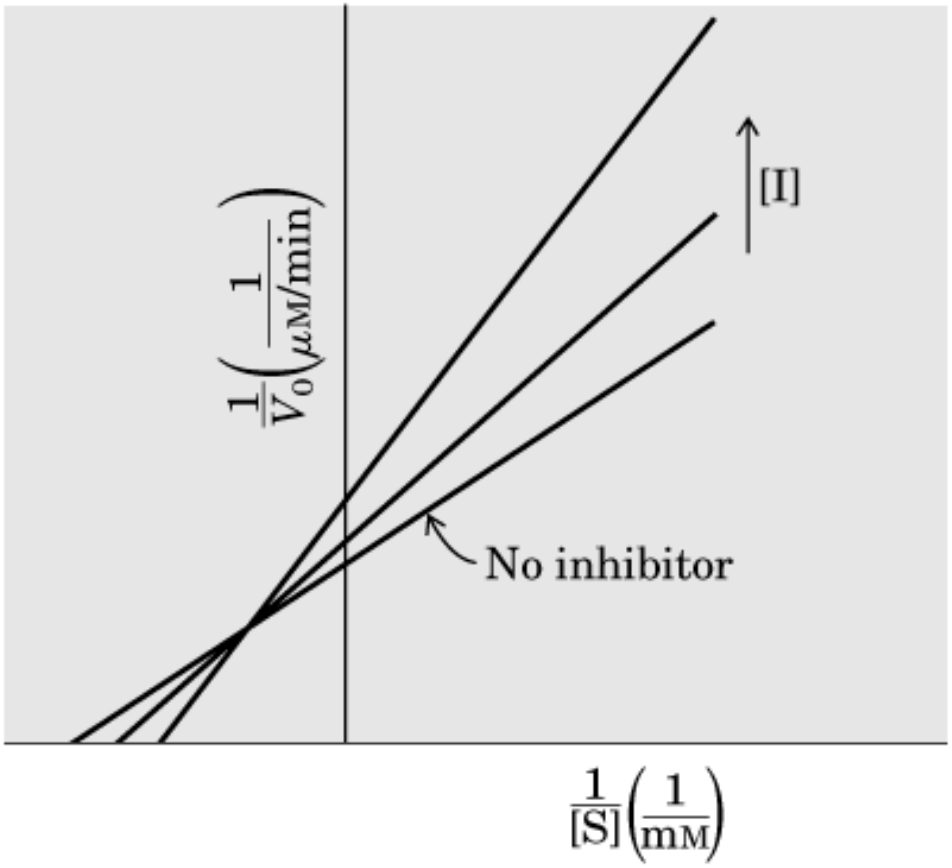
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Michaelis-Menten plot of uninhibited enzyme activity versus noncompetitive inhibition. M Figure 6.12



Mixed or noncompetitive inhibition, plots similar to the sequential binding in ternary complexes

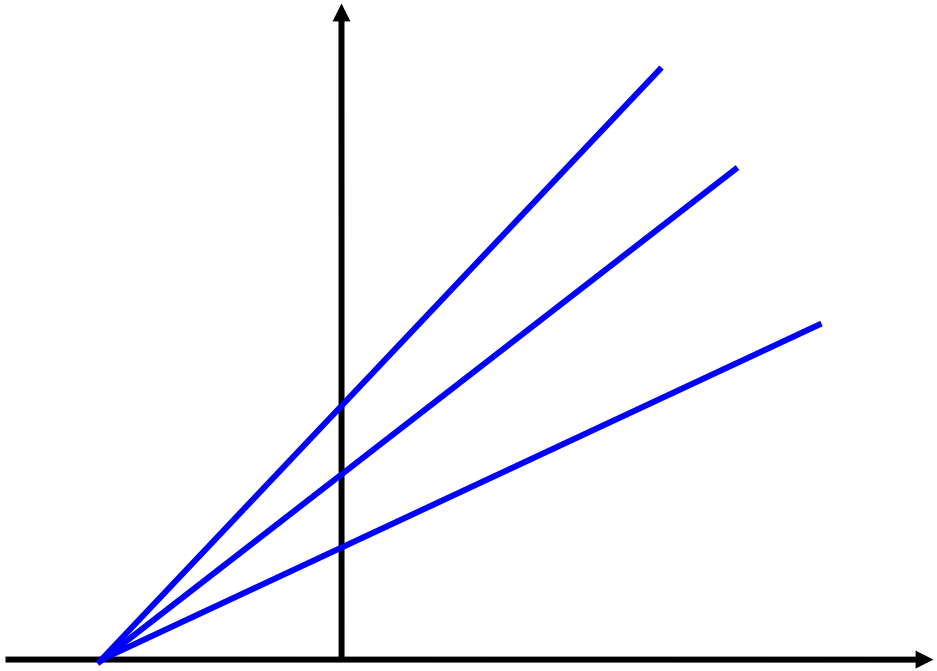
$$\frac{1}{V_0} = \left(\frac{\alpha K_m}{V_{max}}\right)\frac{1}{[S]} + \frac{\alpha'}{V_{max}}$$



Km increased； Vmax decreased

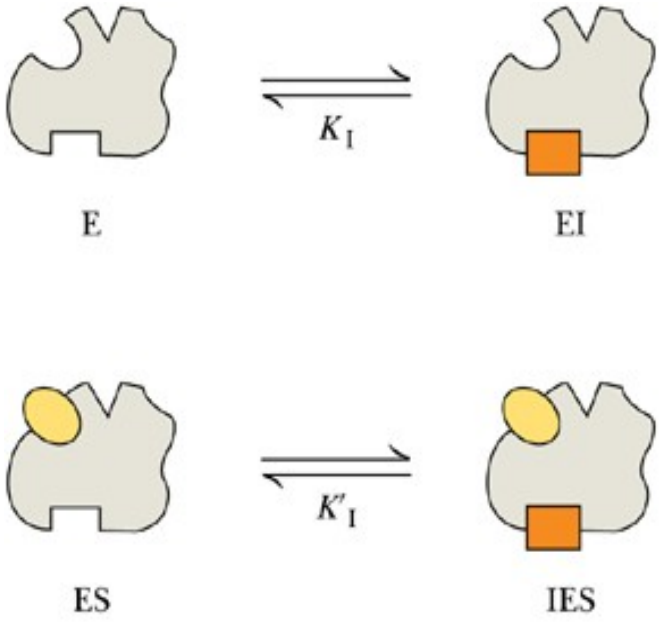
$K_I=K_I'$

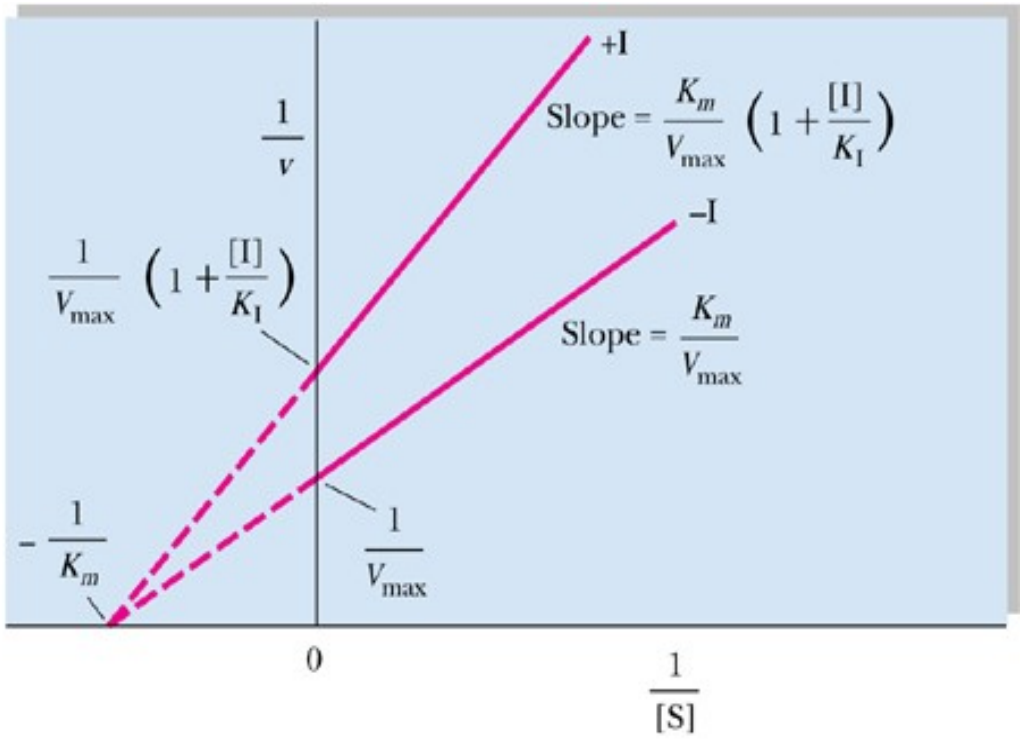
$\alpha=\alpha'$



Km not change； Vmax decreased

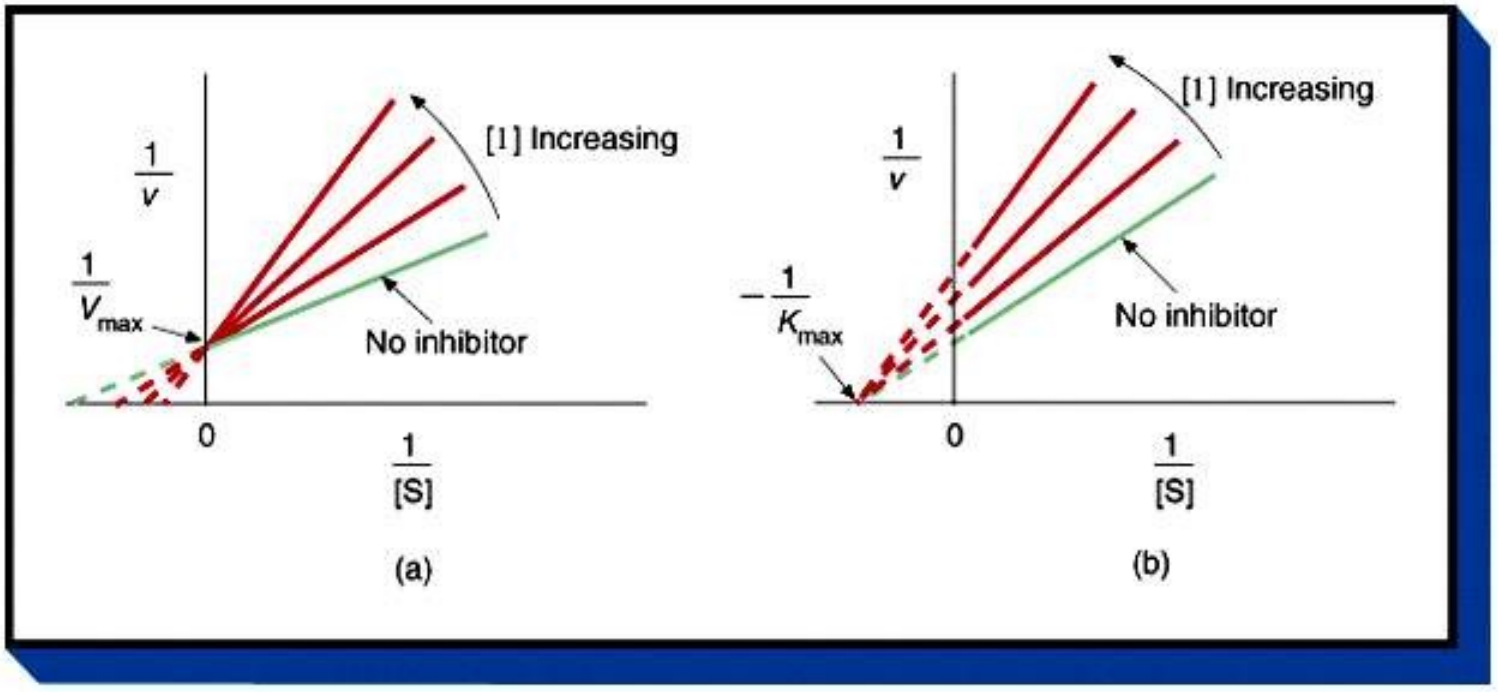
Noncompetitive inhibition



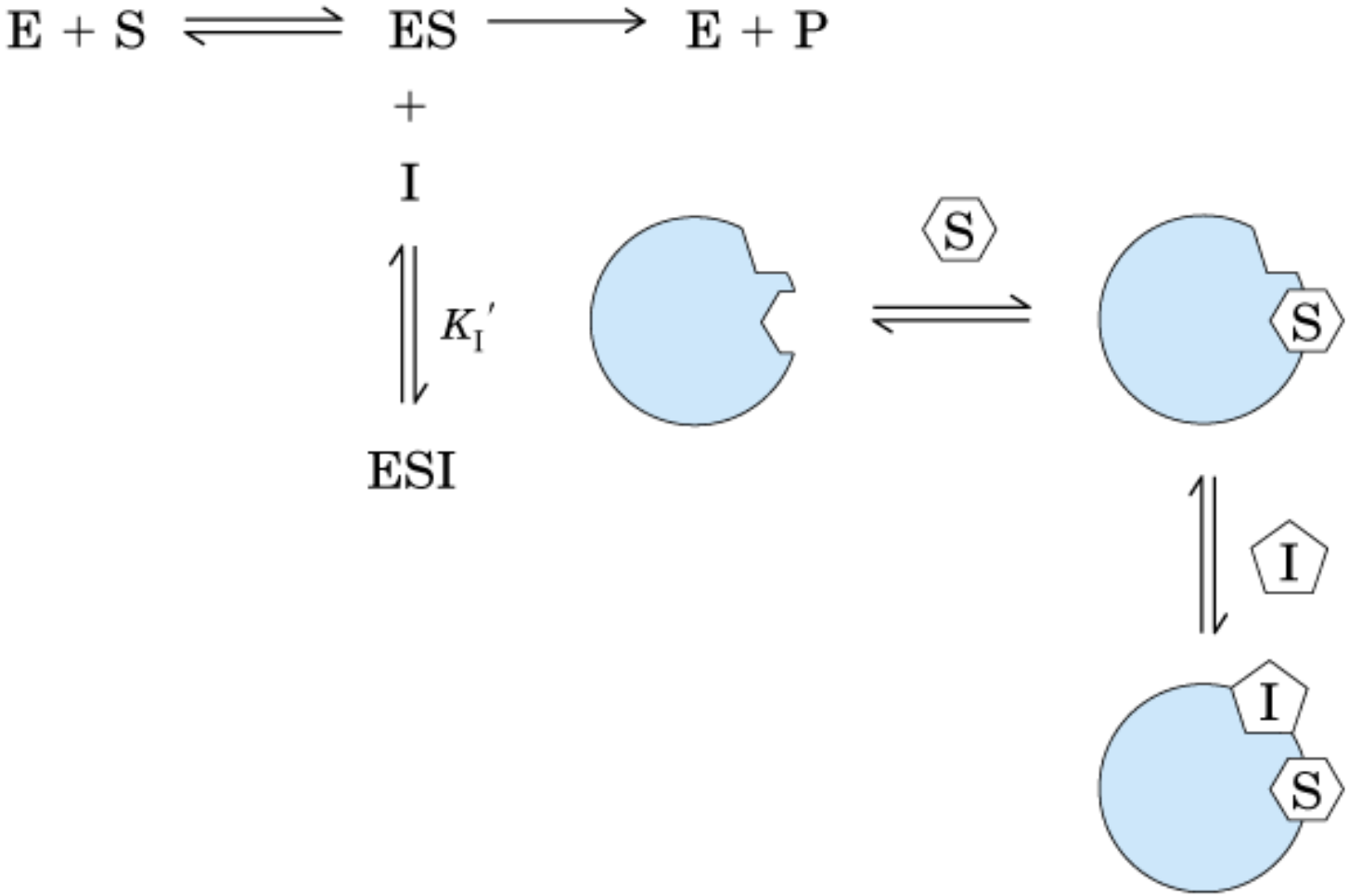


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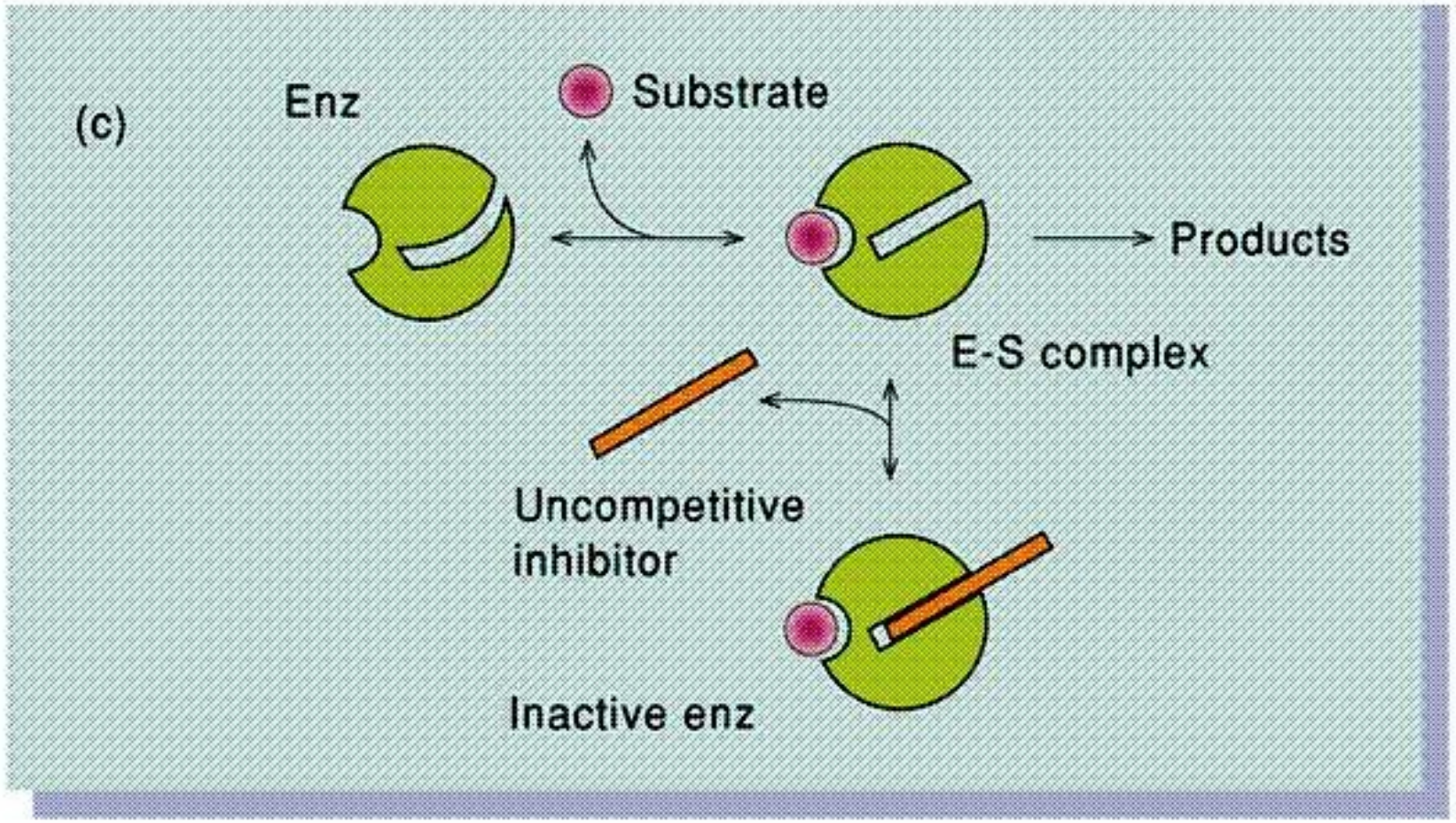
Kinetic analysis of enzyme inhibition. M Figure 6.13



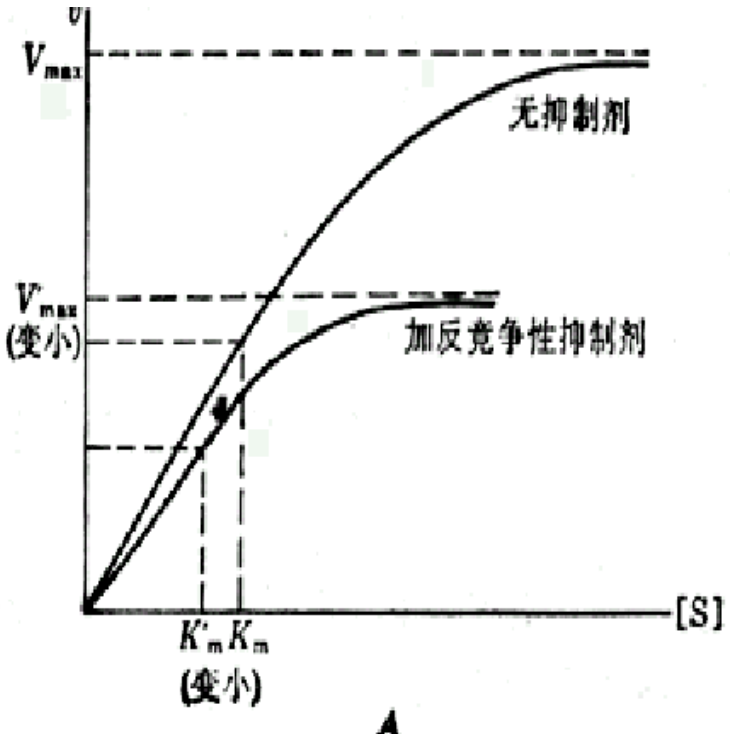
2.10 In uncompetitive inhibition, the inhibitor binds only to the ES complex (unable to bind to free enzyme).



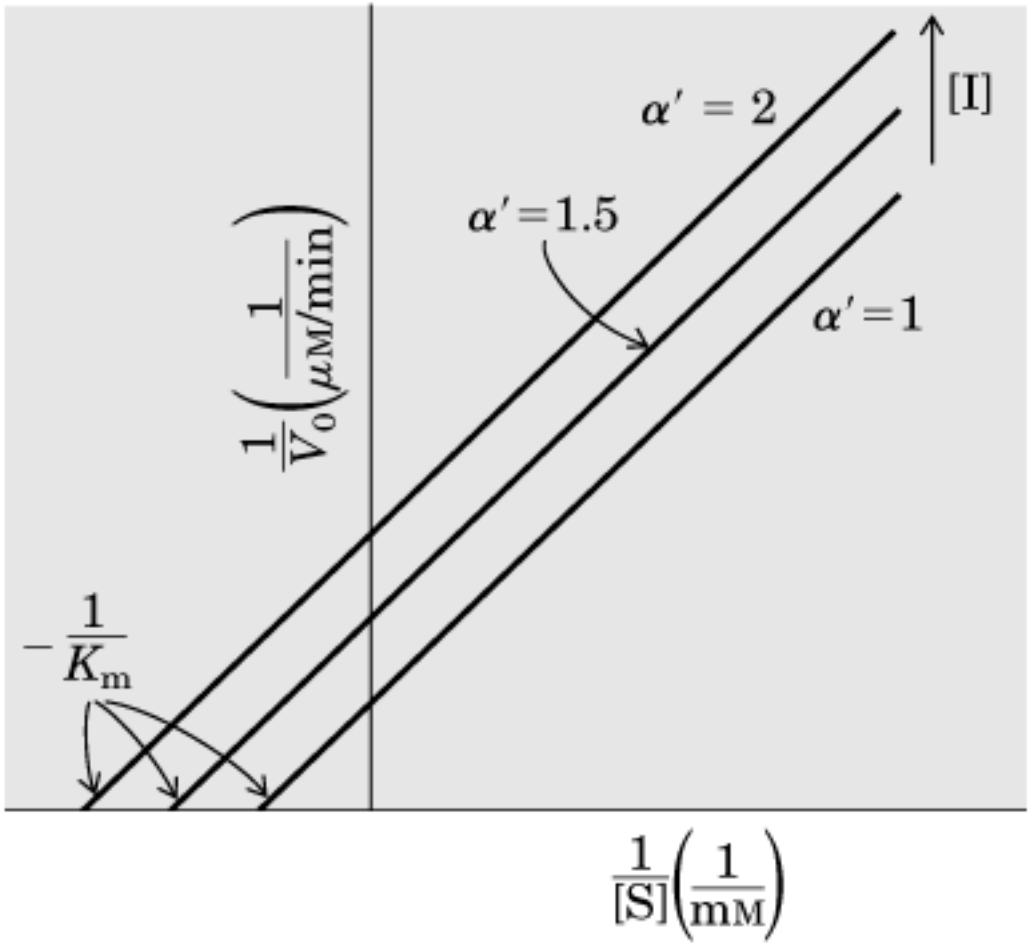
(b) Uncompetitive inhibition


(c)
Enz
Substrate
Products
E-S complex
Uncompetitive
inhibitor
Inactive enz

Uncompetitive inhibition



$$\frac{1}{V_0} = \left(\frac{K_m}{V_{max}}\right)\frac{1}{[S]} + \frac{\alpha'}{V_{max}}$$



Km Decreased； Vmax decreased

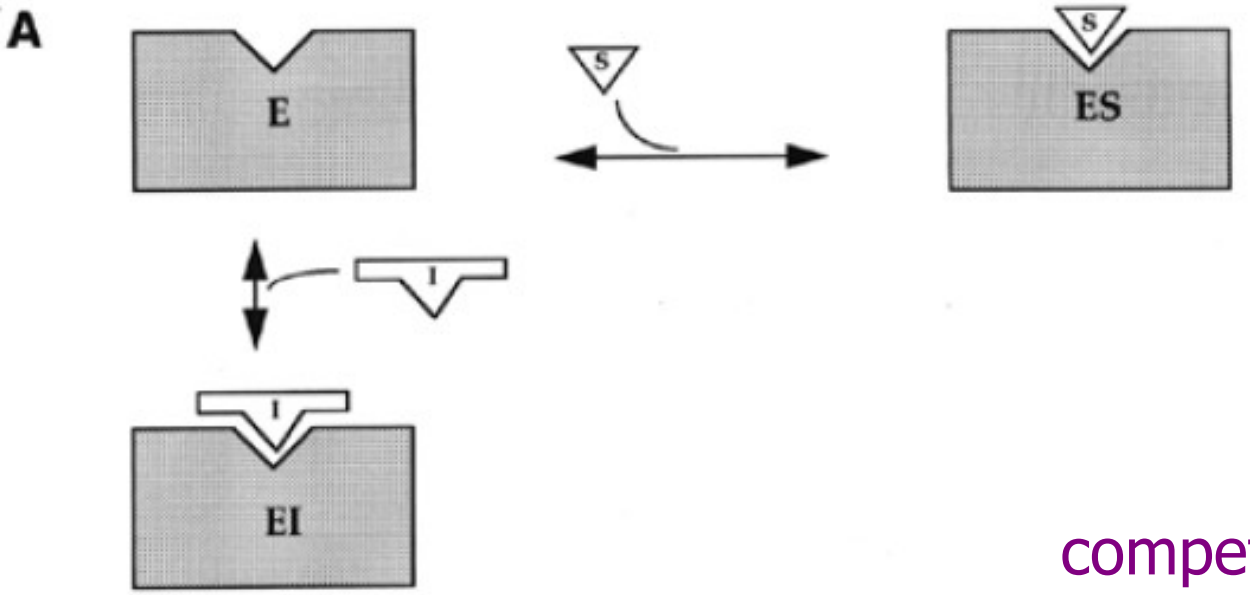
UNCOMPETITIVE INHIBITION

Normalization of the rate equation by total enzyme concentration ($v/[\mathrm{E}_T]$) and rearrangement results in the following expression for the velocity of an enzymatic reaction in the presence of an uncompetitive inhibitor:

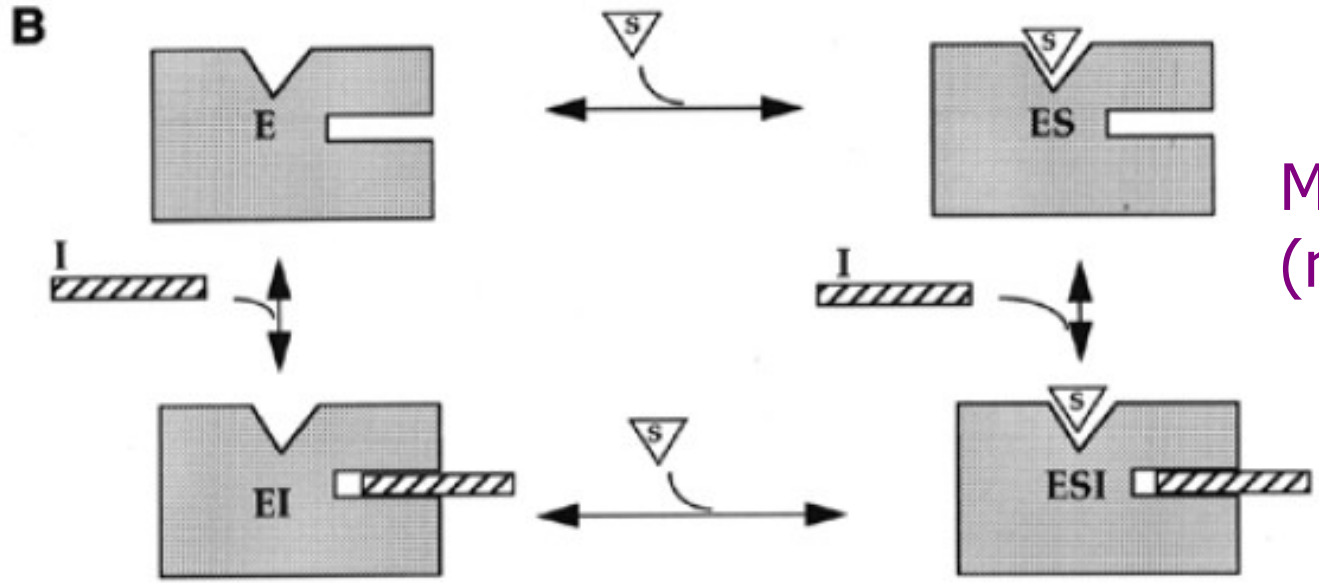
$$v = \frac{V^*_{\max}(\mathrm{S})}{K^*_s + (\mathrm{S})} = \frac{(V_{\max}/\alpha)[\mathrm{S}]}{(K_s/\alpha) + [\mathrm{S}]}$$

where $V^*_{\max}$ and K^*_s correspond, respectively, to the apparent enzyme maximum velocity and apparent enzyme–substrate dissociation constant in the presence of an inhibitor. In the case of uncompetitive inhibition, $V^*_{\max} = V_{\max}/\alpha$ and $K^*_s = K_s/\alpha$, where

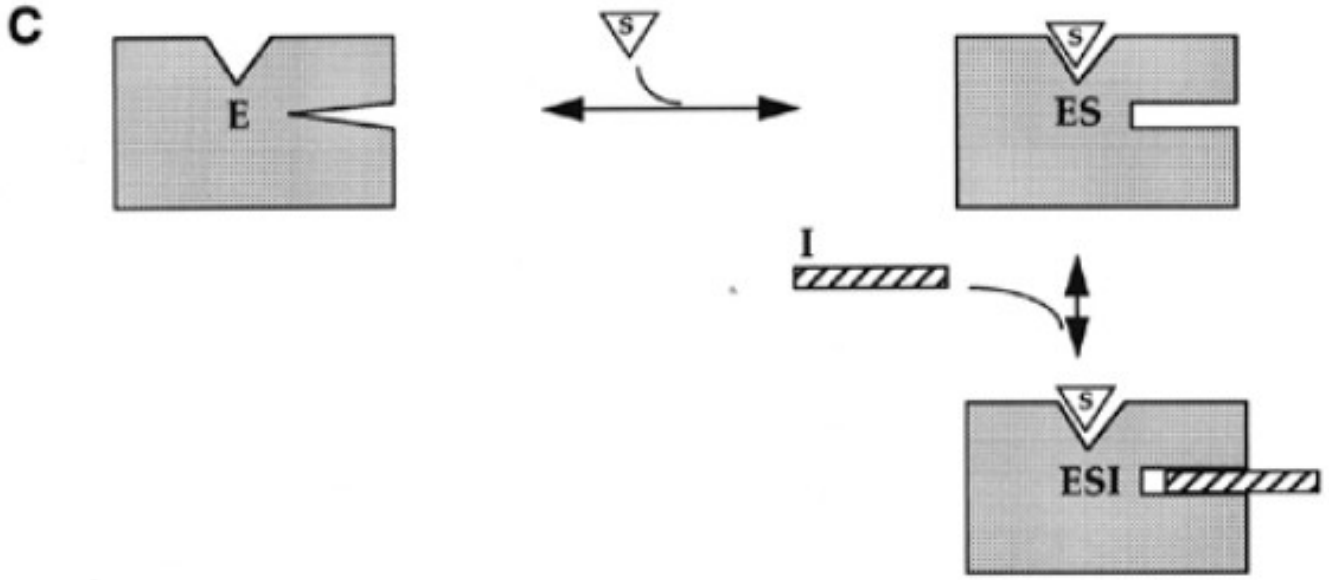
$$\alpha = 1 + \frac{[\mathrm{I}]}{K_i}$$



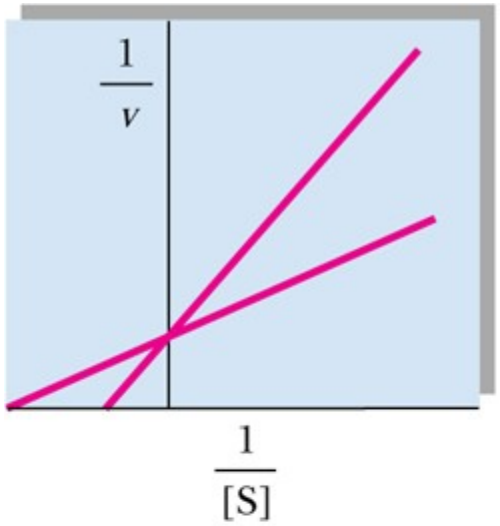
competitive

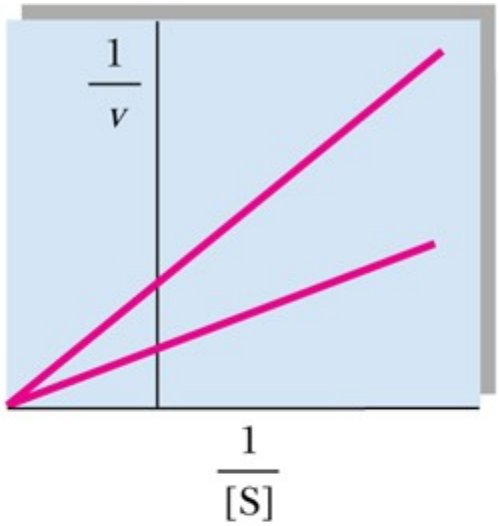


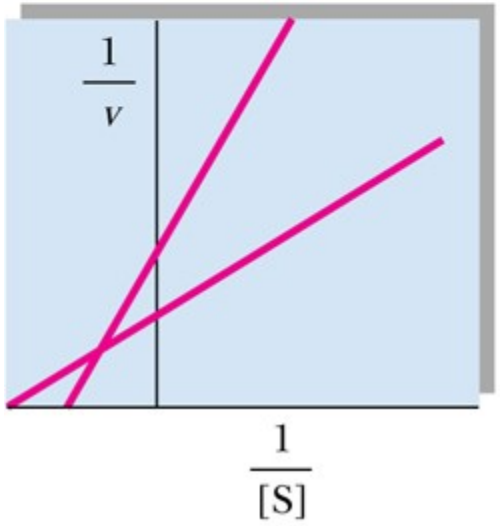
Mixed
(noncompetitive)

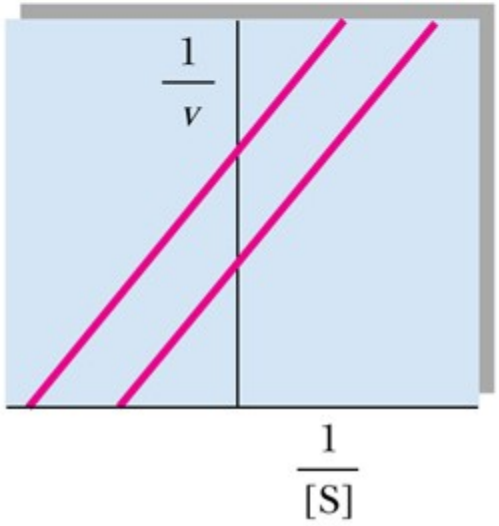


uncompetitive


1/v
1/[S]


1/v
1/[S]


1/v
1/[S]


1/v
1/[S]

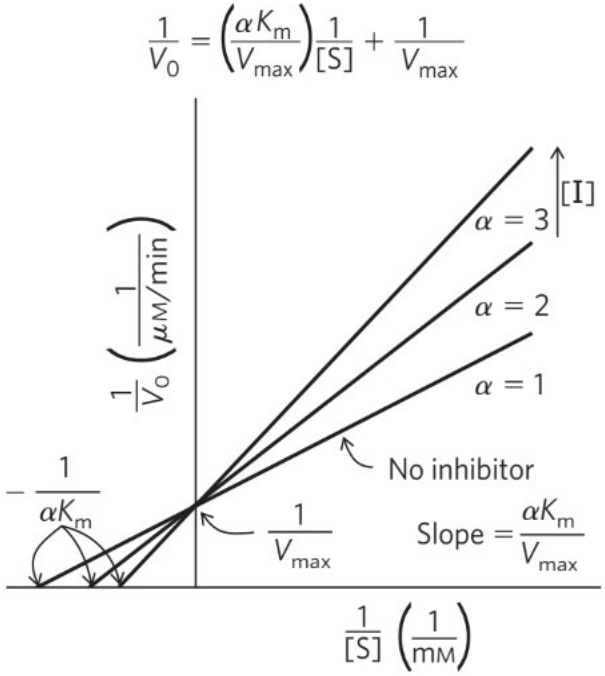


FIGURE 1 Competitive inhibition.

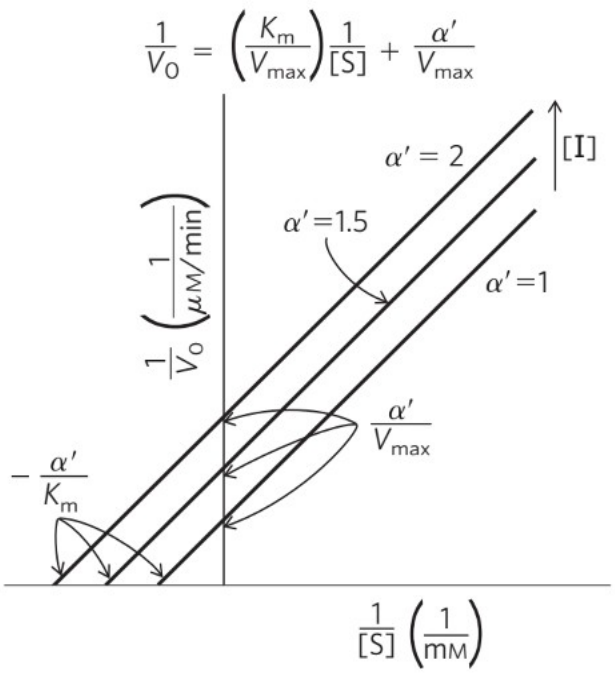


FIGURE 2 Uncompetitive inhibition.

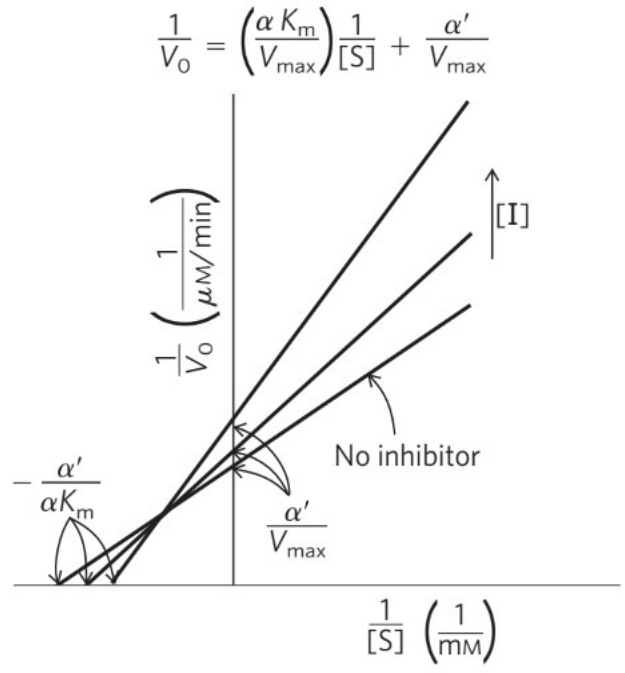


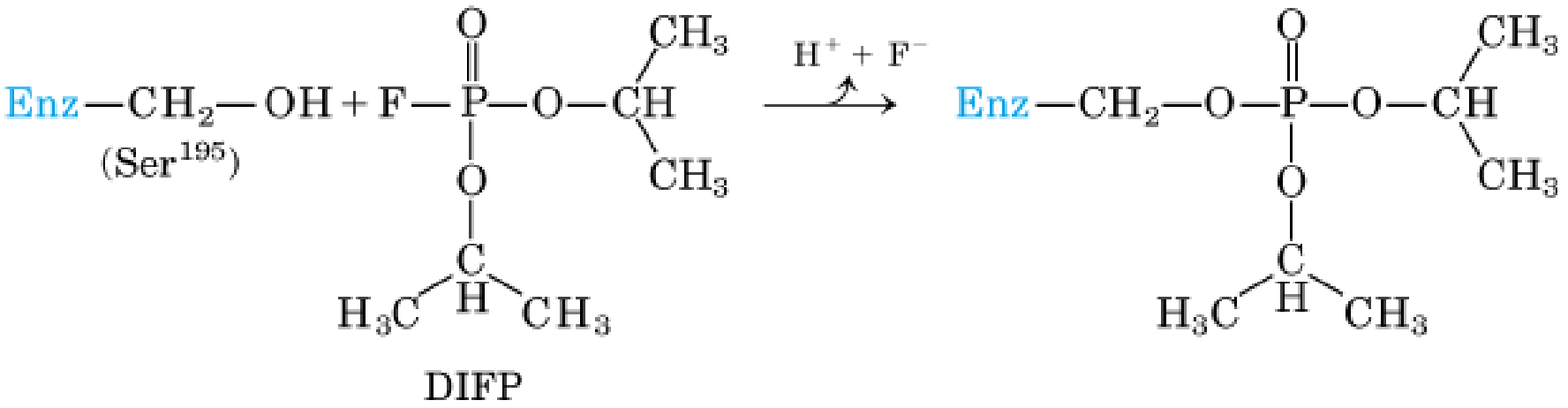
FIGURE 3 Mixed inhibition.

3 Irreversible inhibitors bind very tightly (covalently or noncovalently) to the enzymes.

3.1 Many irreversible inhibitors modify critical catalytic residues covalently, thus inactivating the enzymes.

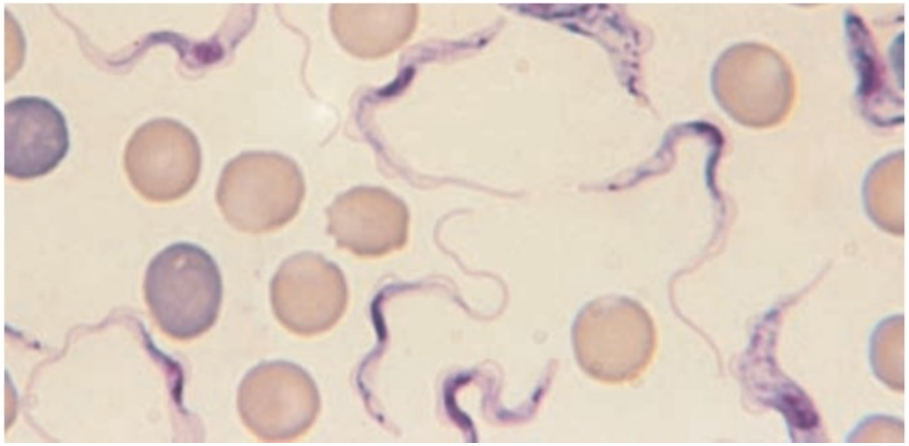
3.2 DiIsopropylPhosphoFluoridate (DIPF, one component of the toxic nerve gases) reacts with a critical Ser residue on acetylcholineesterase.

3.3 Critical catalytic residues in the active site can sometimes be identified using irreversible inhibitors (DIPF on chymotrypsin).



Structure and mechanism of action of diisopropylphosphofluoridate (DIPF) .

Αφριχαν σλεεπινγ σιχκνεσσ



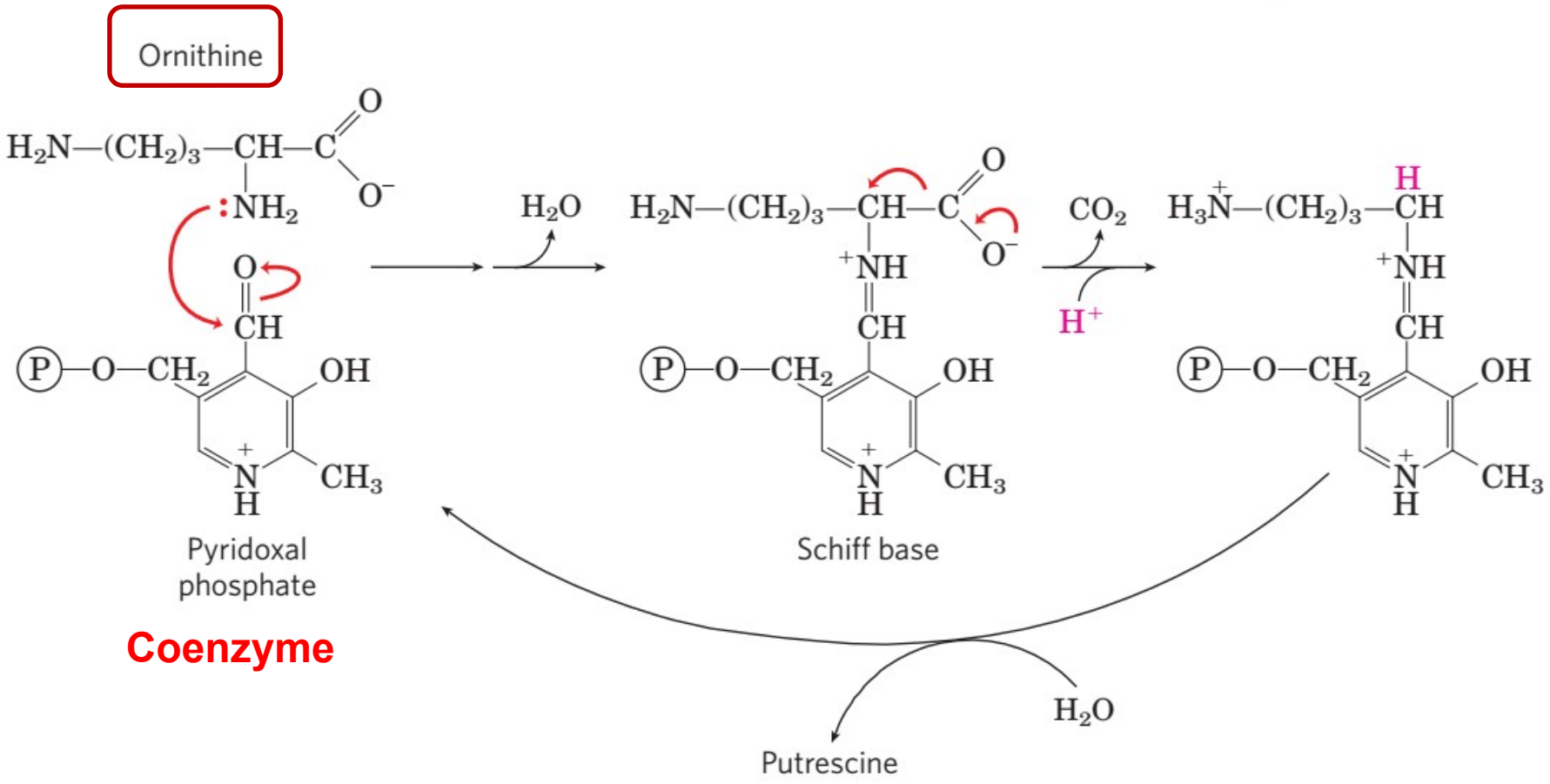


- 百度图片

ΦΙΓΥΡΕ 1 （锥虫） ***Τρψπανοσομα*** *βρυχει ρηοδεσιενσε*, ονε οφ σεϖεραλ τρψπανοσομεσ κνοων το χαυσε Αφριχαν σλεεπινγ σιχκνεσσ.

– *Λεηνινγερ Πριναχιπλεσ οφ Βιοχηεμιστρψ* 6τη Εδιτιον

The polyamines spermine and spermidine

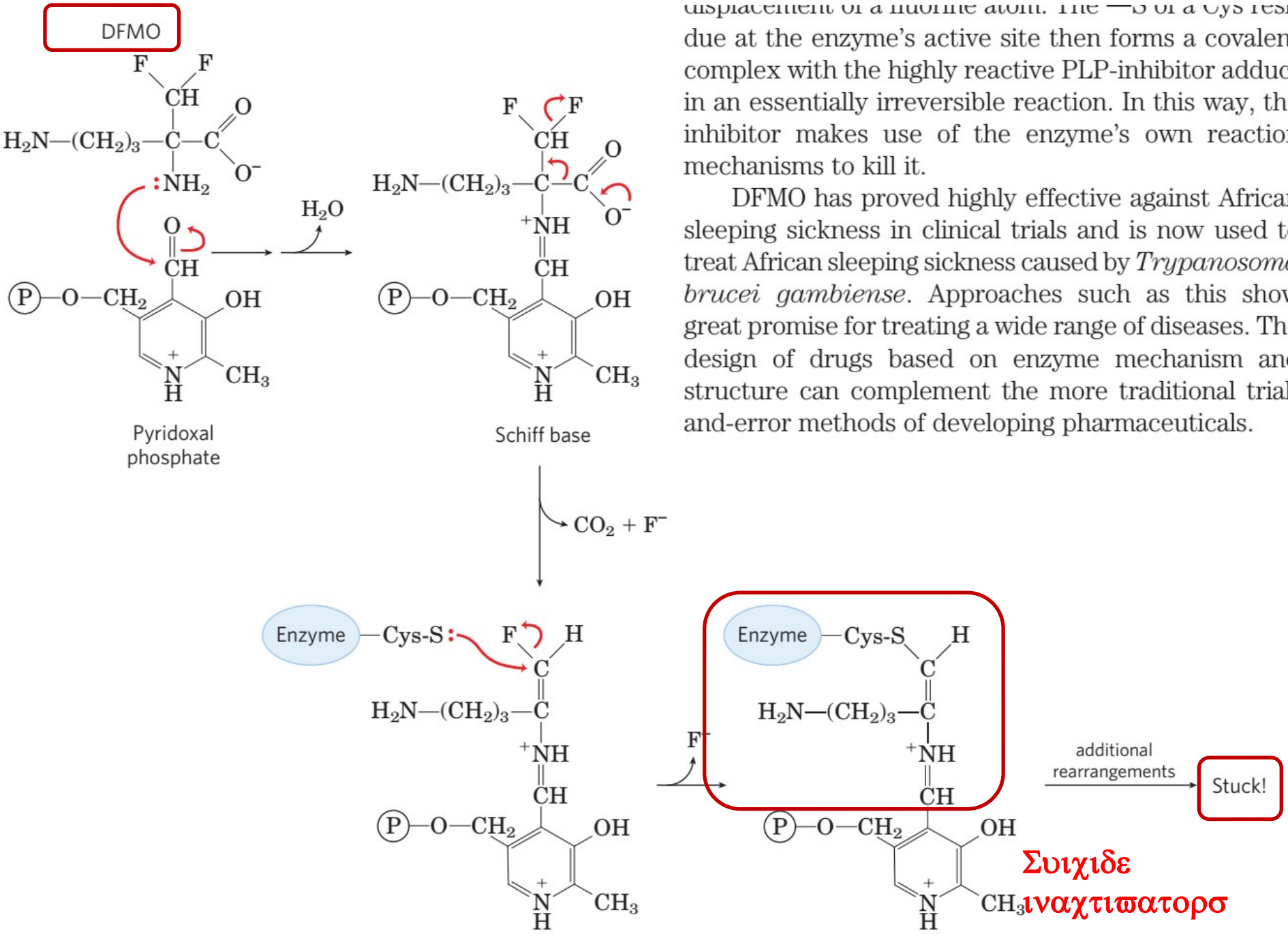


Coenzyme

ornithine decarboxylase

displacement of a fluorine atom. The —S of a Cys resi-due at the enzyme's active site then forms a covalent complex with the highly reactive PLP-inhibitor adduct in an essentially irreversible reaction. In this way, the inhibitor makes use of the enzyme's own reaction mechanisms to kill it.

DFMO has proved highly effective against African sleeping sickness in clinical trials and is now used to treat African sleeping sickness caused by *Trypanosoma brucei gambiense*. Approaches such as this show great promise for treating a wide range of diseases. The design of drugs based on enzyme mechanism and structure can complement the more traditional trial-and-error methods of developing pharmaceuticals.



C5 Regulation of Enzyme activity

The enzymatic activity of some enzymes are precisely regulated in living organisms to meet physiological requirements.

- **Feedback regulation**
- **Allosteric enzyme**
- **Reversible covalent modification**
- **Proteolytic activation**
- **Regulation of enzyme synthesis and breakdown**

1. Feedback inhibition

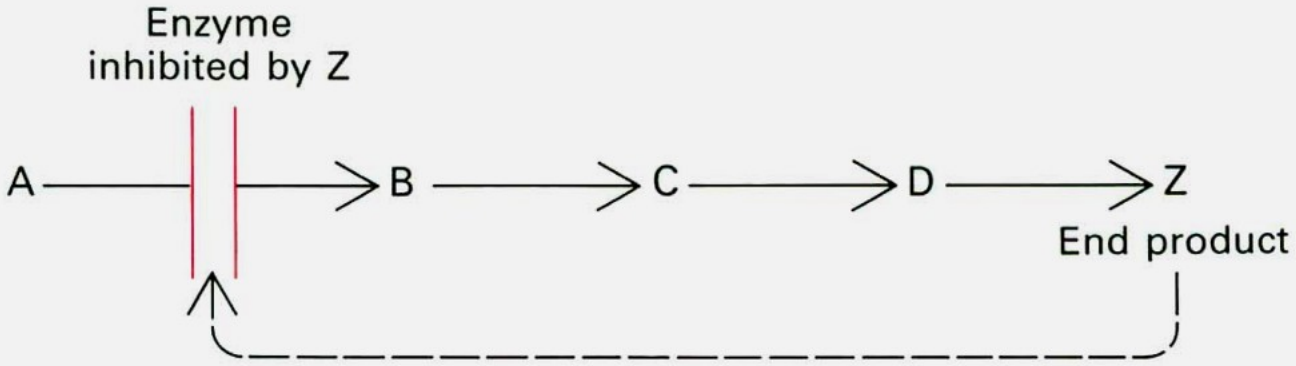
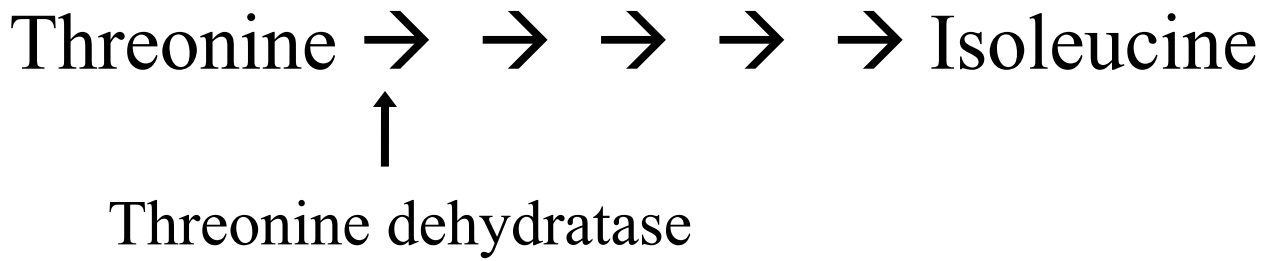


Figure 8-2
Feedback inhibition of the first enzyme in a pathway by reversible binding of the final product.

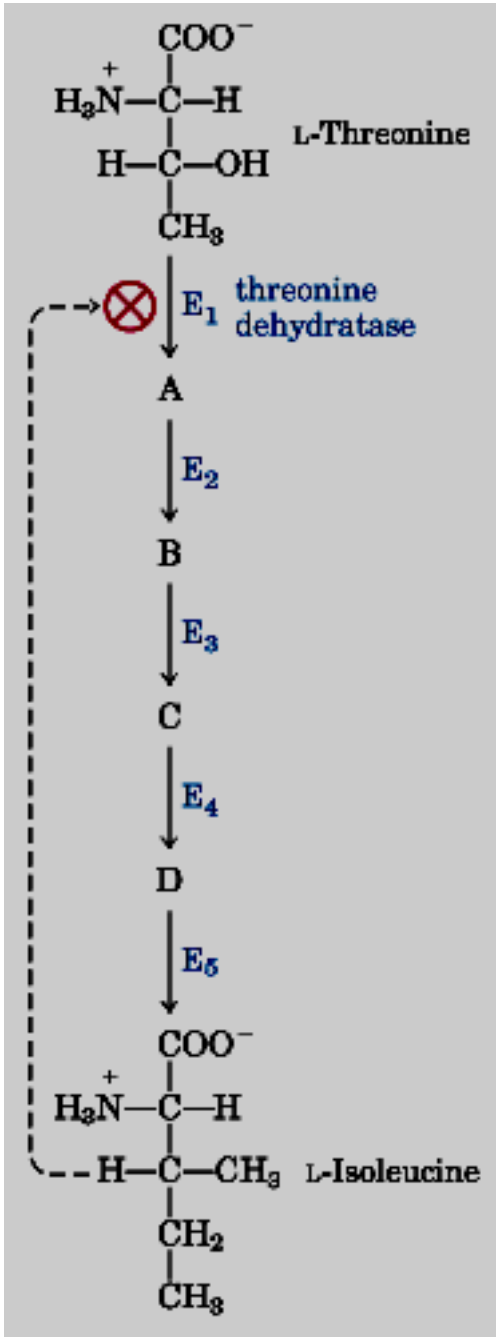
Committed step

In biological systems the rates of many enzymes are altered by the presence of effectors (inhibitors or activators).

Building up of a pathway's end product ultimately slows the entire pathway. This is called feedback inhibition and often takes place at the committed step in the pathway. The committed step is the first step to produce an intermediate which is unique to the pathway in question, and therefore normally commits the metabolite to further metabolism along that pathway.

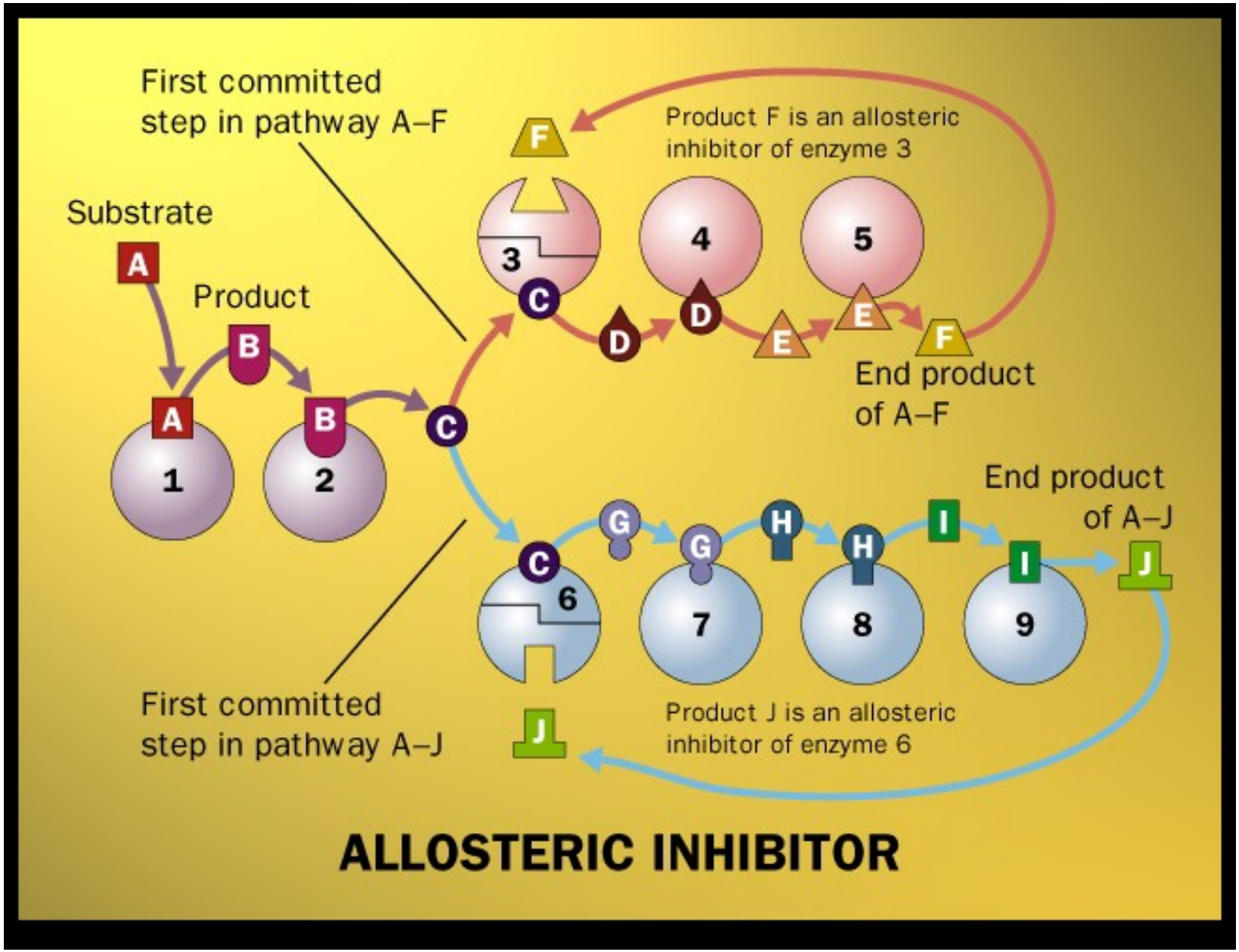


Isoleucine inhibits by binding to the enzyme at a regulatory site, which is distinct from the catalytic site. This inhibition is mediated by an *allosteric interaction*, which is rapidly reversible. When the level of isoleucine drops sufficiently, threonine deaminase becomes active again, and consequently isoleucine is synthesized once more.


L-Threonine
E1
threonine dehydratase
A
E2
B
E3
C
E4
D
E5
L-Isoleucine

As many metabolic pathways are branched, feedback inhibition must allow the synthesis of one product of a branched pathway to proceed even when another is present in excess.

Sequential feedback inhibition



协同反馈抑制

concerted feed back inhibition

协同反馈抑制：在分支代谢途径中，几种末端产物同时都过量，才对途径中的第一个酶具有抑制作用。若某一末端产物单独过量则对途径中的第一个酶无抑制作用

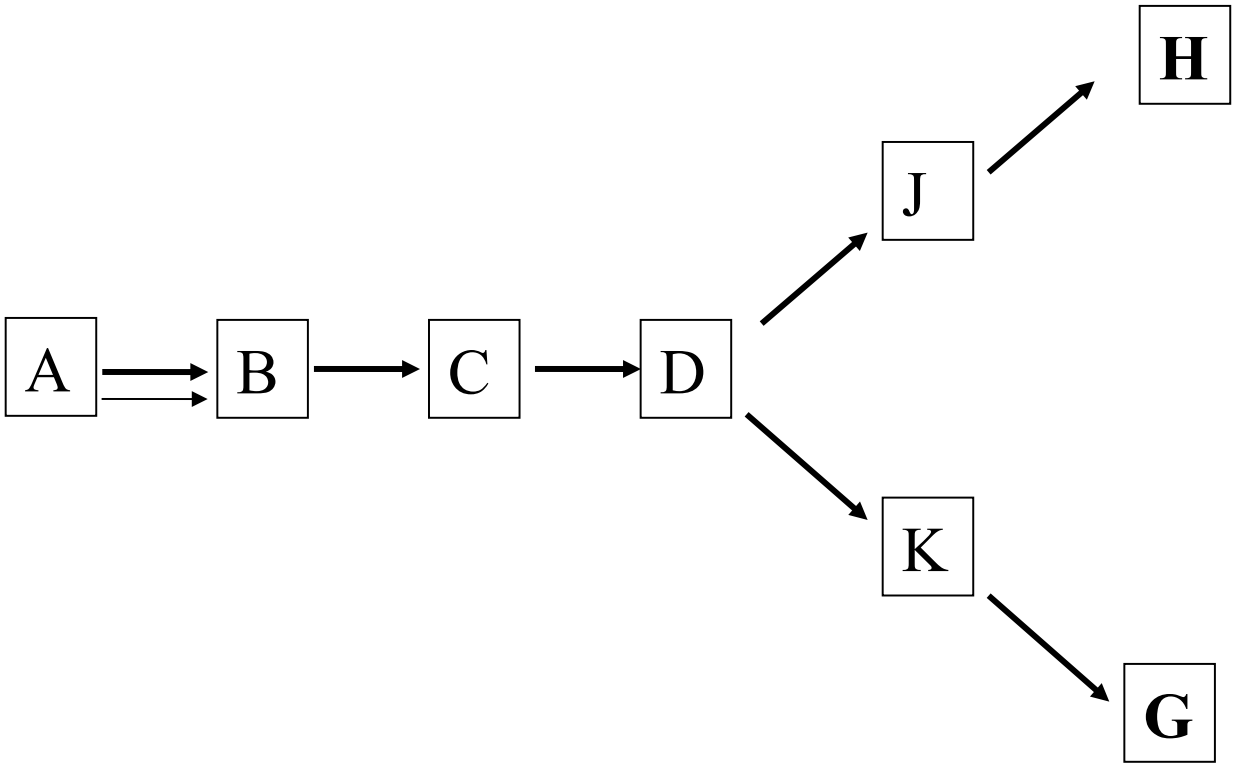
积累 (累积）反馈抑制

cumulative feed back inhibition

积累反馈抑制在分支代谢途径中，任何 1 种末端产物过量时都能对共同途径中的第 1 个酶起抑制作用，而且各种末端产物的抑制作用互不干扰。当各种末端产物同时过量时，它们的抑制作用是累加的。

同工酶反馈抑制

Isozyme feed back inhibition


A
B
C
D
J
H
K
G

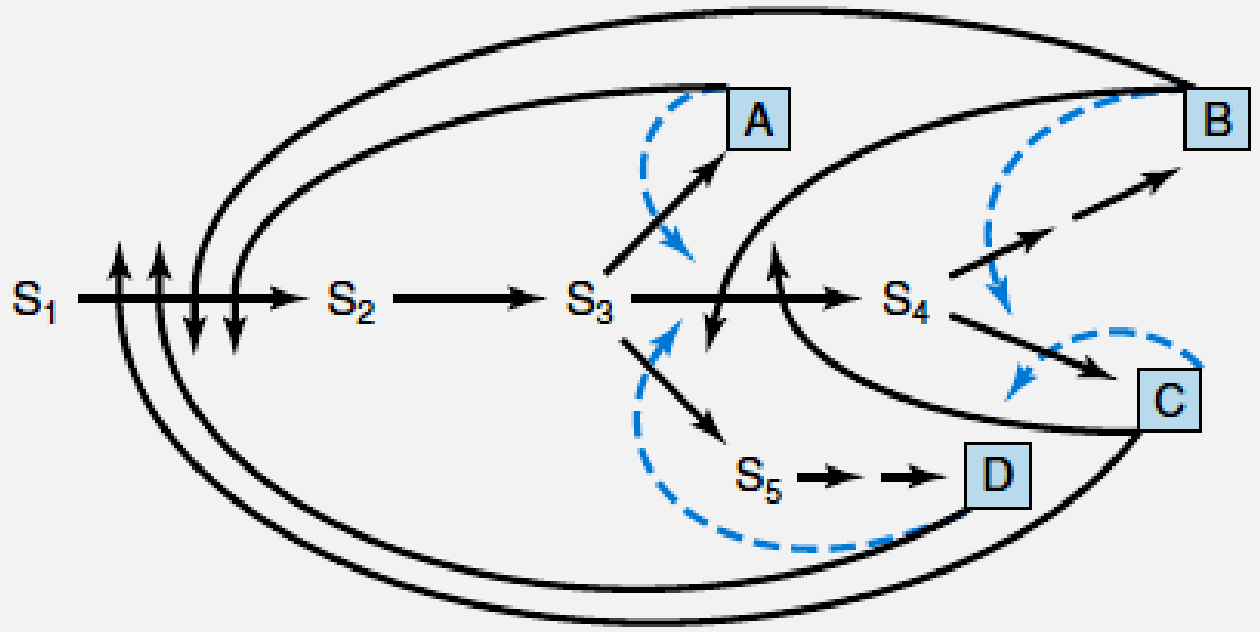


Figure 9–5. Multiple feedback inhibition in a branched biosynthetic pathway. Superimposed on simple feedback loops (dashed, curved arrows) are multiple feedback loops (solid, curved arrows) that regulate enzymes common to biosynthesis of several end products.

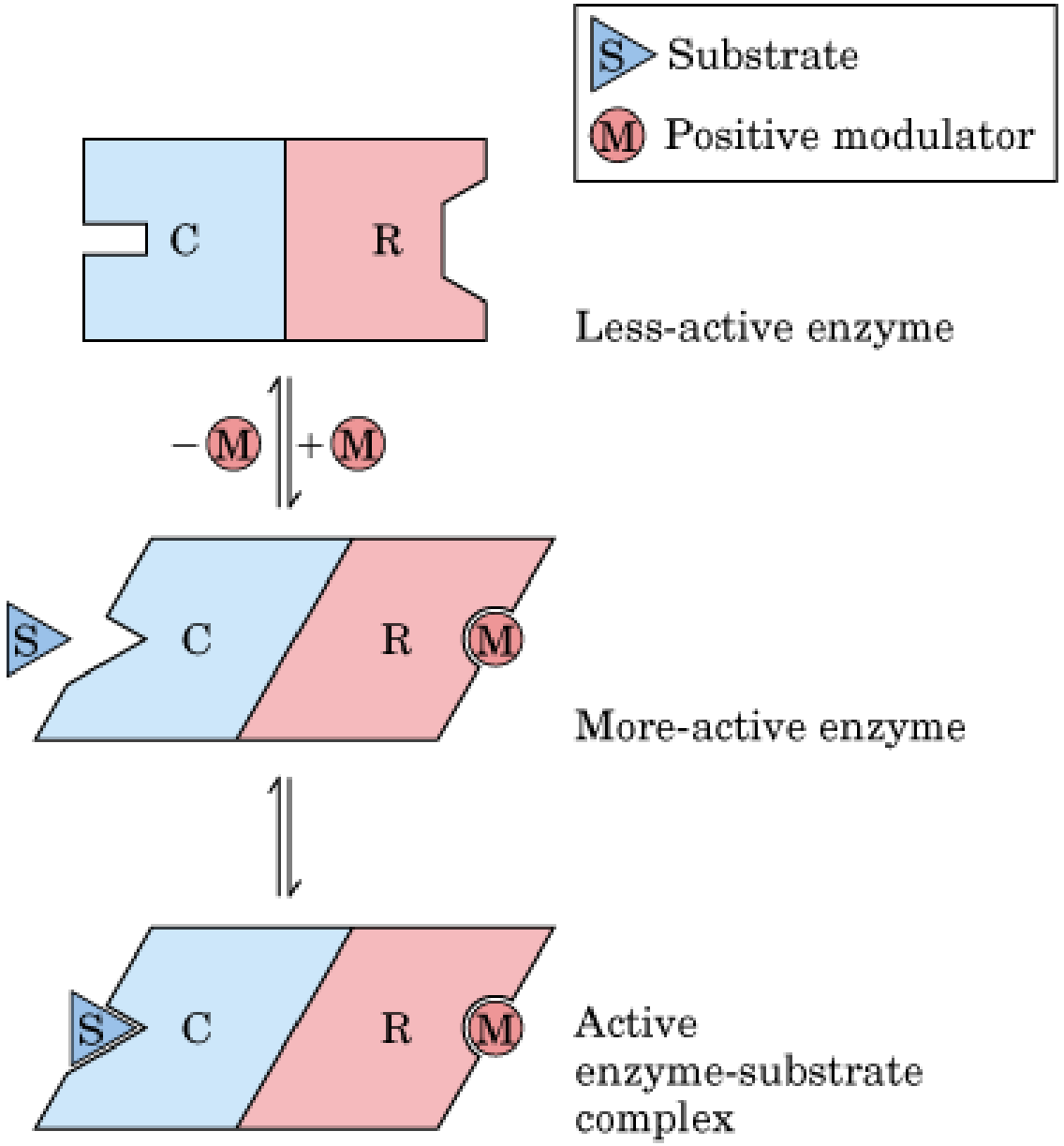
2. Allosteric enzymes (similar to hemoglobin) are regulated by reversible, noncovalent binding of modulators (often being metabolites).

Allosteric effect 见 Section B 3.4.1

2.1The enzyme catalyzing the first step of a synthetic pathway is often an allosteric enzyme.

2.2 (2.5) Allosteric enzymes are often multi-subunit proteins, with one or more active sites on each subunit.

In many allosteric enzymes the substrate binding site and the modulator(M) binding sites are on different subunits, the catalytic (C) and regulator (R) subunits, respectively. Binding of the positive (stimulatory) modulator to its specific site on the regulatory subunits commuicated to the catalytic subunit through a conformational change. This change renders the catalytic subunit active and capable of binding the substrate (S) with higher affinity.


S Substrate
M Positive modulator
C
R
Less-active enzyme
− M
+ M
S
C
R
M
More-active enzyme
S
C
R
M
Active
enzyme-substrate
complex

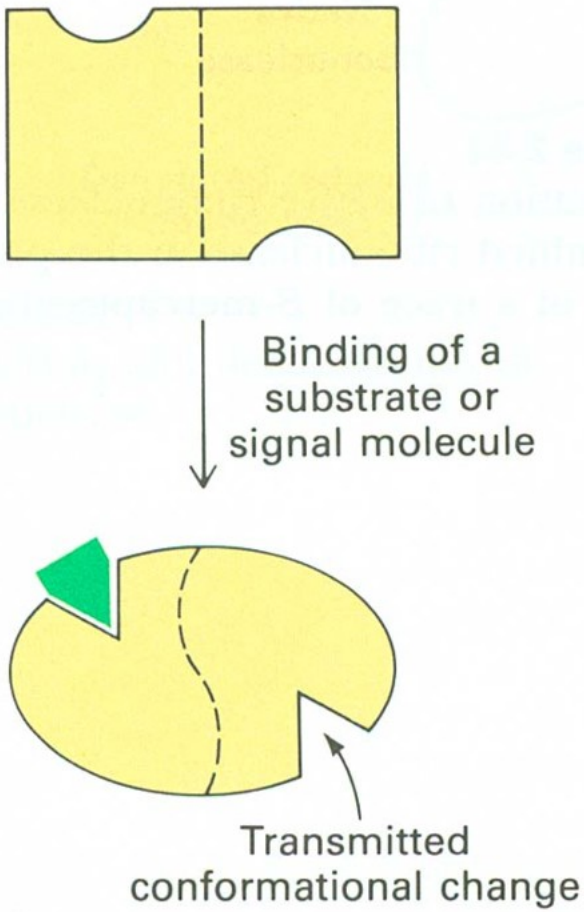
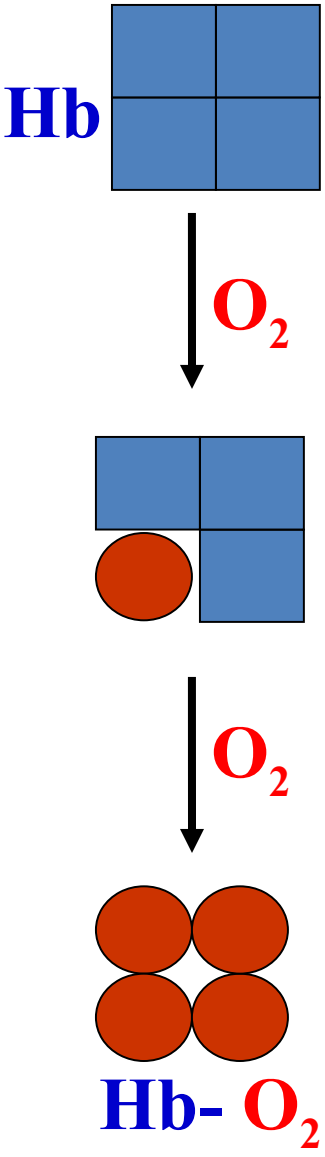
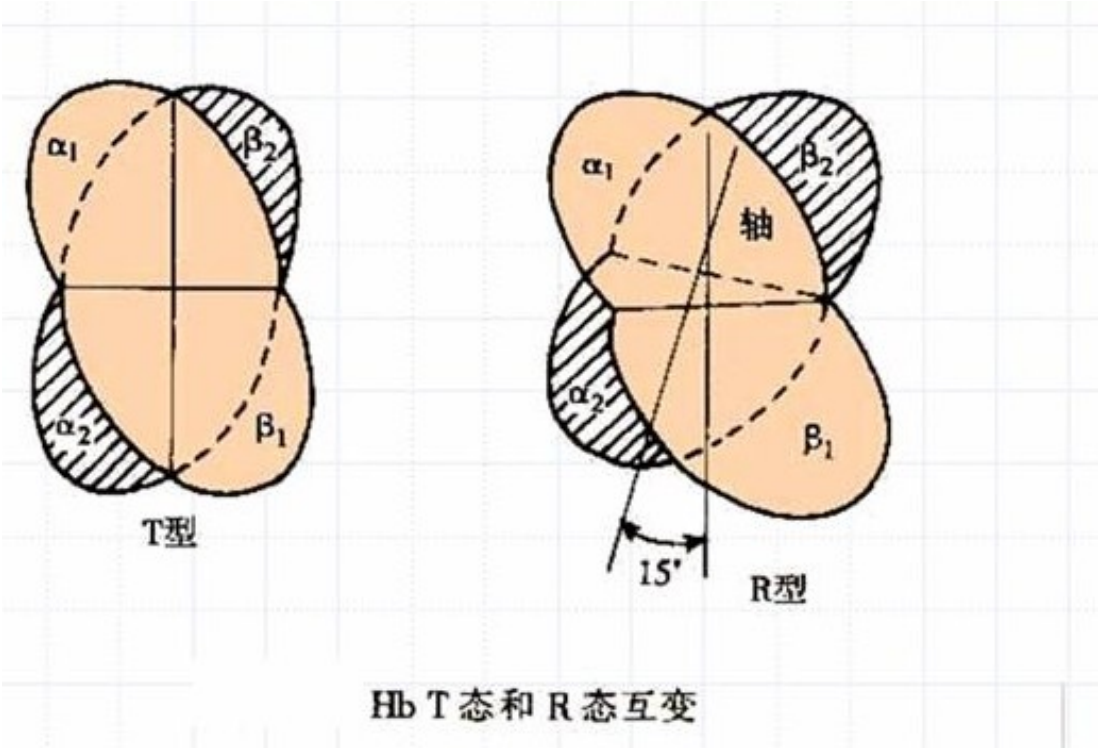


Figure 2-57
Schematic diagram of an allosteric interaction in a protein. The binding of a small molecule or macromolecule to a site in the protein leads to conformational changes that are propagated to distant sites.

2.3 In allosteric enzymes, the binding of a substrate molecule to one active site affects the binding of substrate molecules to other active sites in the enzyme; the different active sites are said to behave cooperatively in binding and acting on substrate molecules.

血红蛋白输氧功能和构象变化





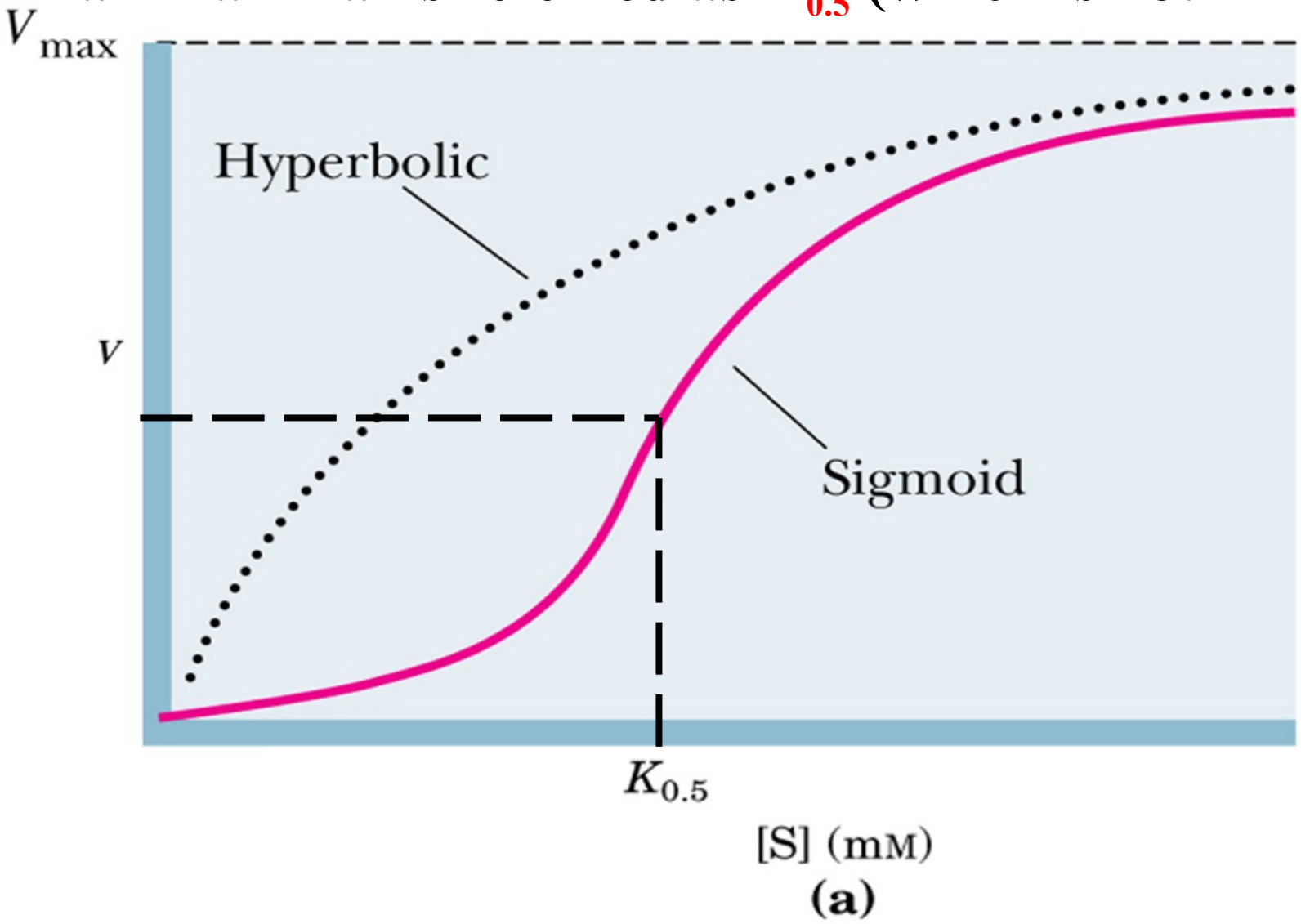
Hb T 态和 R 态互变

to **behave cooperatively** in binding and acting on substrate molecules

2.4(2.7) Allosteric enzymes do not follow the Michaelis-Menten kinetics.

2.4.1 They do exhibit saturation effect, but do not show a hyperbolic curve (usually a sigmoidal curve) when plotting Vo against [S].

2.4.2 The substrate concentration at which Vo is half maximal is referred as $K_{0.5}$ (which is not Km!)

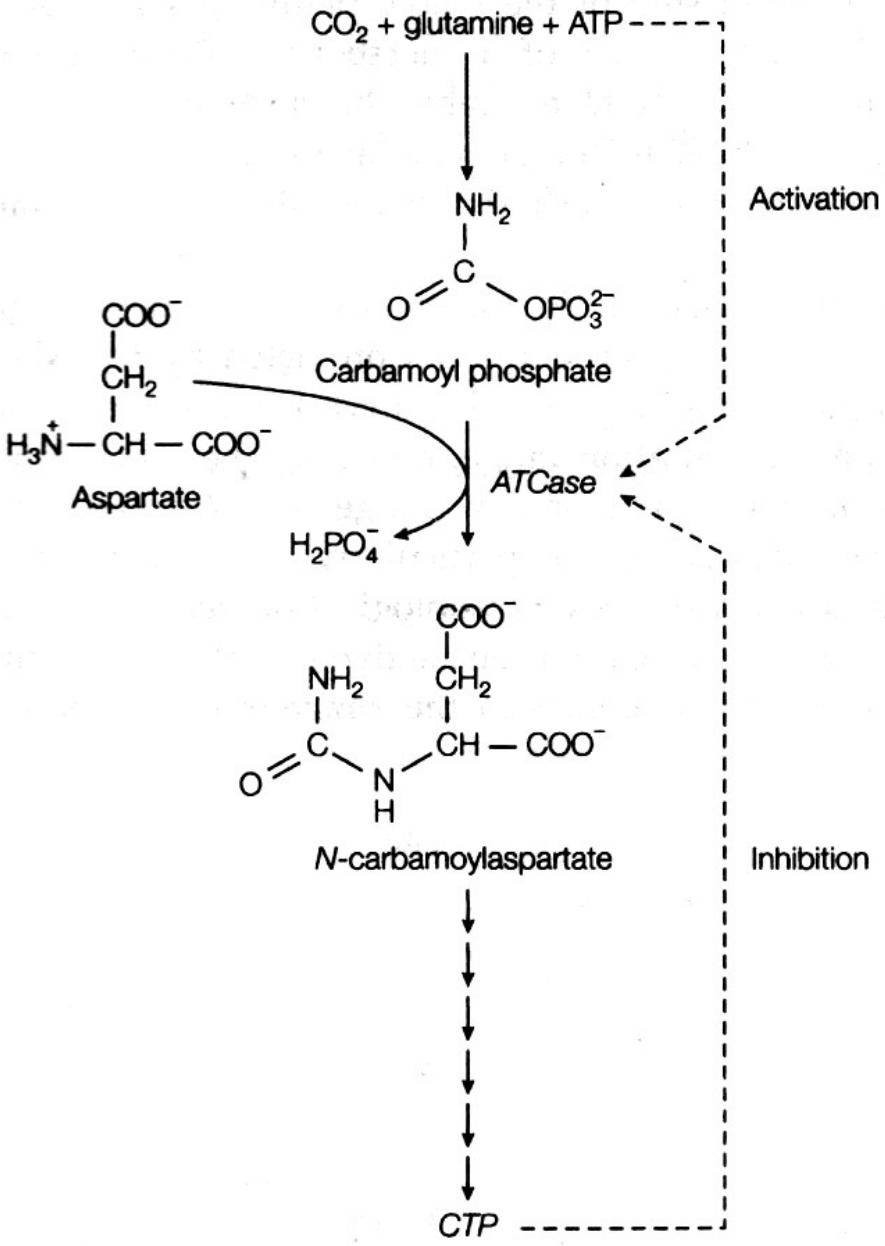


The curve has a steep section in the middle of the substrate concentration range, reflecting the rapid increase in enzyme velocity which occurs cover a narrow range of substrate concentrations.

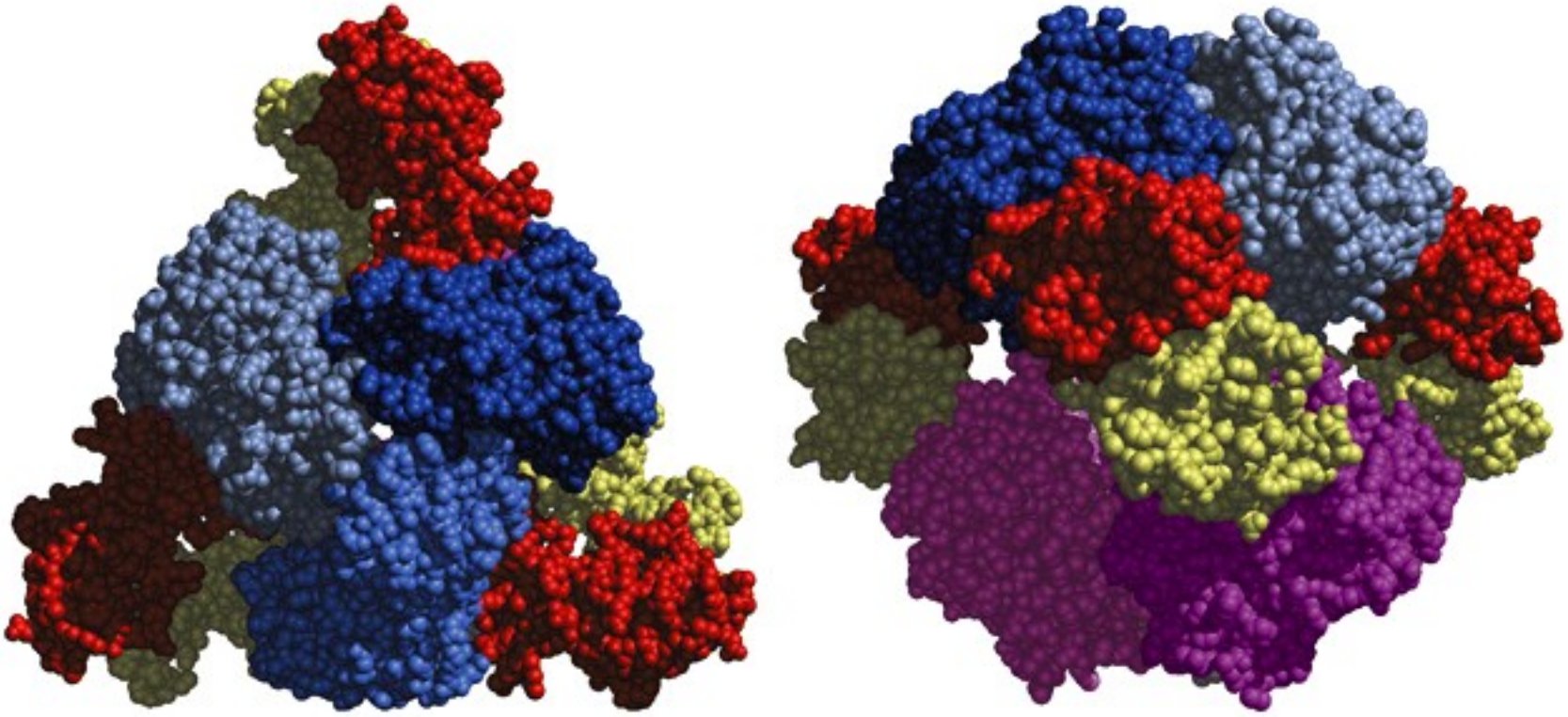
This allows allosteric enzymes to be particularly sensitive to small changes in substrate concentration within the physiological range.

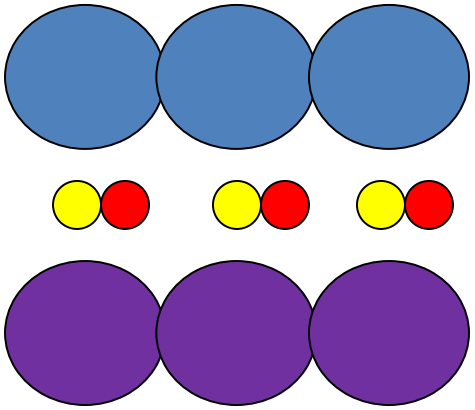
2.5 (2.6) The enzyme activity varies when the concentration of the modulators vary, hence the synthesis pathway is open only when the end product is lacking (closed when the end product is abundant).

For example, threonine dehydratase in the Ile synthesis pathway, and Aspartate TransCarbamoylase (ATCase, the best understood allosteric enzyme) in the CTP synthesis pathway.


CO_2 + glutamine + ATP
Activation
NH_2
C
O
OPO_3^{2-}
Carbamoyl phosphate
COO^-
CH_2
$H_3\overset{+}{N}$ — CH — COO^-
Aspartate
ATCase
$H_2PO_4^-$
COO^-
NH_2
CH_2
C
CH — COO^-
O
N
H
N-carbamoylaspartate
Inhibition
CTP

aspartate transcarbamoylase (ATCase)

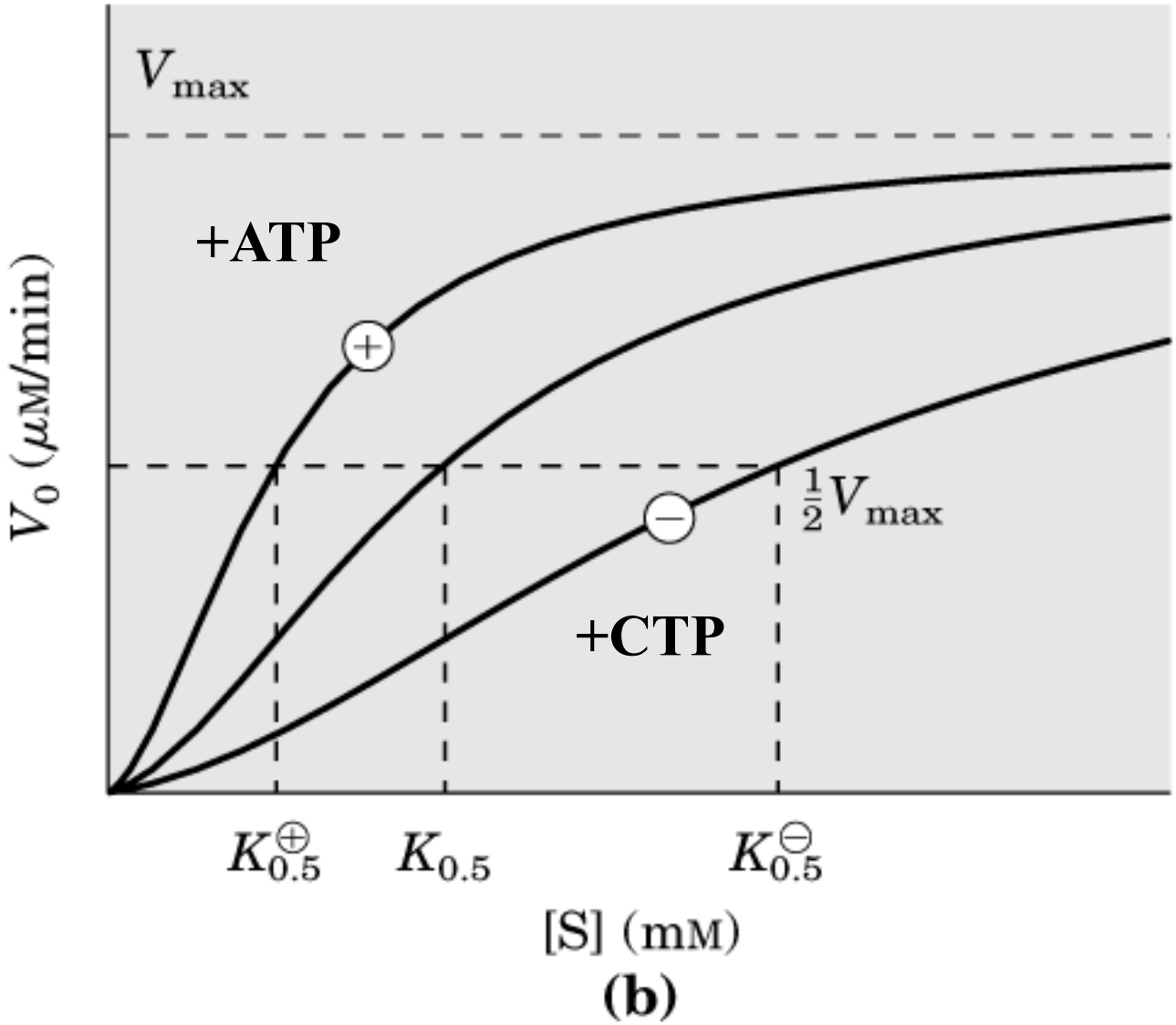




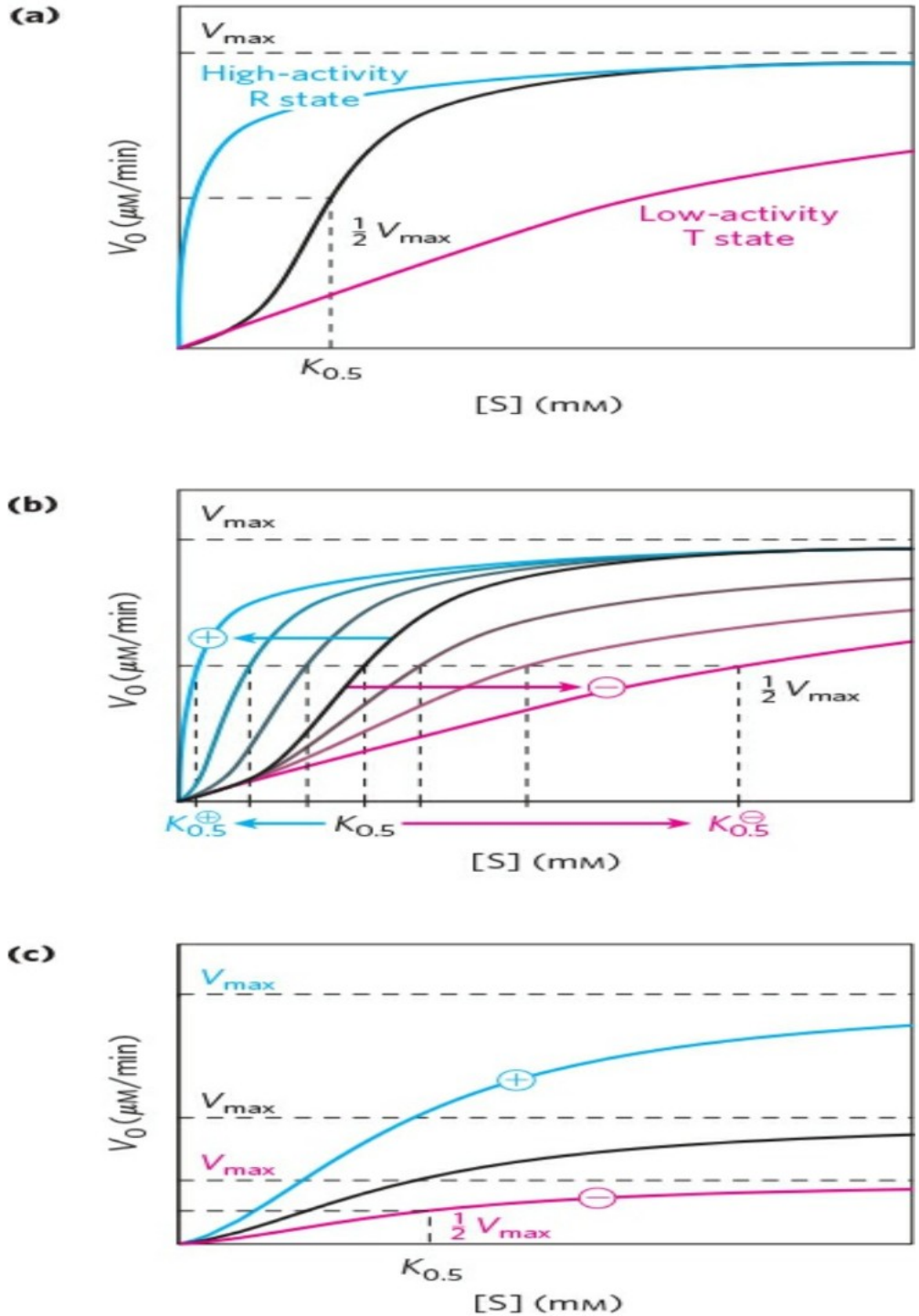
This allosteric regulatory enzyme has two stacked catalytic clusters, each with three catalytic polypeptide chain (in shades of blue and purple), and three regulatory clusters, each with two regulatory polypeptide chains (in red and yellow).

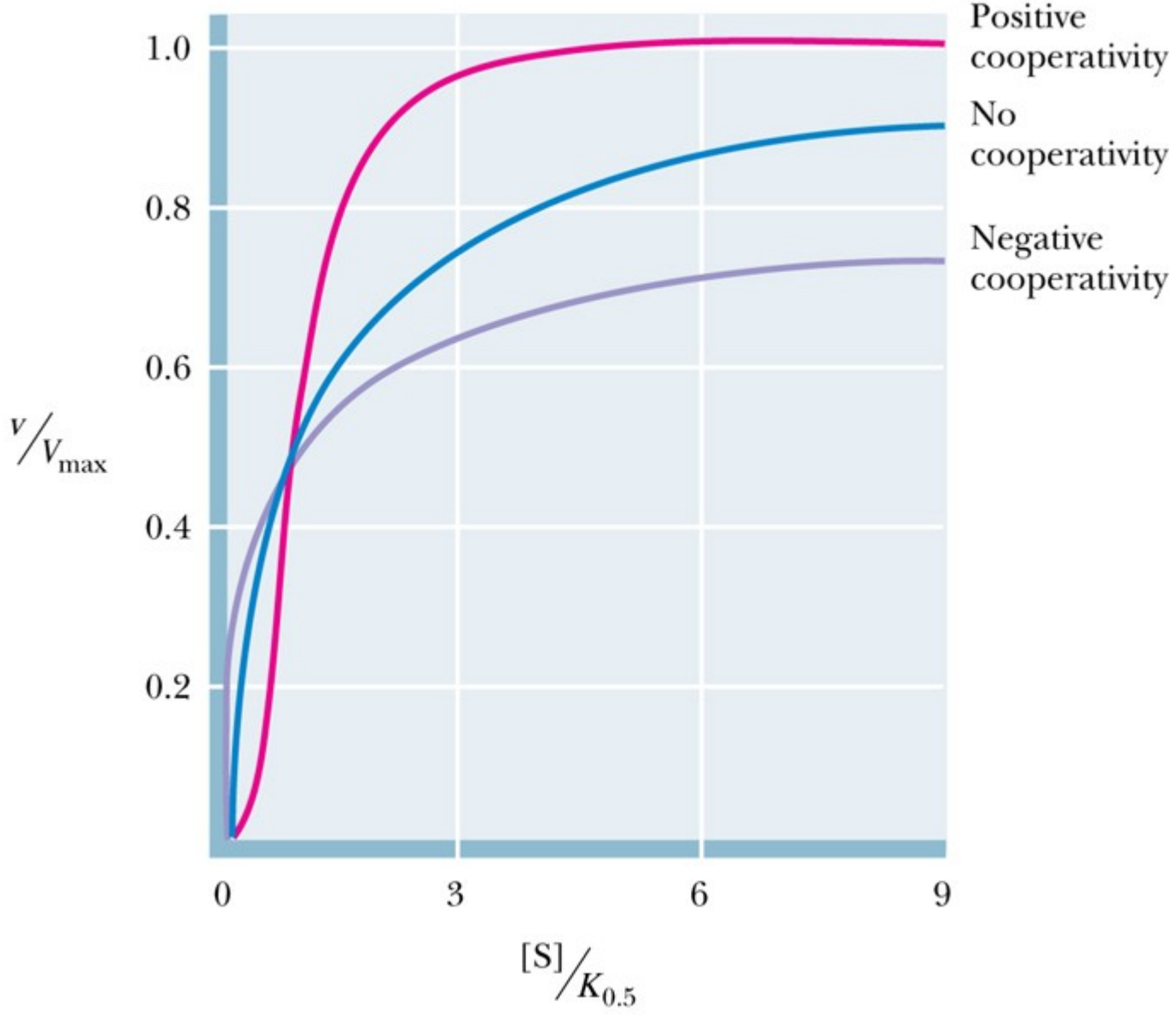
The regulatory clusters form the points of a triangle surrounding the catalytic subunits. Modulator binding produces large changes in enzyme conformation and activity.

aspartate transcarbamoylase (ATCase)



(b)


(a)
V_{max}
High-activity
R state
V_0 (μM/min)
$\frac{1}{2} V_{max}$
Low-activity
T state
$K_{0.5}$
[S] (mM)
(b)
V_{max}
V_0 (μM/min)
$\frac{1}{2} V_{max}$
$K_{0.5}^{\oplus}$
$K_{0.5}$
$K_{0.5}^{\ominus}$
[S] (mM)
(c)
V_{max}
V_{max}
V_{max}
V_0 (μM/min)
$\frac{1}{2} V_{max}$
$K_{0.5}$
[S] (mM)

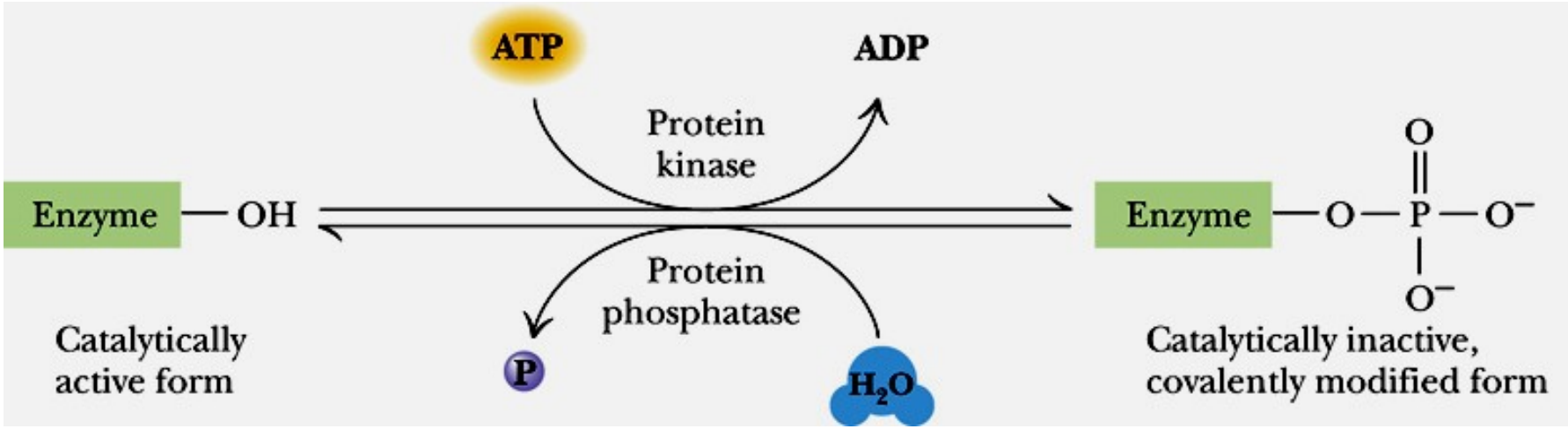

Positive cooperativity
No cooperativity
Negative cooperativity
1.0
0.8
0.6
0.4
0.2
v/V_{max}
0
3
6
9
$[S]/K_{0.5}$

别构酶（allosteric enzymes）的主要特点：

- 寡聚酶
- 别构效应
- S 形动力学曲线
- 酶系统的第一个酶，或代谢分支酶

3. The activity of many enzymes are regulated by reversible covalent modifications.

3.1 Phosphorylation, the most common reversible covalent modification, is a highly effective means of switching the activity of target enzymes.



3.2 Protein kinases （激酶） catalyze the transfer of a phosphate group from an ATP molecule to the side chains of Ser （S）, Thr （T）, or Tyr （Y） residues in proteins.

3.3 Protein phosphatases （磷酸酶） catalyze the hydrolysis of phosphoryl groups attached to proteins, thus reversing the effects of kinases.

3.4 The degree of specificity of protein kinases varies. Some catalyze the phosphorylation of many different target proteins (at sites of conserved sequences, e.g., protein kinase A recognizes a conserved sequence made of Arg-Arg-X-Ser-Z or Arg-Arg-X-Thr-Z). Some phosphorylate a single protein or several closely related ones.

3.5 Phosphorylation is a highly effective means of controlling the activity of proteins for structural, thermodynamic, and kinetic reasons.

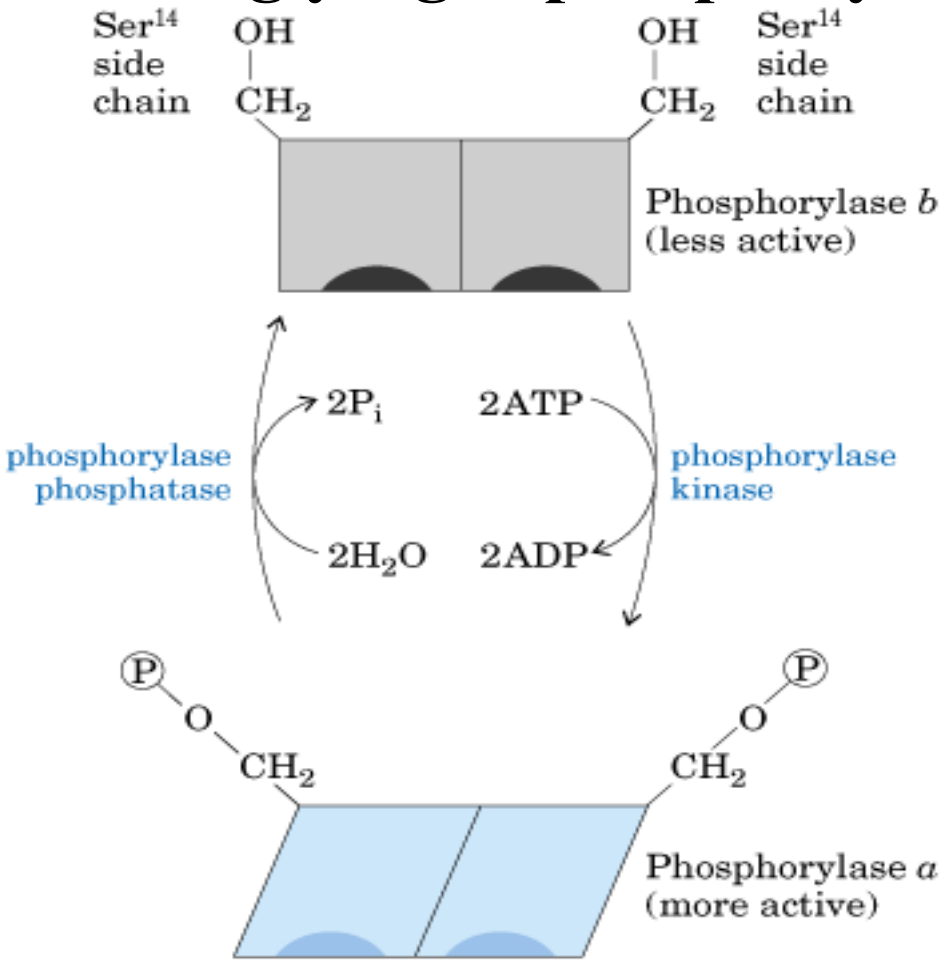
- Glycine (Gly, G)
- Alanine (Ala, A)
- Valine (Val, V)
- Leucine (Leu, L)
- Isoleucine (Ile, I)
- Methionine(Met, M)

- Arginine (Arg, R)
- Lysine (Lys, K)
- Histidine (His, H)
- Aspartate (Asp, D)
- Glutamate (Glu, E)

- Serine (Ser, S)
- Threonine (Thr, T)
- Proline (Pro, P)
- Cysteine (Cys, C)
- Asparagine (Asn, N)
- Glutamine (Gln, Q)

- Phenylalanine (Phe,F)
- Tyrosine (Tyr,Y)
- Tryptophan (Trp, W)

Regulation of glycogen phosphorylase activity



磷酸化酶 / 磷酸解酶
磷酸酶

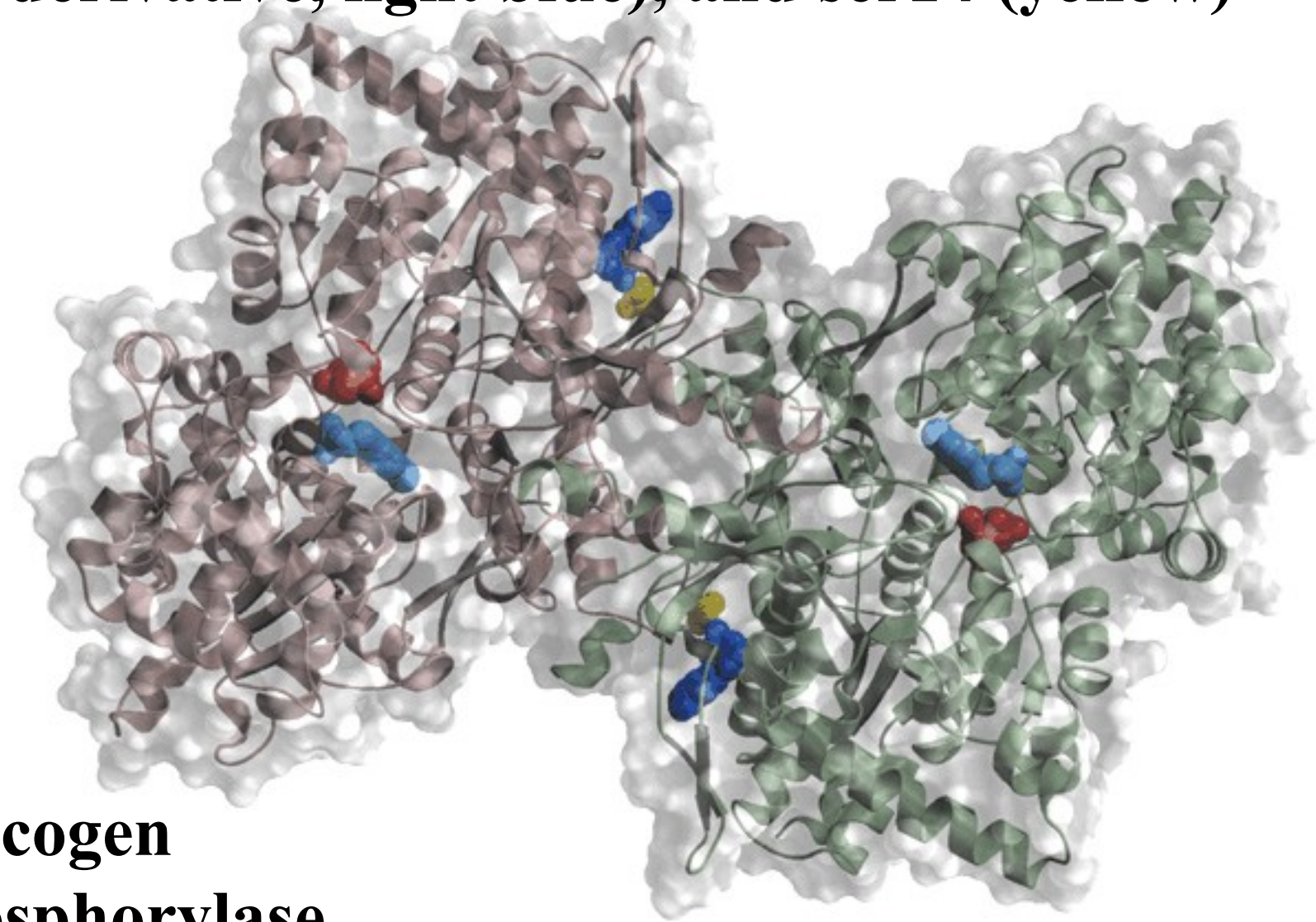
phosphorylase

$$(\text{Glucose})_n + P_i \rightarrow (\text{glucose})_{n-1} + \text{glucose 1-phosphate}$$

Glycogen

Shortened glycogen chain

Glucose (red), AMP (dark blue, allosteric activator), pyridoxal phosphate (PLP, B6 derivative, light blue), and ser14 (yellow)

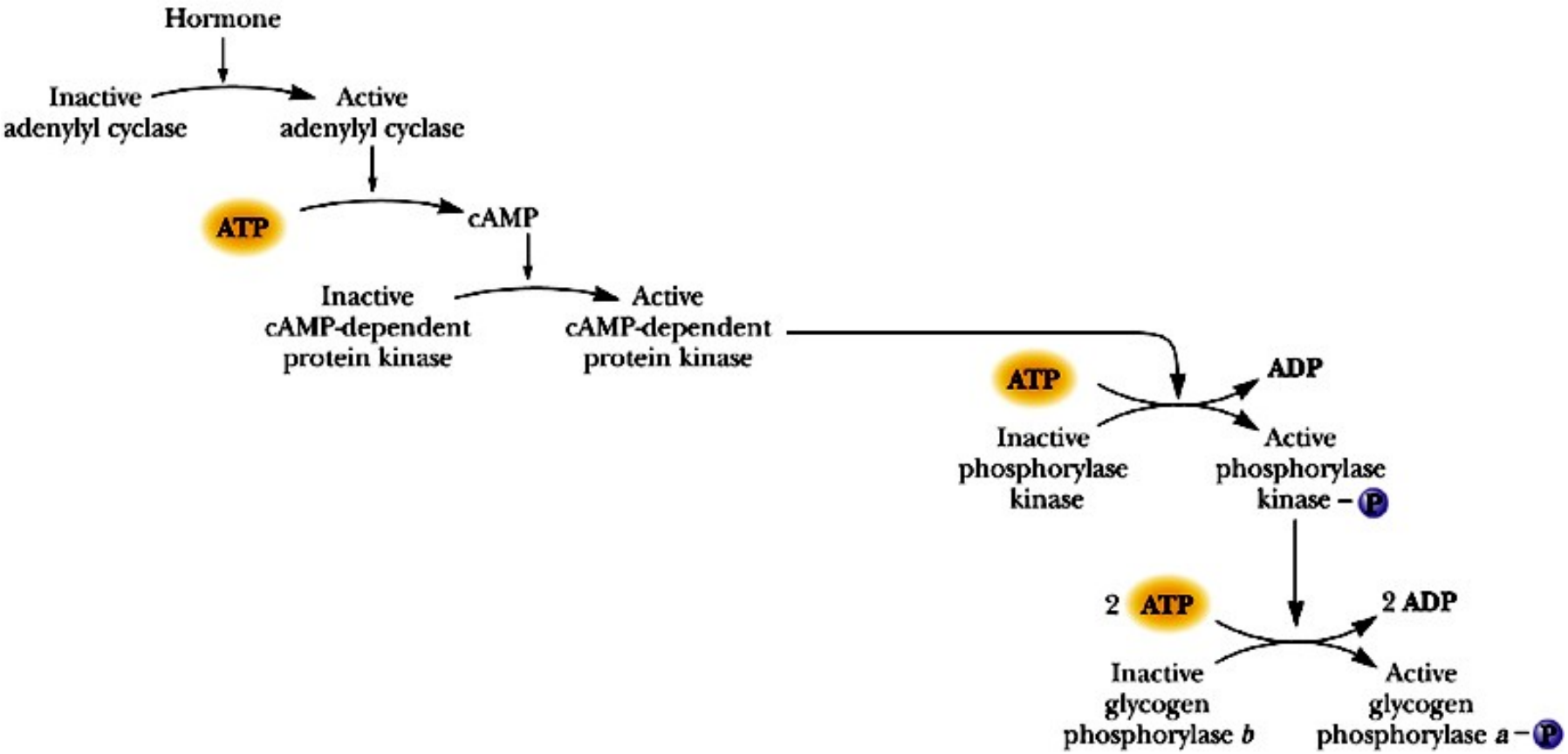


Glycogen phosphorylase

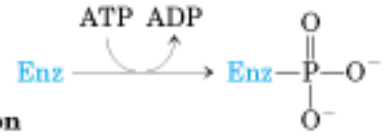
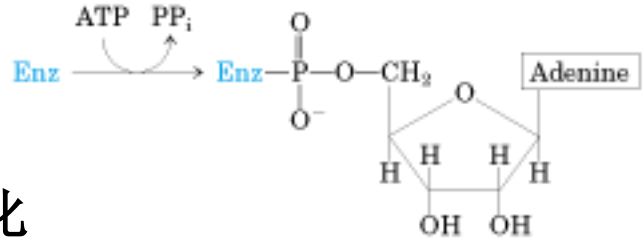
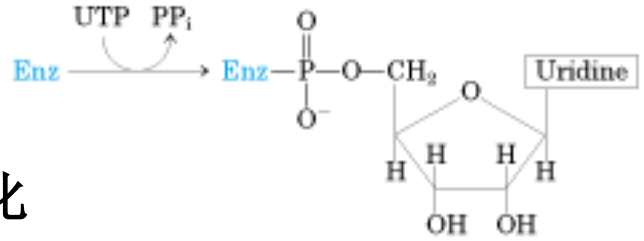
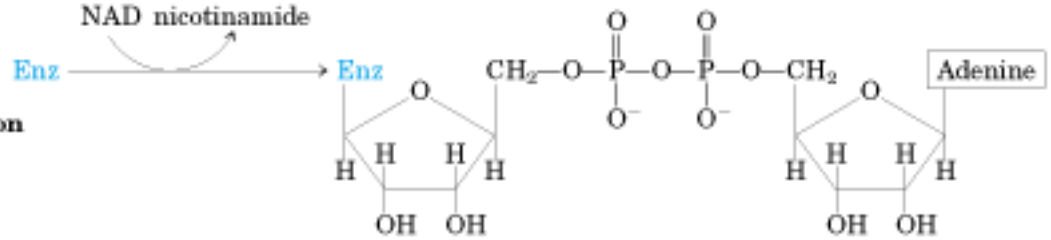
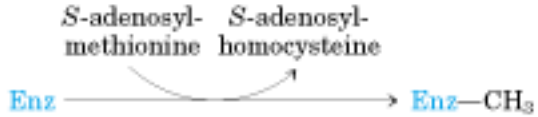
table 8–9

Consensus Sequences for Protein Kinases

Protein kinase	Consensus sequence and phosphorylated residue*
Protein kinase A	–X–R–(R/K)–X–(S/T)–B–
Protein kinase G	–X–R–(R/K)–X–(S/T)–X–
Protein kinase C	–(R/K)–(R/K)–X–(S/T)–B–(R/K)–(R/K)–
Protein kinase B	–X–R–X–(S/T)–X–K–
Ca^{2+}/calmodulin kinase I	–B–X–R–X–X–(S/T)–X–X–X–B–
Ca^{2+}/calmodulin kinase II	–B–X–(R/K)–X–X–(S/T)–X–X–
Myosin light chain kinase (smooth muscle)	–K–K–R–X–X–S–X–B–B–
Phosphorylase *b* kinase	*–K–R–K–Q–I–S–V–R–*
Extracellular signal–regulated kinase (ERK)	–P–X–(S/T)–P–P–
Cyclin-dependent protein kinase (cdc2)	–X–(S/T)–P–X–(K/R)–
Casein kinase I	–(Sp/Tp)–X–X–(X)–(S/T)–B
Casein kinase II	–X–(S/T)–X–X–(E/D/Sp/Yp)–X–
β-Adrenergic receptor kinase	$–(D/E)_n–$(S/T)–X–X–X–
Rhodopsin kinase	$–X–X–(S/T)–(E)_n–$
Insulin receptor kinase	–X–E–E–E–Y–M–M–M–M–*K–K–S–R–G–* *D–Y–M–T–M–Q–I–G–K–K–K–L–P–A–* *T–G–D–Y–M–N–M–S–P–V–G–D–*
Epidermal growth factor (EGF) receptor kinase	*–E–E–E–E–Y–F–E–L–V–*



级联放大反应，应用于信号转到途径

Covalent modification	Amino acid residues known to accept covalent modification
 **Phosphorylation** 磷酸化	Tyr, Ser, Thr, His
 **Adenylylation** 腺苷酰化	Tyr
 **Uridylylation** 尿苷酰化	Tyr
 **ADP-ribosylation** 糖基化	Arg, Gln, Cys, diphthamide (a modified His)
 **Methylation** 甲基化	Glu

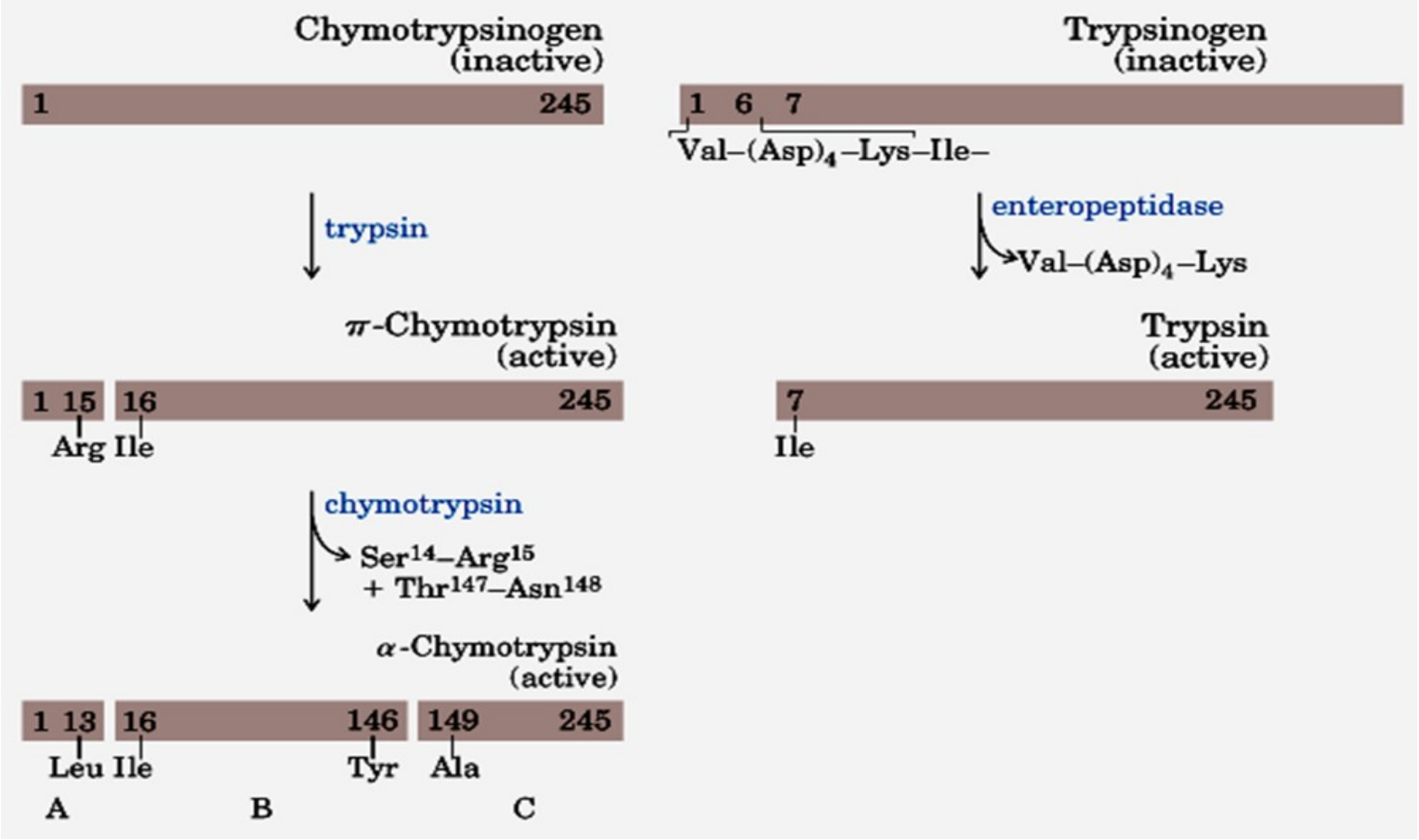
4. Many enzymes are activated by specific proteolytic cleavage （ Proteolytic activation ）.

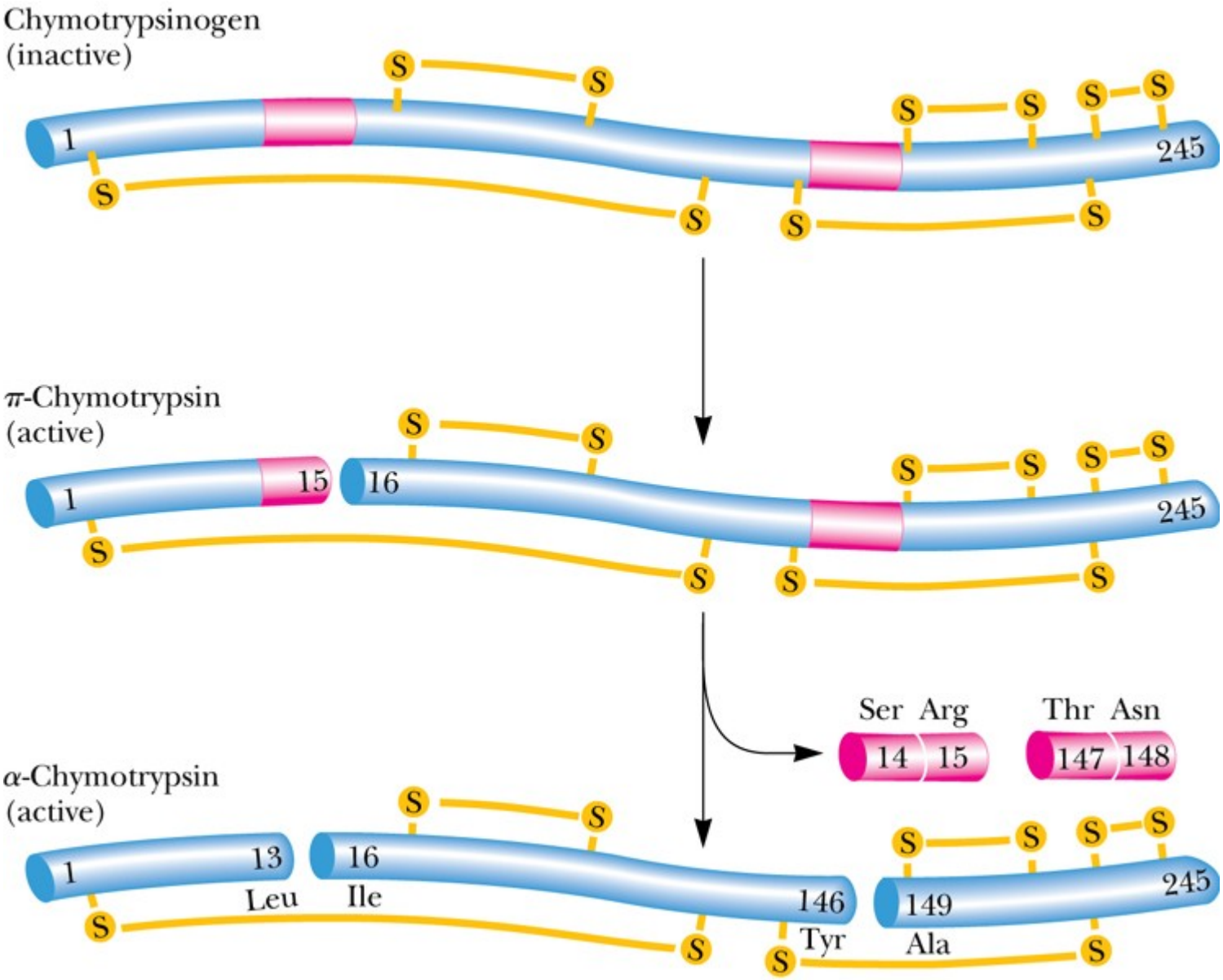
4.1 These enzymes are synthesized as inactive precursors called zymogens or proenzyme.

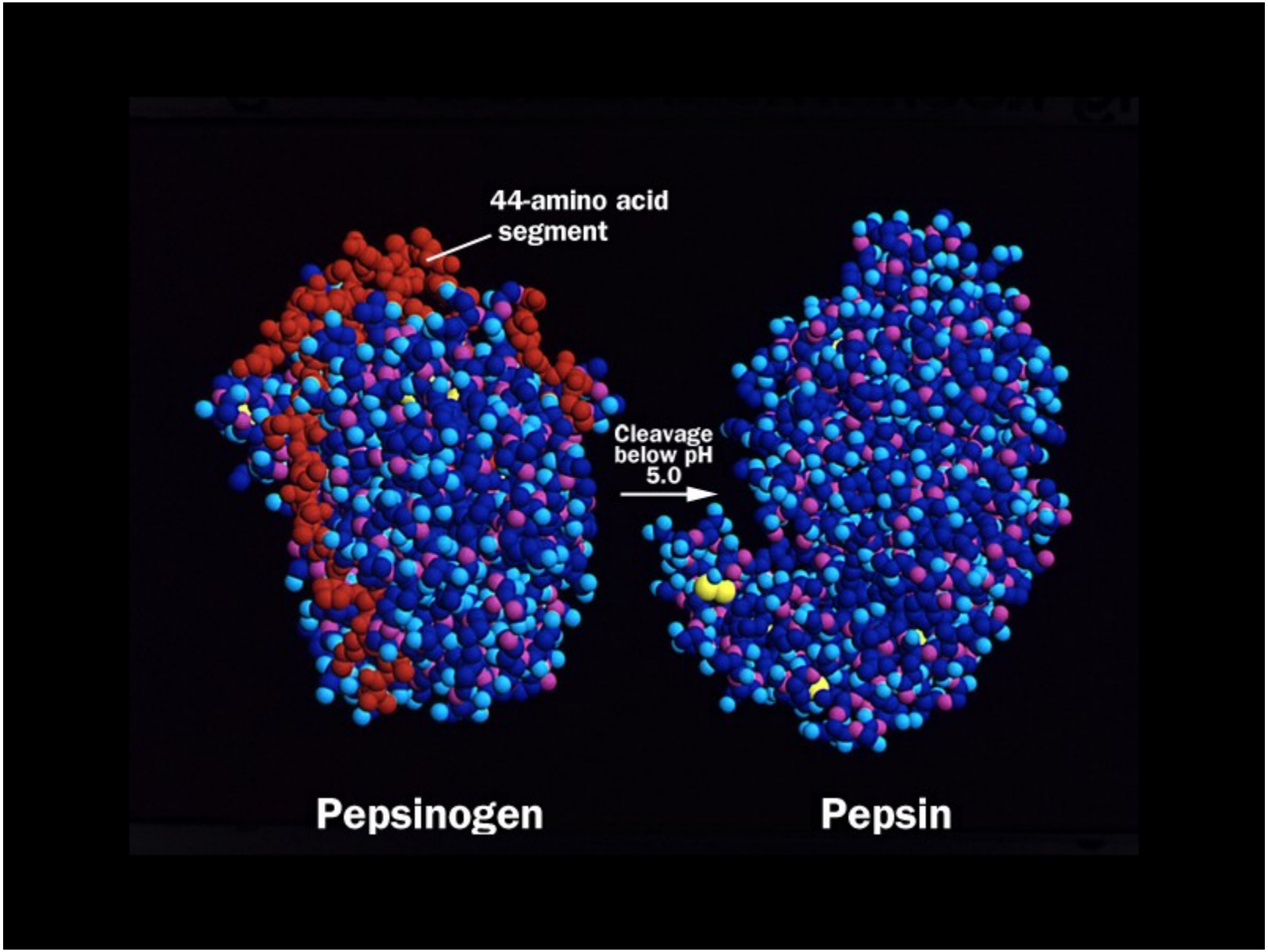
4.2 They are activated by cleavage of one or several specific peptide bonds.

4.3 Proteolytic activation, in contrast with allosteric control and reversible covalent modification, can occur just once in the life of the enzyme molecule.

4.4 The digestive enzymes that hydrolyze proteins are synthesized as zymogens in the stomach and pancreas.




Chymotrypsinogen
(inactive)
1
245
S
π-Chymotrypsin
(active)
1
15
16
245
Ser Arg
14 15
Thr Asn
147 148
α-Chymotrypsin
(active)
1
13
Leu
16
Ile
146
Tyr
149
Ala
245


44-amino acid
segment
Cleavage
below pH
5.0
Pepsinogen
Pepsin



Chymotrypsin

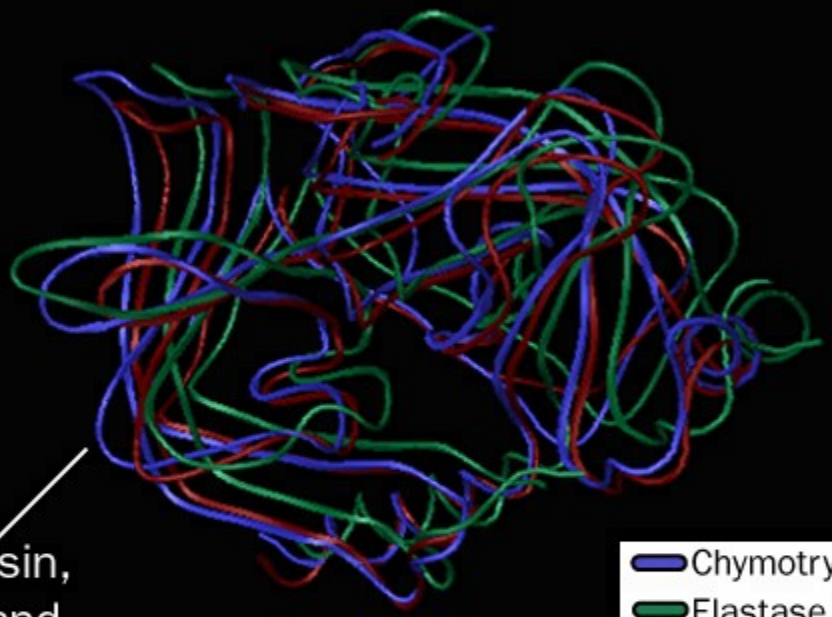


Elastase



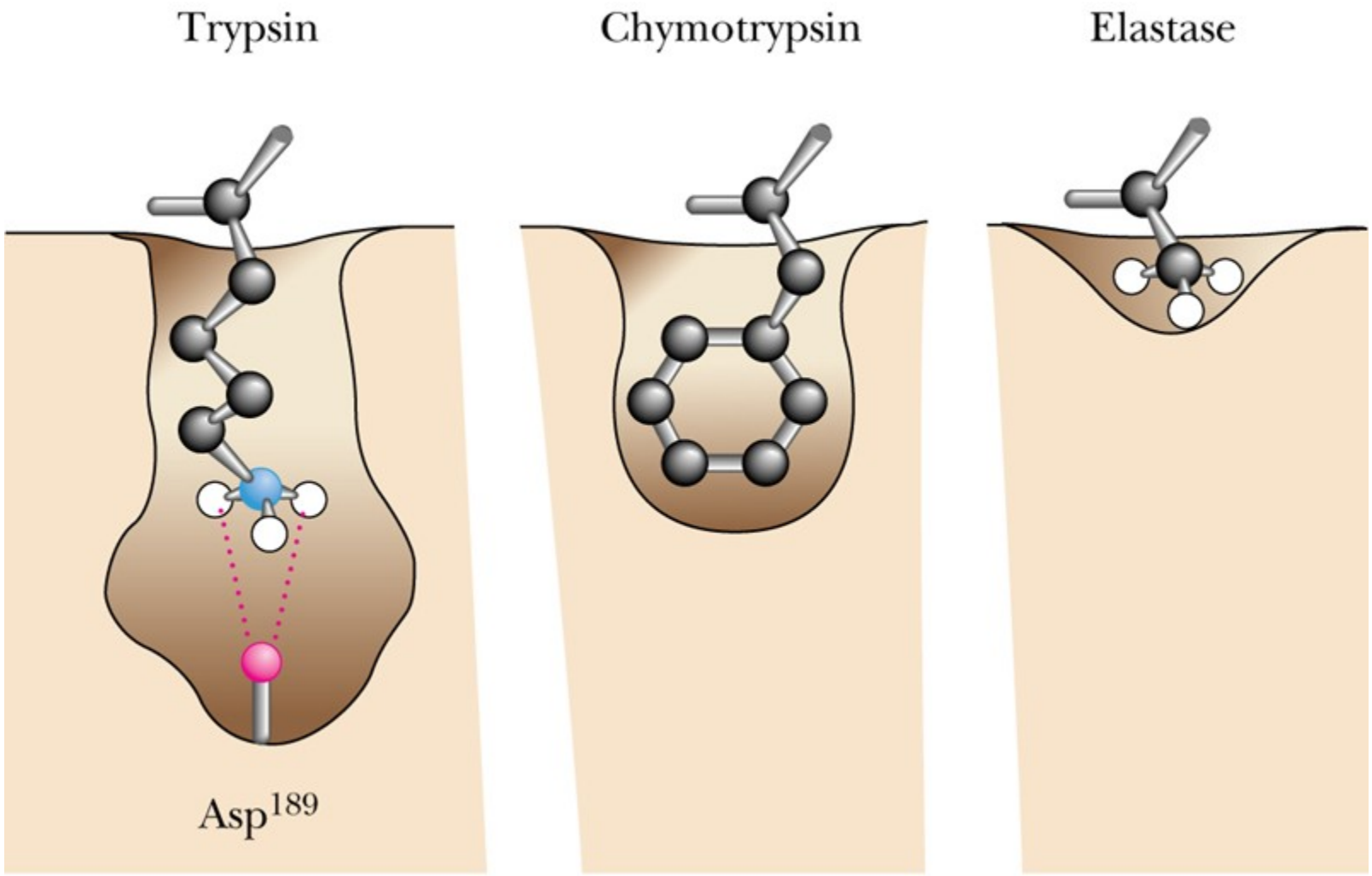
Trypsin

Serine proteases



Chymotrypsin, elastase, and trypsin super-imposed

Serine proteases

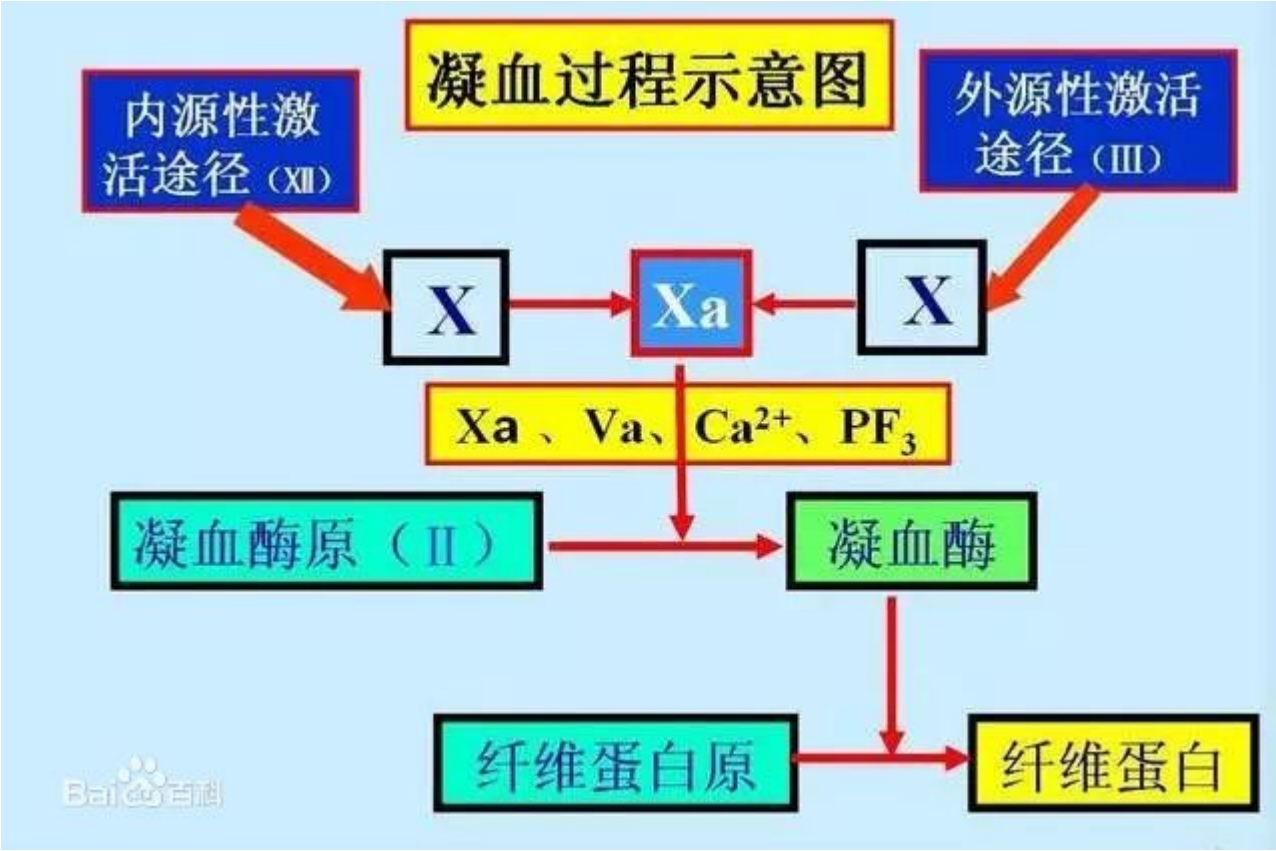

Trypsin
Chymotrypsin
Elastase
Asp189

组织蛋白酶 C	琉基蛋白酶	氨基端双肽，可通过 K，R 或 P 氨基端作为第二或第三个氨基酸封闭	醋酸碘， 甲醛
胰凝乳蛋白酶	丝氨酸蛋白酶	在 F- PheJ- Thr或 Y- Tyr之后	抑肽酶,PMSF,TPCK,a-巨球蛋白
胶原酶	金属蛋白酶	在 P-X-G-P 肽链中 X 之后	EDTA,EGTA,还原剂,但无血清存在
dispase	金属蛋白酶	无特异性	EDTA，EGTA，Hg^{2+}，重金属
内肽酶 Arg-C	丝氨酸蛋白酶	在 R- Arg之后	a 巨球蛋白，TLCK
内肽酶 Asp-N	金属蛋白酶	在 D- Asp和 C-Cys 半胱氨酸之前	EDTA，a 菲咯啉
内肽酶 Glu-C	丝氨酸蛋白酶	在 E- Glu/Gln或 D- Asp之后	a 巨球蛋白，TLCK
内肽酶 Lys-C	丝氨酸蛋白酶	在 K- Lys之后	TLCK，抑肽酶，抑蛋白酶醛肽
肠激酶	丝氨酸蛋白酶	在 D-D-D-D-K-肽链中 K 之后	
Xa 因子	丝氨酸蛋白酶	在 R- Arg之后	PMSF,APMS,大豆胰蛋白酶抑制物
无花果蛋白酶	琉基蛋白酶	无特异性	TPCK，TLCK，a-巨球蛋白
激肽释放酶	丝氨酸蛋白酶	在一些 R- Arg之后	抑肽酶，抑蛋白酶醛肽
木瓜蛋白酶	琉基蛋白酶	长期孵育时具有广泛特异性 arg、lys、gly、L-Citrulline	TPCK，TLCK，抑蛋白酶醛肽 a—巨球蛋白，烷化剂
胃蛋白酶	酸蛋白酶	广泛特异性 phe、trp、tyr 等疏水 aa	胃蛋白酶抑制素
纤溶酶	丝氨酸蛋白酶	在 K- Lys或 R- Arg之后	PMSF,TLCK,抑肽酶,a-巨球蛋白
链霉蛋白酶	混合型	无特异性	B．M．完整药片
蛋白酶 K	丝氨酸蛋白酶	广泛特异性	PMSF，PefablocSc
枯草杆菌蛋白酶	丝氨酸蛋白酶	无特异性	PMSF，a 巨球蛋白，苯甲脒
热溶素	锌金属蛋白酶	在非极性残基之前	EDTA
凝血酶	丝氨酸蛋白酶	在 K- Lys之后	TLCK，PMSF，抑蛋白酶醛肽 抑肽酶，a 巨球蛋白，苯甲脒
胰蛋白酶	丝氨酸蛋白酶	在 K- Lys或 R -Arg之后 arg、lys	TLCK，PMSF，抑蛋白酶醛肽 抑肽酶，a 巨球蛋白

Elastase(胰肽酶 , 丝氨酸蛋白酶类): The specific peptide bonds cleaved are those on the carboxyl side of small, hydrophobic amino acids such as glycine, alanine, and valine . For more on how this is accomplished, see serine protease.

4.5 Blood clotting is mediated by a cascade of proteolytic activation.

血液凝固（ **Blood clotting** ）简称凝血，是血液由流动状态变为凝胶状态的过程，它是止血功能的重要组成部分。凝血过程是一系列凝血因子被相继酶解激活的过程，最终生成凝血酶，形成纤维蛋白凝块。迄今为止，参与凝血的因子共有 12 个。其中用罗马数字编号的有 12 个（从Ⅰ－ⅩⅢ，其中因子Ⅵ并不存在）。



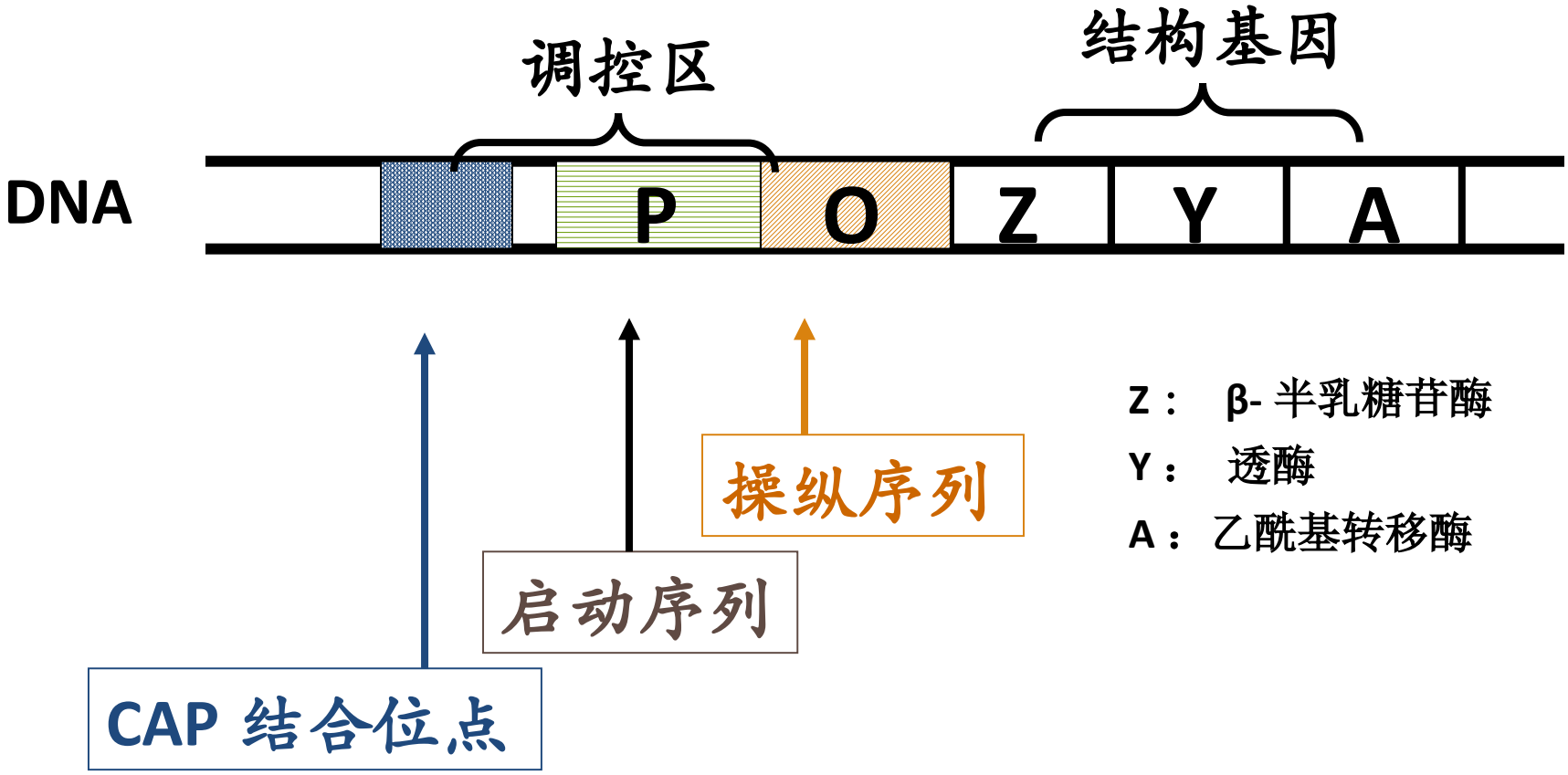
4.6 Some protein hormones are synthesized as inactive precursors (e.g., insulin is derived from proinsulin by proteolytic removal of a peptide).

5. Both the synthesis and degradation of the certain important enzymes are tightly controlled.

5.1 Amount of some enzymes are increased when certain inducers (often enzyme substrates) are present in the cells. This is often seen in bacterium cells.

5.2 The presence of lactose 乳糖 in a culture medium induces a large increase in the amount of β-galactosidase 半乳糖苷酶 (and two other enzymes, galatoside permease 半乳糖苷透过酶 and thiogalactoside transacetylase 硫代半乳糖苷转乙酰酶)

乳糖操纵子

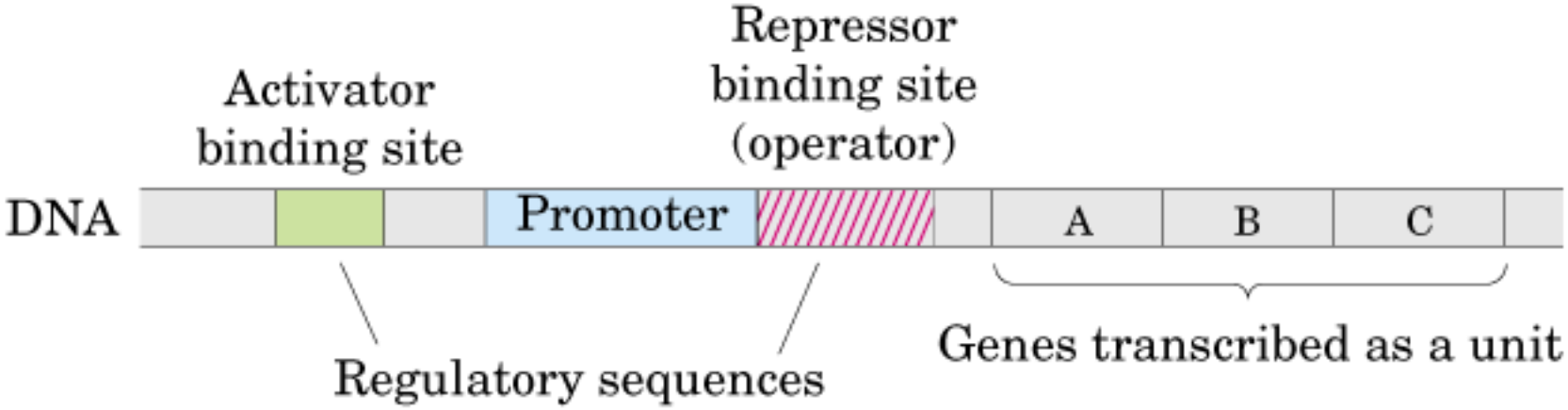


Z： β- 半乳糖苷酶

Y： 透酶

A： 乙酰基转移酶

cAMP—CAP 复合物


Repressor
binding site
(operator)
Activator
binding site
DNA
Promoter
A
B
C
Genes transcribed as a unit
Regulatory sequences

5.3 Once the enzyme protein has been synthesized, the rate of its breakdown can also be altered as a means of regulatory enzyme activity.

问答题

1、影响酶促反应的因素有哪些？它们是如何影响的？

2、试比较酶的竞争性抑制作用与非竞争性抑制作用的异同。

3、什么是米氏方程，米氏常数 Km 的意义是什么？试求酶反应速度达到最大反应速度的 99 %时，所需求的底物浓度（用 Km 表示）

4、举例说明蛋白质的结构和功能之间的相互关系。

5、称取 25 毫克某蛋白酶制剂配成 25 毫升溶液，取出 1 毫升该酶液以酪蛋白为底物，用 Folin- 酚比色法测定酶活力，得知每小时产生 1500 微克酪氨酸。另取 2 毫升酶液，用凯式定氮法测得蛋白氮为 0.2 毫克。若以每分钟产生 1 微克酪氨酸的酶量为一个活力单位计算，根据以上数据，求出（1）1 毫升酶液中含有的蛋白质和酶活力单位数；（2）该酶制剂的比活力；（3）1 克酶制剂的总蛋白含量和酶活力单位数。

名词解释

活性中心　　全酶　　酶原　　活力单位　　比活力
米氏方程　　Km　　诱导契合　　变构效应　　ribozyme
辅酶和辅基

1 . Specific activity

2 . coenzyme

3 . Allosteric enzyme

4 . Proteolytic activation

5 . Inhibitor

6. covalent modification

1 . Specific activity

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美国《科学》杂志社主编 Daniel Koshland 逝世

美国著名的《科学》杂志前主编、杰出的生物化学家 Daniel E.Koshland Jr.于 2007 年 7 月 23 日在一次严重的心脑血管病突发中逝世,享年 87 岁.

Koshland 于 1985~1995 年间担任《科学》杂志主编,他担当起了这一重任,进一步把《科学》杂志变成科学界的重要声音.

Koshland 长期担任加州大学伯克利分校分子和细胞生物学教授,诺贝尔奖得主、加州理工学院(CIT)生物学教授 David Baltimore 说:“Koshland 应用他的深邃化学知识研究生物学的重要问题.”伯克利分校分子和细胞生物学教授 Rolert Tjian 说:“和 Koshland 一起工作是一种愉悦,因为他总是那么努力、激励人、思想敏捷和认真.”

Koshland 于 1941 年毕业于伯克利分校,随即参加了曼哈顿计划,随 G.T.Seaborg 工作.1949 年获芝加哥大学有机化学博士学位.他的最大成就在蛋白质领域,他的“感应适应”(induced fit)理论改变了科学家对一系列蛋白质反应过程的了解,从酶的催化能力到激素的作用.他发表了 400 余篇论文.《科学》杂志现任主编 Don Knnedy 说:“作为一名继承者,我到处可以发现 Dan 对《科学》杂志的影响.我们一起策划了在科学院建 Koshland 科学博物馆.”

江 英 摘自《Science》, July 27,441(2007)《Chemical & Engineering News》, July 30,14(2007)

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